Dental caries

Prepared by

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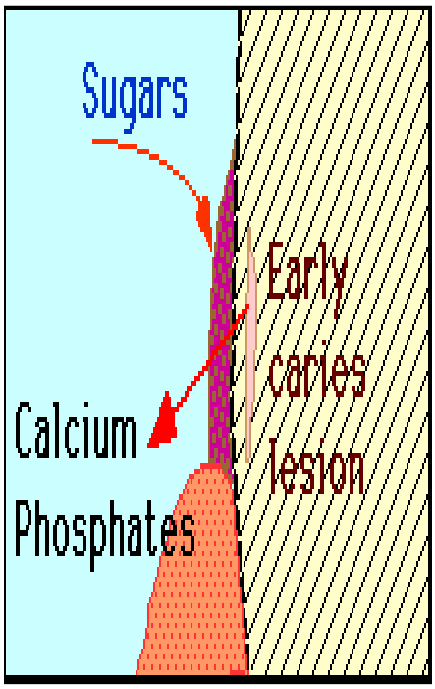
2026-2027

Dental caries

Definition:

- It's a progressive bacterial damage of hard tooth structure exposed to the oral environment, characterized by demineralization of the inorganic portion followed by destruction of the organic substance of the tooth leading to a cavity.
  Tooth+ **Bacteria**+ **CHO** ⟶ Organic acid ⟶ Demineralization of inorganic portion followed by breakdown of organic matrix.

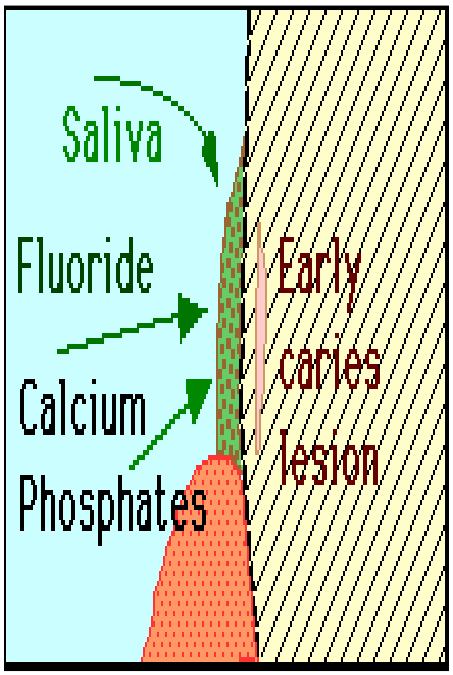
- Dental caries is considered to be a dynamic process consisting of alternating periods of destruction(**demineralization**) and repair (**remineralization).** occurring over a period of time.



Demineralization:

When carbohydrates undergo fermentation by the bacteria, they form acids which start to dissolve the enamel

- an early caries lesion occurs due to loss of Calcium and Phosphates

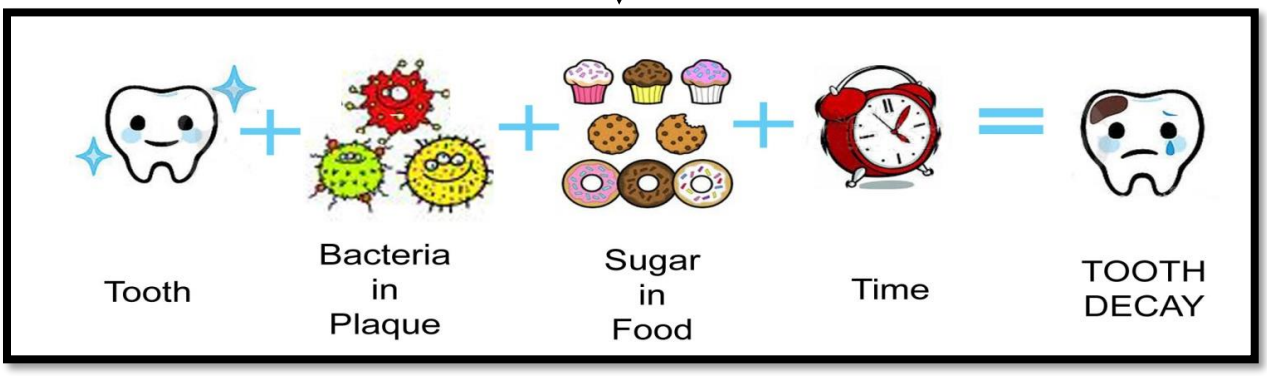


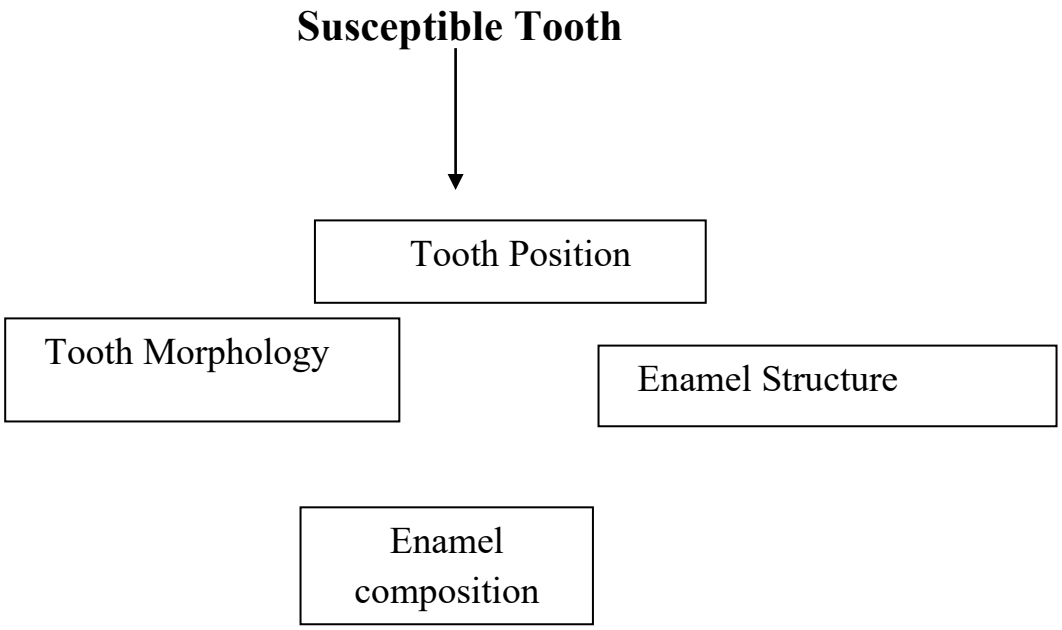
Remineralization:

When sugar consumption has ceased, saliva can wash away sugars and buffer the acids.

Calcium and Phosphates can again enter the tooth.
The process is strongly facilitated by fluorides

Etiology of dental caries





1- Tooth Position:

- Upper teeth are more susceptible to caries than lower teeth due to salivary action.
- Posterior teeth are more susceptible than anterior teeth due to the presence of pits and fissures.
- Malposed and crowded teeth and teeth related to clasps and orthodontic appliances are more susceptible to caries because of the creation of stagnation areas.

2- Tooth Morphology:

(a) Pit and fissure caries:

- Increased number of fissures increases caries.

- Increased depth of fissures increases caries.

- Shape of fissure: I- and inverted Y-shaped fissures are more susceptible to caries than U- and V-shaped fissures.

(b) Smooth surface caries:

- Interproximal surfaces: The contact areas are more susceptible to caries.

- Buccal and lingual surfaces: The cervical third of buccal and lingual surfaces are more susceptible to caries.

3- Enamel Structures:

- Enamel hypoplasia and enamel hypocalcification do not affect the initiation of caries but affect the progression of caries compared to normal teeth.

4- Tooth Composition (Role of fluoride in dental caries):

1- Decreasing demineralization process: Fluoride replaces the hydroxyl group in the hydroxyapatite forming fluorapatite which is less soluble in acids.

2- Increasing remineralization process.

3- Inhibition of bacterial growth.

Role of Saliva in dental caries:

(1) Saliva has a washing effect on the teeth.

(2) Saliva contains bicarbonates and phosphates acting as a buffering effect.

(3) The acquired enamel pellicle which is derived from salivary glycoprotein protects enamel surfaces from being colonized by oral bacteria.

(4) Saliva contains immunoglobulin A(IgA) which plays a role in,

a. Killing or inhibition of bacterial growth.

b. Prevention of adherence of bacteria to tooth surface.

(5) Saliva contains antibacterial substances: These include lysozymes, peroxidase and lactoferrin.

Microorganisms:

- ✓ It is generally agreed that dental caries is caused by acid resulting from action of microorganisms on carbohydrates.
- ✓ The most important organism in dental caries is **Streptococcus mutans.**

- ✓ **S. Mutans** has a role in initiation of dental caries, while **Lactobacillus** is associated with the progression of dental caries.

The characteristic features of Cariogenic bacteria:

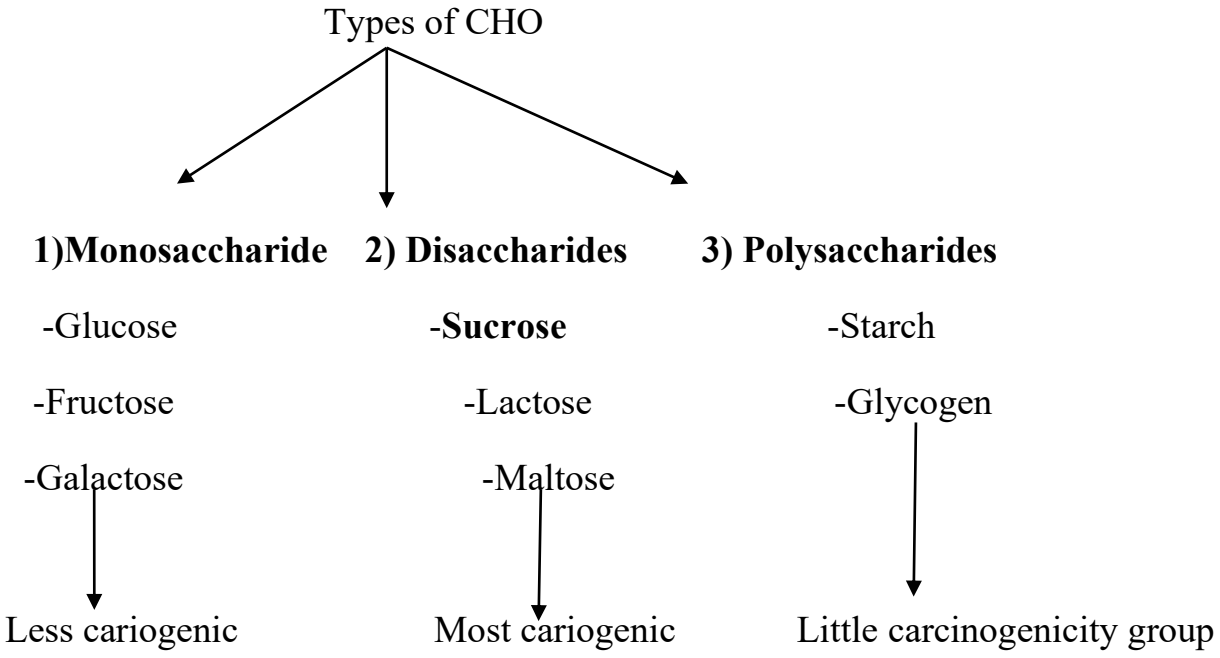
1-**Acidogenic property:** This means that the bacteria have the ability to produce organic acid rapidly from fermentable carbohydrates.

2- **Aciduric property:** This means that the bacteria can live and grow in high acidic media, i.e., low pH levels.

3- **Formation of abundant extracellular polysaccharides,** especially dextran and Levan so these bacteria can produce caries.

Carbohydrates (CHO):

- ✓ CHO are the only component of the diet which can be acted upon by microorganisms in the oral cavity. Proteins and fats undergo little or no change in the mouth.



Sucrose is considered the arch-criminal of dental caries due to

- Sucrose is of low molecular weight and can diffuse rapidly into the dental plaque.
- Sucrose is of low cost, so it is the most common and most frequently consumed carbohydrate (table sugar, drinks, jam, etc.).
- Sucrose is broken down rapidly by the bacteria with the production of organic acids.

Sucrose —Hydrolysis (strept. Mutans)→ Glucose ..... Energy+H2O+organic acid
Fructose.....

- Sucrose, with the help of cariogenic bacteria (Strept. mutans) plays an active role in the building of dental plaque by forming extracellular polysaccharides, mainly Dextran, which forms the main bulk of dental plaque and Levan, which acts as a store for CHO).

Glucose —glycosyltransferase→ Dextran.

Fructose —fructosyltransferase→ Levan.

Dextran	Levan
• Extracellular polysaccharide.	Extracellular polysaccharide.
• Glucose polymer.	Fructose polymer.
• Adhesive.	Less adhesive.
• Forms the main bulk of dental plaque.	Acts as storage.

Dental plaque

Definition: A biofilm that builds up on and around the teeth. Plaque is Sticky and usually colorless. It forms every day on teeth.

Composition:

- Dental plaque
  - microorganisms (60-70%)
  - Extracellular matrix
    - organic content
      - protein
      - CHO
      - lipid
    - inorganic content (Ca, p+Fluorid).

Notes:

- Proteins derived from saliva and gingival fluids include (glycoprotein, amylase. desquamated epithelial cells and Ag).

- CHO which include extracellular polysaccharide(Dextran+Levan).

Mechanism of formation of dental plaque:

1- Acquired enamel pellicle:

- This form is a glycoprotein that is derived from the saliva and adsorbed on the cleaned tooth surface.

Function of enamel pellicle:

- Protect the tooth from being colonized by oral bacteria.
- It is on this component of dental plaque takes place; thus, the pellicle serves as a nutrient for plaque microorganisms.



2- Colonization and growth phase:

Colonization phase

- Bacterial adhesion and colonization on the tooth surface begin within minutes after the pellicle is formed.
- The predominant initial colonizers (pioneer bacteria) are S.sanguis. S. oralis.
- It's believed that some of these pioneer organisms release an extracellular enzyme capable of liberating carbohydrate from pellicle glycoprotein and using it as so a source of energy.

Growth phase (8-48 hours):

- This begins with the ingress and proliferation of secondary invaders Streptococcus mutans.
- **S. mutans** appears in large numbers and has two effects,

1- Fermentation of sucrose and production of organic acid.

2- Synthesis of extracellular polysaccharides (dextrane & Levan) which form the main bulk of dental plaque.

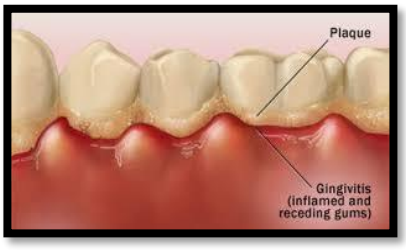
3- Maturation (2 weeks): Maturation of dental plaque by further colonization of other bacteria. Also, there is buildup of plaque substance by polysaccharides which are produced by microorganisms**.**

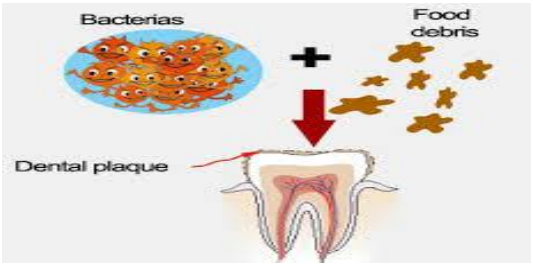
Role of Dental Plaque

1- Acting as a diffusion-limiting membrane by

- Delayed diffusion of salivary buffers into the plaque and
- Delayed diffusion of acids outward into the saliva.

2- Contributes to adhesiveness, bulk and resistance to washing.





Histopathology Features of Enamel Caries

When enamel loss minerals and caries progress, the enamel develops several zones visible under a light microscope.

1**- Translucent zone**:

- This zone lies at the advancing front of the enamel lesion.
- This zone is slightly more porous than sound enamel, having a pore volume of 1% compared with 0.1%of sound enamel.

2- **Dark zone:**

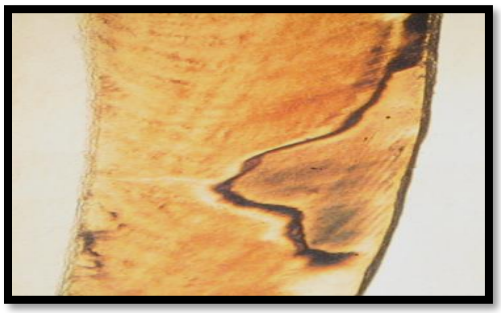
- Lies superficial to the translucent zone.
- Called positive zone as it is always present.
- Pore volume is 2-4%; increased porosity in this zone is due to the degree of demineralization.

3- Body of lesion:

- Forms the bulk of the lesion and lies between the dark zone and the surface zone.
- Area of greater demineralization, having a pore volume of 5-25%.

4-Surface zone:

- Relatively intact.
- Zone of greater mineralization (high fluoride) and pore volume of only 1%.



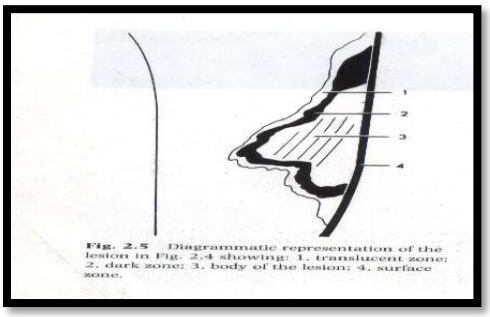


Fig. 2.5 Diagrammatic representation of the lesion in Fig. 2.4 showing: 1, translucent zone; 2, dark zone; 3, body of the lesion; 4, surface zone.

Histopathology of Dentine Caries

- ✓ Once bacteria penetrate enamel, they spread laterally along the DEJ and attack dentine over a wide area. Dentine becomes infected through dentinal tubules, which provide an easy pathway for bacteria.

Five zones that have been recognized in dentine caries include:

1- **Zone of Dentinal sclerosis:** characterized by deposition of calcium salts in dentinal tubules, which finally seals them against further penetration by microorganisms.

2- **Zone of decalcification:**

- Lies in front of bacterial invasion.

- Only a few tubules are invaded by bacteria (**pioneer bacteria**) even before clinical evidence of caries.

3- **Zone of bacterial invasion:**

- Dentinal tubules are packed with the invading bacteria, leading to their confluence.

- Coalescence and breakdown of adjacent tubules lead to formation of "**liquefaction foci".**

4- **Zone of destruction:** The liquefaction foci enlarge and increase in number, resulting in cracks or **clefts** containing bacteria and necrotic dentine.

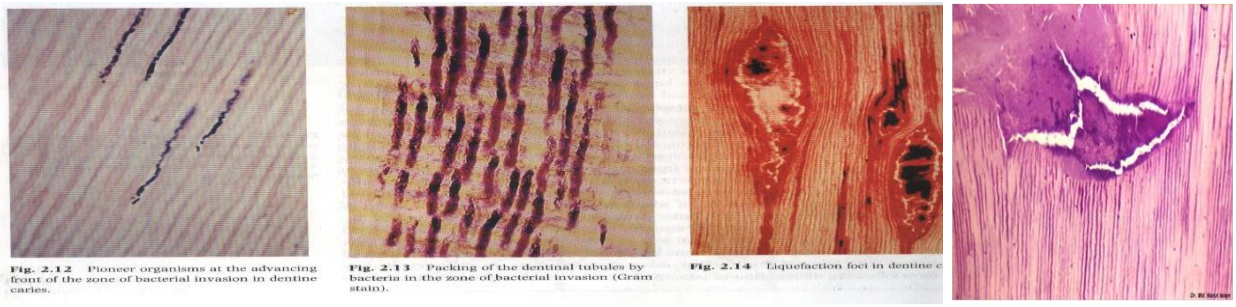
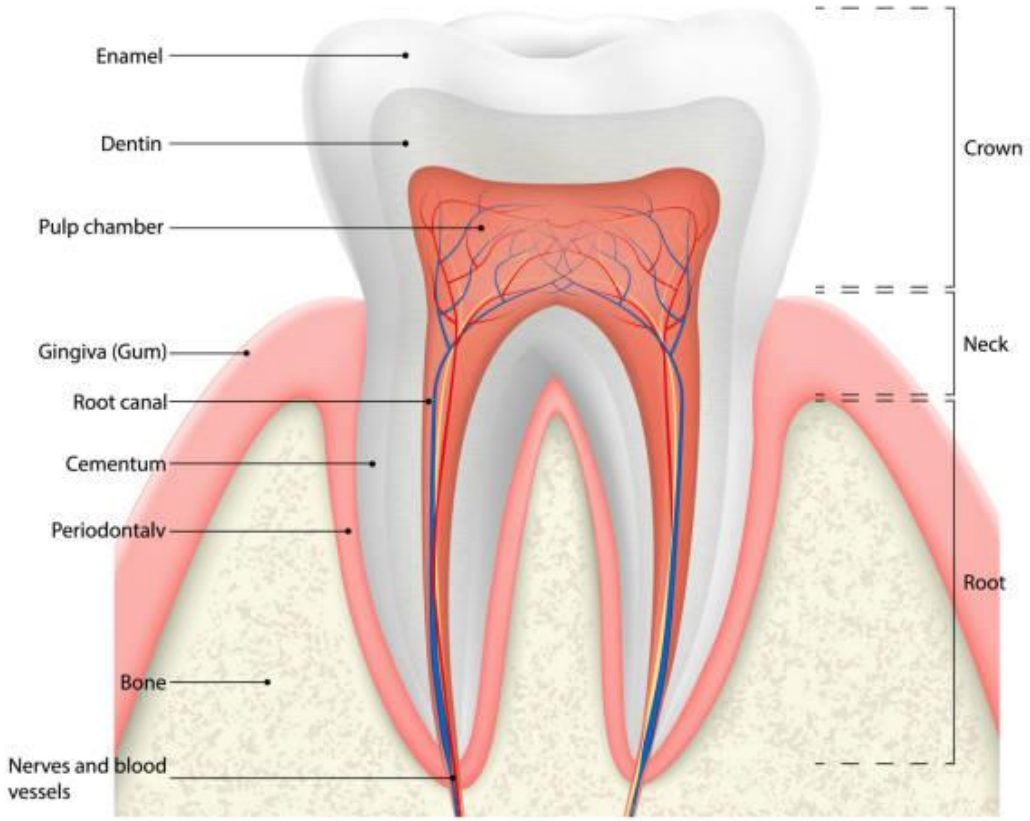


Fig. 2.12 Pioneer organisms at the advancing front of the zone of bacterial invasion in dentine caries.

Fig. 2.13 Packing of the dentinal tubules by bacteria in the zone of bacterial invasion (Gram stain).

Fig. 2.14 Liquefaction foci in dentine c

5-Zone of reactionary or tertiary dentin:



Diseases of the dental pulp

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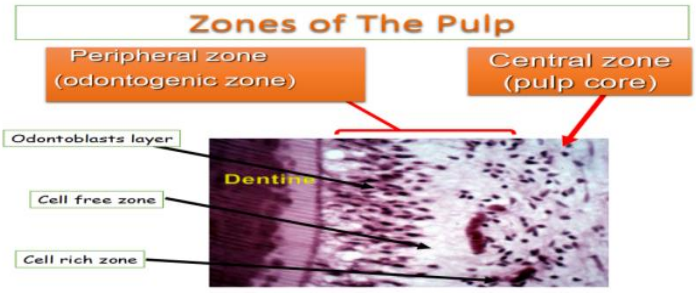
Third Year
2026- 2027

Disorders of the dental pulp

❖ Dental pulp

Dental pulp can be defined as a richly vascularized, innervated and unminerlized connective tissue enclosed by dentin with communication to the periodontal ligament by apical foramen and accessory canals. The embryonic origin of dental pulp from dental papilla which originates from neural crest cells.

During histological examination, four unique zones can be distinguished: Odontoblastic zones: it is representing the peripheral aspect of pulp, which contains the odontoblast cells body. Cell-free zone or zone of Weil: This zone contains many bundles of fibers, numerous capillaries and nerves but does not contain any cells. Cell-rich zone: it contains several fibroblasts which represent the predominant cells type of pulp and also contains undifferentiated mesenchymal cells which can be differentiated into different pulp cells. Pulp core: This zone mainly contains the main blood vessels in the pulp and Raschkow plexus of nerves .



Dental pulp like other connective tissues throughout the body; it reacts to noxious stimuli by an inflammatory response. This response is not significantly different from that seen in other tissues, the final result can be different because of certain anatomical features of the pulp which includes: -

1-The pulp is enclosed within the calcified walls of the dentin, which precludes the excessive swelling of the tissue that occurs in hyperemic and edematous phases of inflammation in other tissues.

2-Absence of collateral circulation.

3-Lack of propriceptors making pain localization to correct tooth is a diagnostic challenge.

Causes of pulpal disease:

I-Living irritants (Bacterial)

Bacteria can gain access to the pulp through dental caries, fracture of the tooth exposing the pulp, or haematogenous infection (Anachoresis).

II- Non living irritants:

1-*Physical*

a-Thermal

- Heat from cavity preparation
- Exothermic heat generated during cement setting

b. Electrical (galvanic current from dissimilar metallic fillings)

2-Mechanical:

a-Traumatic accident

b-Attrition and abrasion

c- Cracked tooth syndrome; particularly in posterior teeth, which may develop cracks or split under masticatory stresses.

3.Chemical:

a- Erosion

b- Inappropriate use of acidic dental material, for example, sterilizing agents, monomers present in composite restoration, or arsenic present in silicate cement.

4. ***Idiopathic:*** Aging, internal and external resorption.

5. ***Barodontalgia:*** It's defined as tooth pain resulting from change in barometric pressure. "Baro" means pressure, and "odontalgia" means tooth pain. Is a condition in which there is toothache in persons flying at high altitudes and underwater divers. It is usually associated with recently filled teeth or teeth affected by hyperemia.

N.B: Dental ***attrition, abrasion, and erosion*** are types of dental wear. However,

abrasion (caused by contact between teeth and other materials), **attrition** (tooth-to-tooth contact), and **erosion** (dissolution of hard tissues by acidic substances).

Classification of Pulpal diseases

I. Pulpitis (Pulp inflammation).
II. Necrosis.
III. Pulp Calcifications/ Pulp Stones/ Denticles.
IV. Internal resorption

Pulpitis (Pulp inflammation)

Pulpitis refers to the inflammation of the dental pulp within the tooth and it is the most common cause of dental pain.

Dental caries remains the primary cause of pulpitis worldwide. Other causes (less common than caries) including trauma, restorative procedures, periodontal diseases, and iatrogenic factors.

Classification of Pulpitis:

(1) According to severity

- a-Acute (symptomatic).
  b-Chronic (Asymptomatic).

(2) According to extent of pulp involvement

a-Focal or Partial Pulpitis.

b-Total or Generalized Pulpitis.

(3) According to presence or absence of direct communication between dental pulp and oral environment

a- Open pulpitis.

b- Closed pulpitis: It may occur from trauma or excessive orthodontic forces. Here carious lesion is absent.

(4) According to recover from the insult and damage

a- Reversible Pulpitis.

b- Irreversible Pulpitis.

Focal Reversible Pulpitis

(Pulp Hyperaemia)

Is a reversible inflammation of the dental pulp, causing temporary sensitivity to stimuli like cold or sweet foods, which resolves quickly once the stimulus is removed. It represents the earliest stage of pulpal inflammation from an irritant like a decayed cavity, and can be healed by removing the cause, such as treating the decay.

Pulp hyperaemia is due to active dilatation of pulpal blood vessels.

Aetiology: Any of previously mentioned etiological factors can cause pulp hyperaemia.

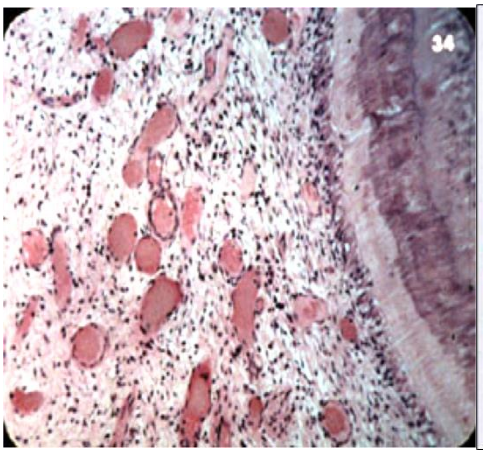
Clinical features:

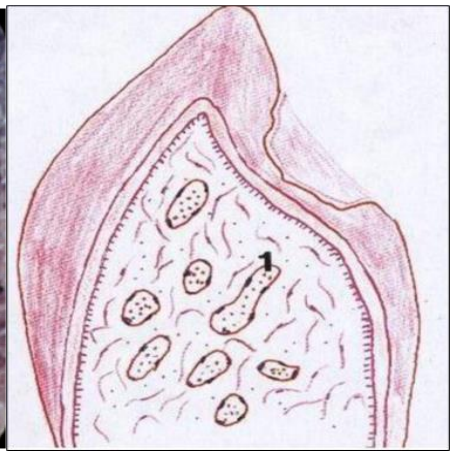
- The tooth is sensitive to thermal changes particularly cold application. It is usually caused by cold, sweet and sour stimuli.
- The pain is sharp shooting pain.
- The pain disappears when the stimulus has been removed.
- The tooth is more sensitive to electrical pulp tester.
- In focal reversible pulpitis, the pain is typically easily localized to a specific tooth because it is triggered by a stimulus and resolves quickly once the stimulus is removed.
- However, the inability to localize the tooth perfectly can still occur, as the dental pulp lacks proprioceptive fibers and nerve fibers from a single tooth may branch to innervate other teeth, making pain sometimes diffuse.

Histopathological Features:

1. Dilated blood vessels which carry more blood to the area of injury.
2. Intact odontoblastic layer.
3. Increased permeability of the capillary walls leads to fluid exudates in the pulp tissue (oedema) that press on nerve endings causing pain.
4. Mild inflammatory cell infiltrate- mainly lymphocytes and neutrophils.

5. Presence of reactionary dentine.





Treatment:

Removal of the cause if possible

Prognosis:

Good prognosis for pulp recovery with appropriate dental intervention, such as placing a filling.

Sequelae:

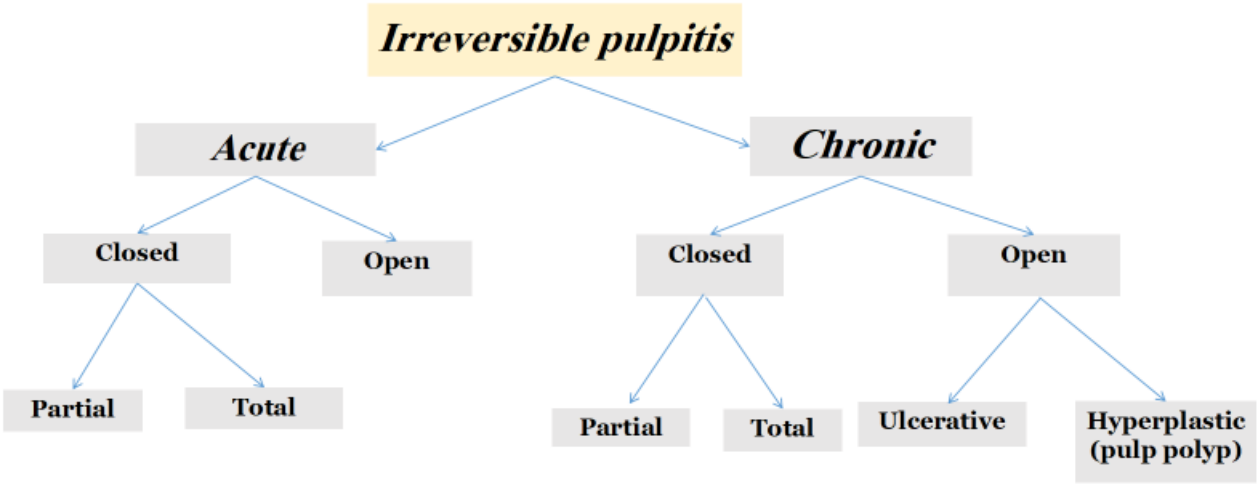
Regression (if treated before pulp damage).

Irreversible pulpitis (acute or chronic).

Irreversible pulpitis:

The process of irreversible pulpitis may be acute or chronic pulpitis.

Acute pulpitis may be a progression of focal reversible pulpitis or may present as an acute exacerbation of an already established chronic pulpitis.



Acute pulpitis

Irreversible condition characterized by acute, intense inflammatory reaction in pulp tissue.

Mode of development:

- Sequelae of focal reversible pulpitis.
- Acute exacerbation of chronic pulpitis
- As denovo condition, where inflammation is acute from beginning.

Types:

- Partial acute pulpitis confined to one horn of coronal pulp.
- Total acute pulpitis involves the whole pulp.

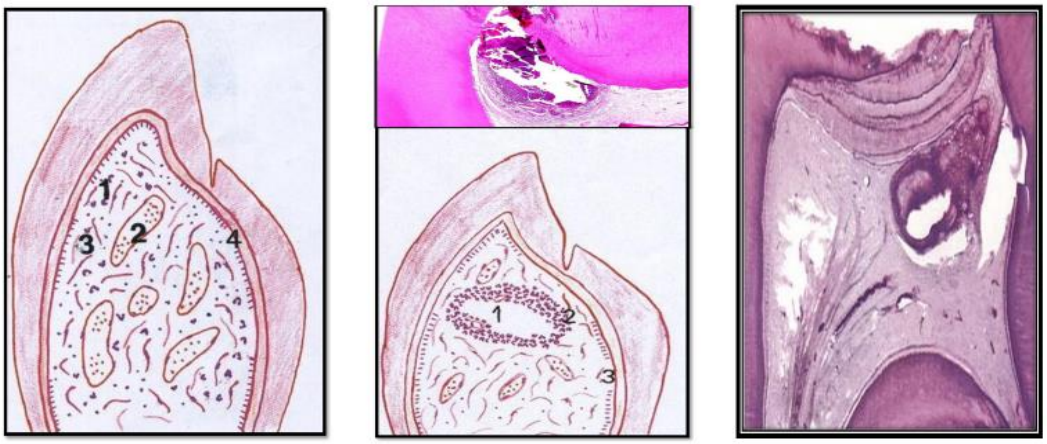
Clinical features:

- The patient with early irreversible pulpitis presented with sharp, sever pain (lancinating, throbbing) on thermal stimulation, and the pain continues after removal of the stimulus. Cold is the most uncomfortable, although heat or sweet and acidic food can cause pain.
- Later on, the pain may be spontaneous or continuous and may be exacerbated when the patient lies down.
- The tooth responds to electric pulp testing at lower levels of current.
- Mobility and sensitivity to percussion are negative.

- At early stage, the pain often can be localized easily to the individual affected tooth. With time the patient discomfort is increasing and can no more be able to identify the offending tooth.
- In the later stages of irreversible pulpitis, the Pain increases in intensity and experienced as throbbing, which keeps the patient awake at night. At this point heat increases the pain, while cold may produce relief. why?
- As the pulpal inflammation progresses, heat intensifies the response because it has an expansible effect on blood vessels. Cold tends to relive pain because of its contractile effect on vessels, reducing the intra-pulpal pressure.

Histopathological features:

1. Destruction of the odontoblastic layer.
2. Dilatation of blood vessels (Vasodilatation).
3. Edema in the connective tissue.
4. Acute inflammatory cells infiltration (polymorphonuclear leukocytes).
5. Focal areas of tissue destruction occur leading to formation of pus. Pus consists of necrotic tissues, dead and alive polymorphs and dead and alive bacteria. PMNLs in concentric layer around the pus form pulp abscess.
6. Numerous abscesses appear. Finally fuse together leading to total liquefaction of pulp forming suppurative pulpitis.



Treatment:

Root canal therapy (RCT) or tooth extraction.

Chronic Pulpitis

Chronic pulpitis is a chronic inflammation of dental pulp with little or no symptoms.

Aetiology:

1. It may follow acute pulpitis or arise de novo.
2. In case of de novo chronic pulpitis, the causative agents are usually of low virulence and the body resistance is high.
3. The general etiologic factors for pulpitis can also cause chronic pulpitis.

Types:

- It may be closed or open.

  The open may be ulcerative or hyperplastic (pulp polyp).

Clinical features:

- Pain is not a feature of chronic pulpitis due to diminished exudative inflammatory activity and decrease in intra-pulpal pressure.
- In some cases, patients may complain of intermittent dull pain.
- The reaction to thermal changes and EPT is reduced.
- The tooth is not sensitive to percussion.

Histopathological Features:

- The pulp tissue exhibits increased fibroblastic activity (to wall-off the infection).
- The pulp infiltrated with chronic inflammatory Cells (plasma cells, lymphocytes, and macrophage), fibroblasts, + new blood vessels.
- Adjacent odontoblasts are destroyed.
- Incomplete calcific barrier around the inflammatory focus.

Treatment:

Root canal therapy (RCT) or tooth extraction.

Chronic Hyperplastic Pulpitis (Pulp Polyp)

It is a hyperplasia of a chronically inflamed pulp, characterized by hyperplasia of connective tissue of the pulp in the form of polypoid mass which fungated from exposed pulp chamber in response to persistent, low-grade mechanical irritation and bacterial invasion of the pulp.

For pulp polyp to develop,4 prerequisites should be fulfilled

1) A good sheltered area for protection of the newly formed tissue.
2) A wide apical foramen for good blood supply.
3) A wide pulp exposure to allow protrusion of new tissue.
4) Young aged patient for good proliferative power.

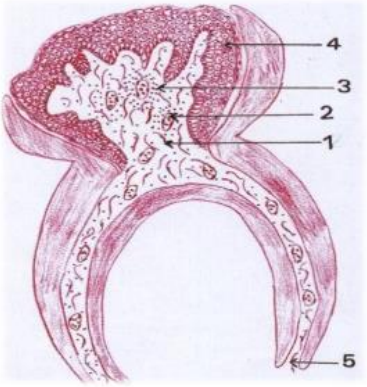
Clinical features:

- It occurs almost exclusively in children & young adults and involves teeth with large open carious cavity.
- Usually occurs in deciduous molars or the first permanent molar due to excellent blood supply and large root opening.
- There is a piece of tissue extruded from the pulp and present in a large carious lesion.
- The lesion is relatively asymptomatic (insensitive to manipulation because it contains fewer nerve endings) but may cause discomfort during mastication due to the pressure of the food bolus.
- It bleeds easily because of rich network of blood vessels.
- Should be differentiated from gingival polyp which is attached to the gingiva and is more sensitive to manipulation with less tendency to bleeding if compared to pulp polyp.

Histopathological features:

1) The pulp chamber is transformed into granulation tissue containing newly formed fibroblasts, collagen fibrils, blood capillaries and chronic inflammatory cells.

2) The granulation tissue is covered by stratified squamous epithelium derived from gingiva or desquamate epithelial cells of the oral mucosa or tongue.



1. Collagen fibers.
2. Blood vessels.
3. Chronic inflammatory cells.
4. Stratified squamous epithelium.
5. Wide apical foramen.



Treatment:

Pulp polyp (chronic hyperplastic pulpitis) treatment depends on the tooth's condition, but generally involves surgical removal of the polyp followed by a root canal (pulpectomy) if the tooth is restorable, or extraction if too much tooth structure is lost.

Pulp Necrosis

Pulp necrosis is the death of the dental pulp typically caused by infection from a deep cavity, crack, or trauma. As this is associated with bacterial infection called **pulp gangrene.**

Etiology:

- It follows either pulpitis (acute or chronic) or traumatic injury to apical blood vessels.
- In sickle cell anemia, blockage of pulp vessels by defective RBC (cannot flow smoothly through the tiny pulp blood vessels) results pulp necrosis.

Clinical features:

- It is associated with foul odor when pulp is opened for endodontic treatment.
- Pain is absent (asymptomatic).

- Discoloration of the tooth is the first indication that the pulp is dead (due to heamolysis of red blood cells).

Histopathological features:

The pulp chamber and canals appear empty or contain structureless mass of necrotic pulp tissue. i.e. no cell, no fibers, no blood vessels.

Treatment:

RCT (root canal therapy) or tooth extraction.

Pulp Calcifications

Pulp calcifications are mineralized masses in the dental pulp and are generally classified as pulp stones or pulp canal obliteration (PCO).

Pulp stones are nodular, calcified masses appearing in either or both the coronal and root portion of the pulp. The larger calcification is called denticles. Pulp stones may form either due to some injury or natural phenomenon.

Classification of pulp stone:

1.On the basis of structure

a. True denticles: These are rare and they are similar in structure to dentin in that they have dental tubules and surround by odontoblasts.

b. **False denticles:** These stones do not exhibit dentinal tubules but appear instead as concentric layers of calcified tissue.

2.On the basis of location

a. Free: These are entirely surrounded by pulp tissue.

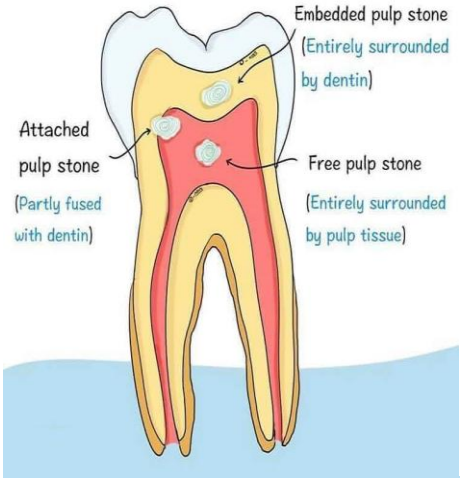
b. Adherent(attached): Partly fused with dentin.

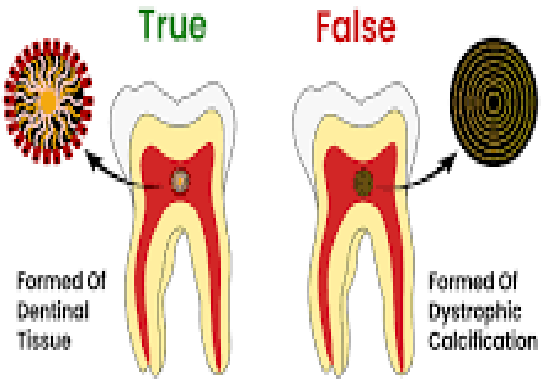
c. Embedded: These are entirely surrounded by dentin.

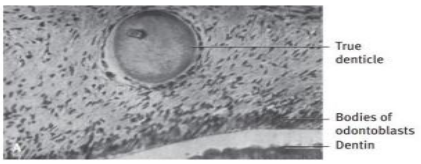
3.On the basis of the size

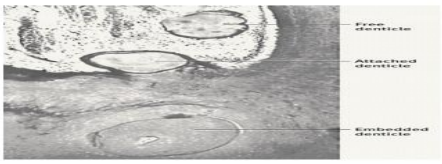
a. Diffuse

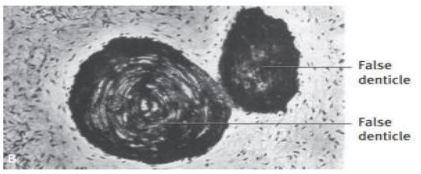
b. Fine

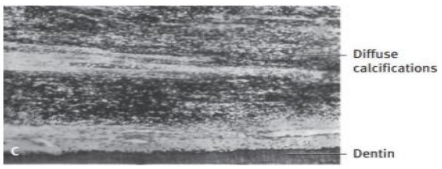












Clinical Features:

- No clinical significance. Usually, it is discovered on a radiograph only as radio opacity.
- Sometimes, it may cause pain, from mild to severe pain.
- Difficultly during root canal therapy.
- Associated with some conditions, like dentin dysplasia type II, and Ehlers-Danlos syndrome.

Treatment:

- No treatment is necessary if the tooth is asymptomatic.
- The removal of pulp calcification (pulp stones) is present another challenge in endodontic treatment, as they block the access to the root canals. The best treatment for this condition is the combination of magnification and ultrasonic tips.

True pulp stone	False pulp stone
✓ It result from undeferentated mesenchymal cell due to inductive effect of remanent of HERS.	✓ It occures at area of heamorrage or degenerated cells
✓ It contains organic matrix with dentinal tubules and surround by odontoblasts .	✓ It consist of concentric layer of calcific material with no tubular structure
✓ Small in size	✓ Large in size

Hertwig's epithelial root sheath =HERS

Pulp obliteration

or pulp canal obliteration (PCO)

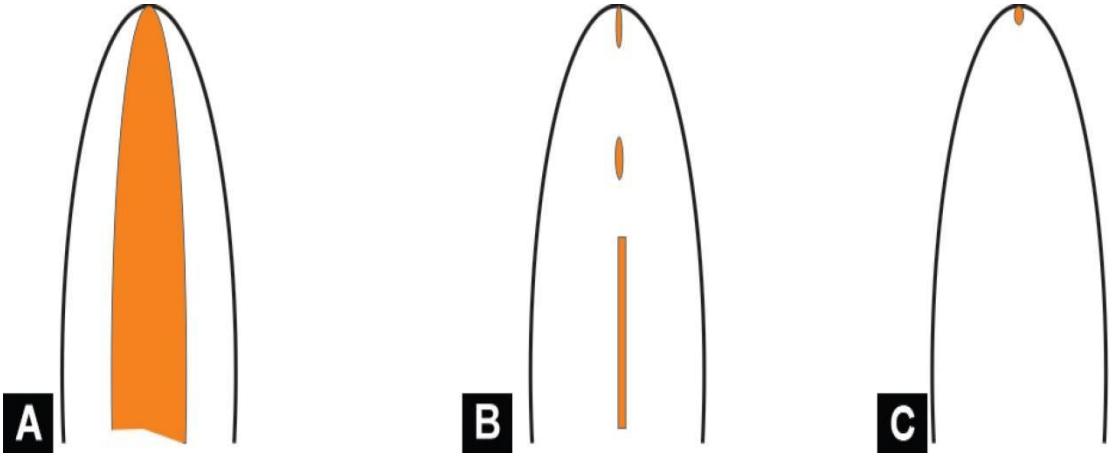
Pulp obliteration, or pulp canal obliteration (PCO), is the excessive, rapid formation of dentin within the pulp chamber and root canals, reducing or eliminating the pulp space.

Causes of pulp obliteration:

1. Trauma this is the most common cause, often following concussion or subluxation injuries, especially in young, developing teeth.
2. Aging: The process can naturally occur as part of aging.
3. Systemic conditions such as secondary hyperparathyroidism.
4. Dental anomalies such as dentinogenesis imperfecta and dentinal dysplasia.
5. Medications: Long-term use of certain medications, such as glucocorticoids (GCs).

Clinical features:

- A common sign is a yellowish or grayish discoloration of the affected tooth's crown due to increased dentin thickness.
- Routine radiographs will show a reduction in the size of the pulp chamber and narrowing or apparent complete obliteration of the root canal.
- Teeth with PCO often show lack of response to pulp vitality tests.



Progressive obliteration of pulp canal space: (A) Normal root canal; (B) Partially obliterated root canal; (C) Completely obliterated root canal

Internal resorption /pink tooth

It is an idiopathic slow or fast progressive resorptive process occurring in the dentin of the pulp chamber or root canal of teeth.

The condition is more frequently observed in male than in female subjects.

Etiology:

Unknown but is usually secondary to pulpitis.

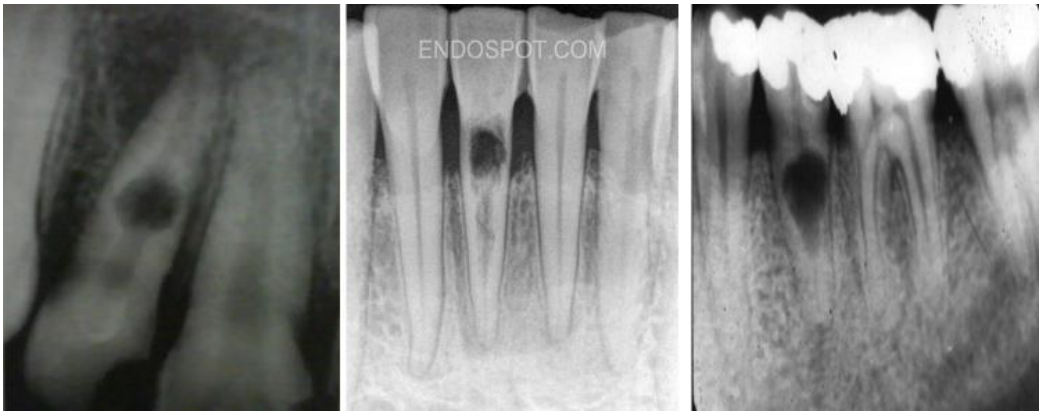
History of trauma either a blow or restorative procedures.

Clinically:

In case of coronal resorption, the presence of a pink spot in the coronal portion of the affected tooth is due to presence of granulation tissue just under the enamel.

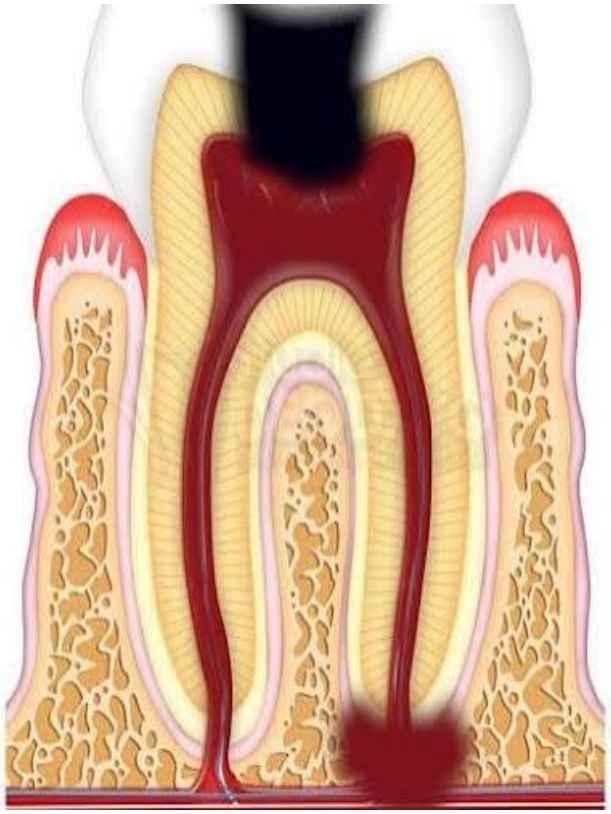
Radiographically:

Symmetrical, round, well defined radiolucency continuous with the pulp space. Pulp space appears enlarged; root canal outline is distorted.



Age change in the pulp:

1. Decreased pulp size gradually with age due to production of secondary dentine.
2. Decreased vascularity.
3. Decreased cellularity.
4. Increased collagen fibers.
5. Increased pulp stones.



Diseases of Periapical Tissues

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Third Year

2026- 2027

Diseases of Periapical Tissues

The periapical tissue is composed of connective tissue, blood vessels, nerves, periodontal ligaments and the alveolar bone that collectively support and anchor the tooth.

Periapical tissue diseases are inflammatory conditions that affect the periapical tissue and surrounding structures, particularly around the apical portion of the root.

- Inflammation in the periapical part of the periodontal ligament is similar to that occurring elsewhere in the body. A particular feature of inflammation at this site is that the adjacent bone and occasionally the root apex may resorb. The periapical vascular network has a rich collateral circulation, which greatly enhances the ability of tissue to heal if the cause of inflammation is removed.
- Other feature is that the patients are often able to locate the symptoms to a specific tooth due to stimulation of the proprioceptive nerve endings in the periodontal ligament.
- In periapical periodontitis the pain is induced by pressure, percussion, or biting.

Etiology of periapical diseases:

- ✓ 1. Spread of bacterial infection following pulp necrosis.
- ✓ 2. Occlusal trauma.
- ✓ 3. Inadvertent endodontic procedures.
- ✓ 4. Infection through the gingival crevice.

Periapical diseases include

1. Acute periapical periodontitis.
2. Chronic periapical periodontitis (Periapical granuloma).
3. Periapical cyst (Apical periodontal cyst, Radicular cyst).
4. Periapical abscess (Dentoalveolar abscess).

Periapical Periodontitis(Apical periodontitis)

Definition:

It is an inflammation of periodontal tissues around apex of a non-vital tooth.

Types:

1. Acute periapical periodontitis.
2. Chronic periapical periodontitis (commonly presenting as a periapical granuloma).

Complications / Sequelae of periapical periodontitis

The consequences of periapical periodontitis depend on **the virulence of pathogens, the host response, and the effectiveness of any dental treatment received. If untreated, periapical periodontitis may progress to:**

1. Periapical granuloma (chronic inflammatory tissue at the apex)
2. Chronic periapical abscess.
3. Periapical(radicular) cyst.
4. Acute exacerbation of chronic periapical periodontitis (sudden flare-up of symptoms, leading to acute periapical abscess).
5. Spreading of acute periapical abscess into adjacent tissues, resulting in:

a. Osteomyelitis of the jaws (if the infection spreads into bone marrow spaces)

b. Cellulitis

c. Fascial space infections

d. Extraoral sinus tract formation (in severe chronic cases).

Acute Apical Periodontitis

It is acute periapical inflammation in periapical part of periodontal ligament.

Clinical features:

- Slight extrusion of the tooth from the socket.
- Tooth is tender on percussion due to inflammatory edema collected in the periodontal ligament.
- Most important feature is severe pain on slight pressure during mastication; due to external pressure, forcing of edema fluid against already sensitized nerve endings results in severe pain.
- Thermal changes do not induce pain.

Radiological features:

The radiographic appearances are often normal as there is generally insufficient time for bone resorption to occur. If radiographic changes are present, they consist of **slight widening of the periodontal ligament** and the lamina dura around the apex.

Histologic features:

1. Vascular dilatation.
2. Inflammatory cell infiltrate, predominantly neutrophils (PMNs).
3. Oedema in the periodontal ligaments situated in the confined space between the root apex and the alveolar bone, pus is not necessarily present.

Periapical Granuloma

Periapical granuloma is a localized mass of chronically inflamed granulation tissue at the apical region of non-vital tooth.

It is the most common periapical lesion.

Etiology\ pathogenesis:

- Usually develops in response to persistent low-grade infection from a necrotic\ infected pulp.
- Microbial product and inflammatory mediators exit through the apical foramen. This produces chronic inflammation in the periapical tissues.
- Osteoclast activation causes periapical bone resorption.
- The resorbed bone is replaced by vascular granulation tissue containing chronic inflammatory cells, forming a peri apical granuloma.
- It may also develop following an in adequately treated\ drained acute apical abscess.

Clinical features:

- Usually asymptomatic (most common)
- Sometimes a history of mild, dull, intermittent pain or slight discomfort during biting or chewing on solid foods.
- The patient may feel that the tooth is slightly extruded from the socket.
- May be slightly tender to percussion.
- No response to pulp vitality tests (due to pulp necrosis).
- May discovered incidentally on radiograph during routine check-up.

Radiological features:

The periapical granulomas are usually **well-defined radiolucency**, but there are some situations when the lesions can appear ill-defined as in early stages, during acute exacerbation (acute abscess), or when healing\ inflammatory response is poor like in immunosuppression.

Histopathological Features:

1- Fibrous connective tissue capsule firmly attached to cementum.

2- Granulation tissue consists of newly formed blood vessels, fibroblasts, and collagen fibers.

3- Chronic inflammatory cells (lymphocytes, macrophages, and plasma cells).

4- Cholesterol clefts (slit like spaces) because the xylol used in the preparation of a histologic section dissolves the cholesterol crystals, leaving empty spaces (cholesterol clefts).

5- Russell bodies (immunoglobulin inclusions in plasma cells) may see.

6- Macrophages may accumulate to phagocyte the cholesterol clefts and called foam cells (foamy macrophages).

7- Foam cells might be found, which represent macrophages that contain lipoid material.

8- Multi nucleated giant cells could be seen around cholesterol clefts.

9-Epithelial cell rest of Malassez can be seen. They play a role in changing granuloma into cyst later on.

May also contain:

- ✓ Oral epithelium migrating from the periodontal pocket.
- ✓ Respiratory epithelium if the lesion is near the maxillary sinus.

Treatment:

1 -Tooth extraction.

2- Root canal therapy (RCT).

3- Endodontic treatment with apicectomy and curettage of the granuloma.

Sequalae of periapical granuloma:

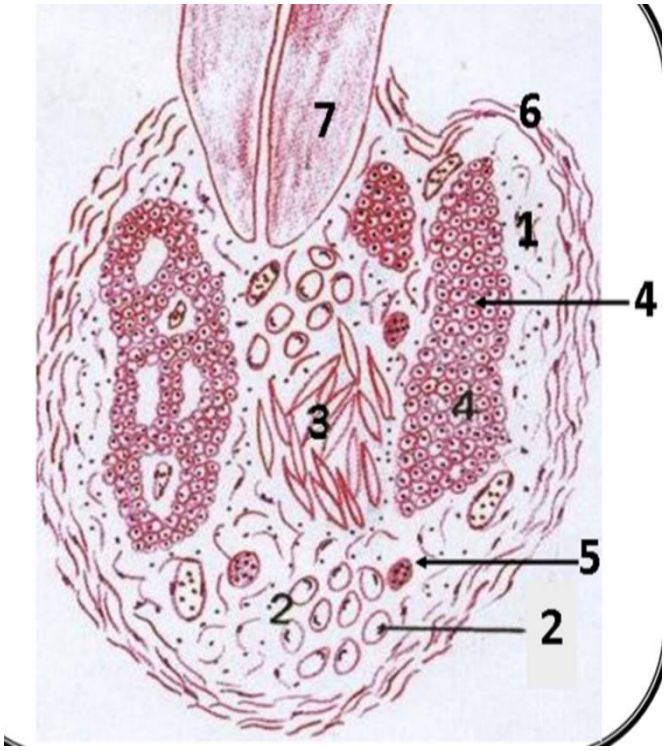
1- Periapical granuloma may continue to enlarge with continued bone resorption.

2- Acute exacerbation to an acute periapical periodontitis.

3- A suppuration to form an acute periapical abscess.

4- Formation of a radicular cyst.

5- Low grade irritation may cause osteosclerosis (bone apposition) or cementum apposition (hypercementosis).



1. Granulation tissue.
2. Foam cells.
3. Cholesterol clefts.
4. Epithelial rests of Malassez.
5. Foreign body giant cells.
6. Fibrous CT capsule.
7. Apical root.

Periapical abscess

A periapical abscess is a localized collection of pus at the apex of a tooth root, resulting from bacterial infection of the pulp and periapical tissues. It may present as acute or chronic, with distinct clinical and radiographic features.

Acute Periapical Abscess

It is a localized suppurative inflammation (collection of pus) in the periapical area at the root apex of non-vital tooth.

Synonyms: Acute abscess, Acute dentoalveolar abscess, Acute periapical abscess, Acute radicular abscess.

Etiology

1. Extension of infection from pulp tissue.
2. May follow faulty endodontic treatment.
3. My result from acute exacerbation of periapical granuloma.
4. The causative organisms are pyogenic bacteria.

Clinical features:

1) Severe throbbing Pain, tooth is painful even to touch.
2) Tooth is usually extruded from its socket and mobile.
3) Localized swelling & erythematous change in overlying mucosa.
4) Tenderness on percussion and palpation.
5) Pain increases with systemic manifestations (fever, malaise & Lymphadenopathy).
6) No response to the pulp tester (non-vital tooth)
7) Pus discharge sometimes through sinus tract if chronicity develops.

Radiographic Feature:

Hazy or ill-defined radiolucent area around the apex of tooth.

Histopathological features:

1. Central area of liquefactive necrosis with pus (PMNLs, necrotic debris, bacteria)
2. Peripheral acute inflammatory infiltrate (PMNLs) with dilated vessels and edema.
3. Early tissue destruction and bone resorption at the apex.

Treatment:

- Systemic antibiotics if the periapical abscess is severe or has already spread to other parts of the body such as the neck to help bring the infection under control.
- Draining of pus through the root canal followed by endodontic treatment if the tooth is to be preserved.
- Extraction of the tooth.

Phoenix abscess is an acute exacerbation of a pre-existing chronic periapical lesion, usually a periapical granuloma, resulting in acute suppurative inflammation and abscess formation.

- **Complication of periapical abscess:**

1) Facial Cellulitis
2) Ludwig's angina
3) Osteomyelitis
4) Septicemia
5) Meningitis, brain abscess, cavernous sinus thrombosis

Facial cellulitis

- Facial cellulitis is an acute, diffuse, non-cicumscribed bacterial infection of the skin and subcutaneous tissues of the face, characterized by erythema, warmth, swelling, and tenderness. Usually associated with malaise and fever.
- If untreated, it may lead to systemic involvement or severe complications, including orbital or intracranial spread.
- Cellulitis is more frequent in immunocompromised patients as diabetes mellitus, organ transplant and AIDS, malnutrition and alcoholism
- Infection spreads most commonly from mandibular third molars, whose apices are closely related to several facial spaces.
- Cellulitis arising from dental infection and spreading through soft tissues of head and neck can take various forms.
- Mostly, infection spreads through tissue spaces like canine space, infratemporal space, pharyngeal space, buccal space, submental and submandibular space etc.

Two especially dangerous forms of cellulitis

1. Cellulitis associated with mandibular teeth into submandibular and cervical tissues may cause (**Ludwig's angina**).
2. Cellulitis associated with maxillary teeth towards the eye may cause (**Cavernous sinus thrombosis**)

Clinical features:

1-Diffuse swelling, pain, fever, and malaise.

2- The swelling is tense and tender, often described as 'brawny' meaning swollen and hard, with a characteristic broad-like firmness.

3-The overlying skin is taut, shiny and edematous.

4-Limit opening of the mouth and often cause dysphagia.

5-Systemic manifestations include, fever, toxemia, leukocytosis and raised serum C-reactive protein (CRP) level.

6-Regional lymphadenitis.

- **Treatment:**

1-Immediate aggressive antibiotic treatment to prevent further spread of infection.

2-Remove the cause of infection.

3-Surgical drainage.

Ludwig's angina

- Ludwig's angina is a severe cellulitis involving submandibular, sublingual, and submental spaces bilaterally, resulting in life-threatening airway obstruction.
- In 70% cases develops from spread of infection from mandibular teeth.
- Increased prevalence in immunocompromised patients like AIDS, aplastic anemia, organ transplantation etc.

Clinical features:

1) Bilateral Submandibular Swelling: There is swelling on both sides under the chin, described as firm or brawny (indurated).
2) Odynophagia, drooling and trismus.
3) Tongue displacement(elevation) woody tongue.
4) Difficulty eating, swallowing(dysphagia), and breathing(dyspnea).
5) High fever, rapid pulse, and fast respiration.
6) Extension of inflammation to pharynx, larynx and suffocation and death.

Treatment:

1. Maintenance of the airway, tracheostomy may be indicated.
2. Antibiotic therapy.
3. Surgical drainage.

Cavernous sinus thrombosis (CST)

It is a rare, life-threatening blood clot that forms in the cavernous sinuses, which are located at the base of the brain, behind the eyes. It typically results from a bacterial infection that spreads from the face, teeth, ears, nose, or sinuses.

The clot can block blood flow, increase pressure and potentially damage the brain, eyes, and surrounding nerves, and if left untreated, can lead to serious complications like sepsis or even death.

Clinical features:

1. Patients may experience symptoms such as:
2. Severe headache, eye pain and fever.
3. Double vision.
4. Periorbital edema, proptosis (bulging eyes).
5. Chemosis (swelling of the conjunctiva).
6. Ophthalmoplegia (paralysis of eye movements) due to the pressure on cranial nerves.

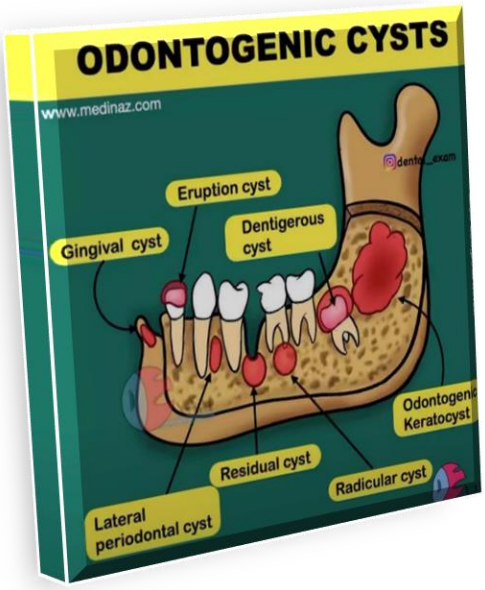
Treatment:

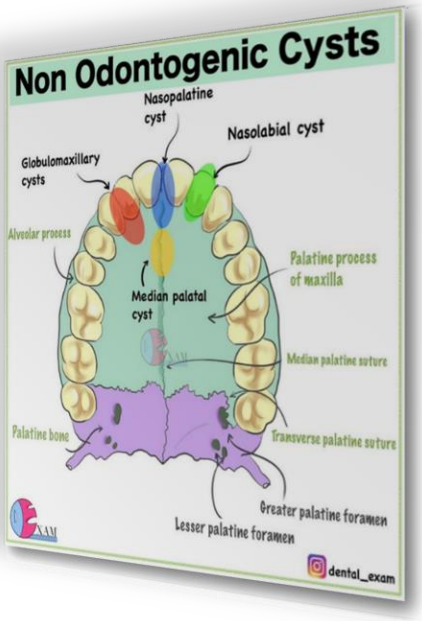
1. High dose of penicillin.

2. Extraction and drainage (if fluctuant).

3. Corticosteroid and anticoagulant to prevent thrombosis and septic emboli formation.

Cysts of the JAW (2026)





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2026

CYSTS OF THE ORAL AND MAXILLOFICIAL REGION



- Classify and identify cysts of the jaws and soft tissue
- Understand the etiology and pathogenesis of jaw cysts
- Recognize the clinical presentation of jaw cysts
- Interpret radiographic findings
- Understand the principles of treatment
- Manage complications and recurrence

CYSTS OF THE ORAL AND MAXILLOIFIAL REGION

Definition of cyst

The cyst is defined as a pathological cavity filled with fluid or semi-fluid, or gaseous contents and not created by the accumulation of pus". It may have an epithelial lining called **(a true cyst)** OR a lack of epithelial lining called **(a pseudo-cyst)**.

Complication of cysts

- **Infection:** Cyst is liable to be infected can cause osteomyelitis, facial cellulitis, etc.
- **Pathological fracture** of bone.
- **Disfigurement** of face.
- **Neoplasm** may arise from the epithelial lining.
- Some cyst e.g. dentigerous cysts may **prevent the eruption** of the involved tooth.

Parts of cyst

generally, the cyst is formed from:

1. **Epithelial lining:**

   Most of the cysts are lined wholly or in part by epithelial lining. This lining can be a stratified squamous epithelium, pseudostratified epithelium, or simple epithelium.

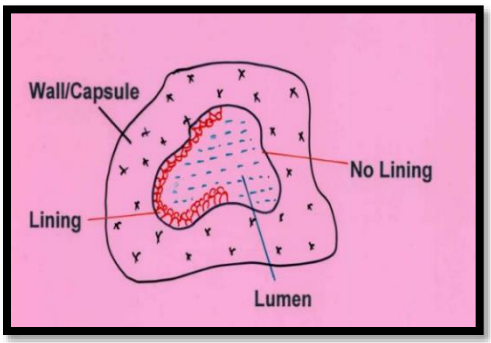


FIGURE 1: SCHEMATIC OF TYPICAL CYST

2. **Cystic cavity or lumen:** the cystic cavity, generally consists of:
   - Water and electrolytes, Breakdown products of degenerating epithelial, inflammatory, .connective tissue cells and Serum proteins.
3. **Fibrous tissue capsule:** which is formed from connective tissue cells such as fibroblasts and fibrocytes as well as collagen fibers.

Pathogenesis of cyst

The formation of a cyst takes place in three phases:

I. **Initiation:** the factors depend on the type of cyst

II. **Cyst formation**

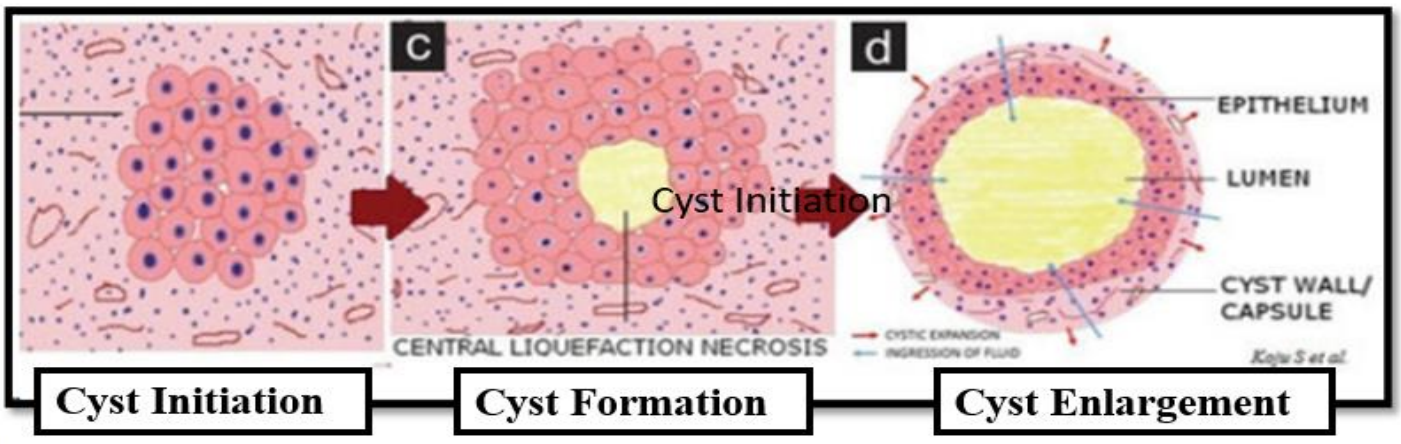
III. **Cyst expansion** or enlargement

I. **Cyst Initiation**: An unknown stimulus promotes the rapid increase of epithelial rest cells via cytokine synthesis. The factors that are suggested to be responsible for the cyst initiation phase are mainly genetics, loss of immunological surveillance, inflammatory mediators, or some local factors like decreased oxygen tension along with increased CO2 tension. Initiation is followed by the rapid growth of the epithelial cells and the development of a cystic lesion.

II. **Cyst Formation:** Nutritional deficiency theory

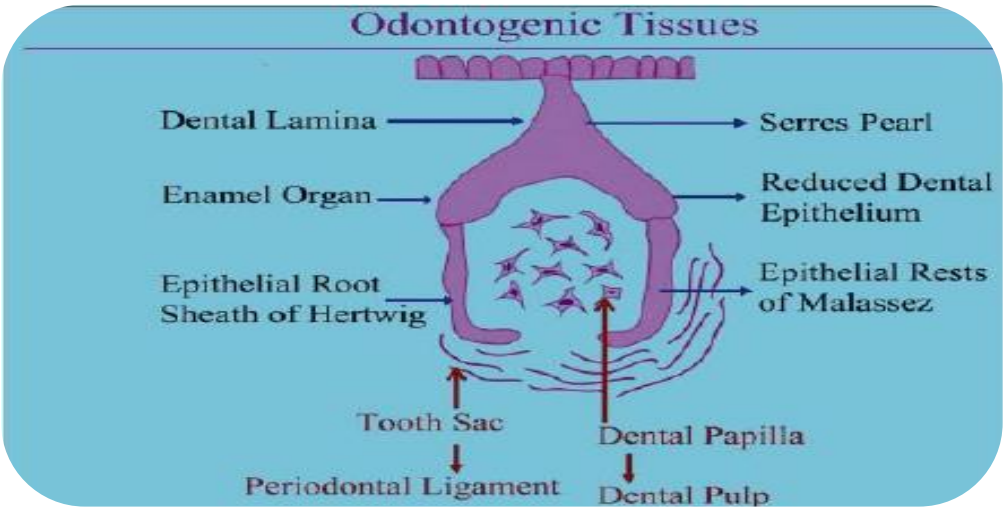
After initiation, proliferating epithelial cells form a mass inside the sac, and the innermost central cells become away from nutrients/blood supply, as they are far from the source of nutrients. The innermost cells do not get adequate blood supply, and so there is ischemic ***liquefactive necrosis*** in the center, leading to cavity formation, which is surrounded by growing epithelial cells.

III. **Cyst Enlargement or Expansion:** Once formed, the cyst continues to enlarge slowly, over the months. The process is similar for all epithelial-lined cysts with some variations. The basic mechanism for enlargement is the same for most cysts.



The Origin of Epithelial Lining of Cyst

- **The epithelial lining of (Odontogenic Cysts)** ⟹ originates from residues of the tooth-forming organ. *There are three types:*
  1. **Epithelial rests of Malassez:** Formed from the fragmentation of the epithelial root sheath of Hartwig.
  2. **The epithelial rests of Serres:** Persisting after the dissolution of the dental lamina.
  3. **The reduced enamel epithelium:** Derived from an enamel organ and covers the fully formed crown of an unerupted tooth.
- **The epithelial lining of (Non-odontogenic Cysts)** ⟹ originates from epithelial cells entrapped between developing facial processes or the epithelial lining of a fistulous tract after abscess healing.



Type	Source	Origin of cell rest	Cyst variety
Odontogenic rests	Cell rests of malassez	Epithelial hertwig's root sheath	Periapical (radicular) cyst
	Reduced enamel epithelium	Enamel organ	Dentigerous cyst
	Rests of dental lamina (rest of Serres)	Epithelial connection between mucosa and enamel organ	Glandular odontogenic cyst, Lateral periodontal cyst, Odontogenic keratocyst, and Gingival cyst of newborn & adult
Nonodontogenic Rests	Remnants of nasopalatine duct	Paired nasopalatine duct-vestigial	Nasopalatine canal cyst and Fissural cysts
	Remnants of maxillary sinus epithelium		Cysts of the maxillary sinus

WHO (2022) Classification of cysts of the jaws

A. Epithelialized cysts

1) Odontogenic cysts

a) Inflammatory Cysts

1. Radicular (Periapical) cyst
   - i. Residual Cyst
   - ii. Lateral radicular Cyst
2. Inflammatory Collateral Cysts:
   - i. Mandibular buccal bifurcation cyst
   - ii. Paradental cyst

b) Development Cysts

1. Dentigerous cyst
2. Eruption cyst
3. Odontogenic Keratocyst (OKC)
4. Orthokeratinized odontogenic cyst
5. Lateral periodontal cyst and botryoid odontogenic cyst
6. Gingival cyst of the adult
7. Gingival (alveolar) cyst of the newborn
8. Glandular odontogenic cyst
9. Calcifying odontogenic cyst (COC)

2) Non-odontogenic cysts

1. Post surgical ciliated cyst
2. Nasopalatine duct (incisive canal) cyst
3. Nasolabial cyst
4. Median palatal cyst

B. Non-epithelialized primary bone cysts

1. Solitary bone cyst
2. Aneurysmal bone cyst
3. Stafine's idiopathic bone cavity

N.B: Cysts of Soft tissues

1. Dermoid & epidermoid cyst
2. Lymphoepithelial cyst
3. Thyroglossal duct cyst
4. Salivary gland cysts

The General Features of Cyst

1) **Origin:** in the case of true cysts:
   - **Odontogenic cysts:** arising from epithelial remnants of the odontogenic epithelium (e.g epithelial rests of Serres, reduced enamel epithelium, or epithelial rests of Malassaz).
   - **Non-odontogenic cysts:** epithelial remnants other than tooth-forming cells.
2) **Clinically:**
   - **Age:** second and third decades (middle age) **Except:**
     - ✓ Inflammatory radicular cysts occur ***at any age*** but ***more in adults***.
     - ✓ Eruption cysts occur in ***children***.
     - ✓ Gingival cysts of newborns occur in ***newborns***.
     - ✓ Gingival cysts of adults occur in adults ***over 40 years***.
   - **Site:** Mandible more than Maxilla.
     - ✓ When present in the mandible, it occurs more posteriorly.
     - ✓ When they occur in the maxilla, they present more in the anterior region.
   - **Symptoms:**
     - ❖ **In case of jaw cysts:**
       - ✓ Cysts are **symptomless** and discovered by routine x-ray.
       - ✓ Cysts are **painless** unless **infected** (**except** aneurysmal bone cyst which is **painful** and tender upon motion).
       - ✓ Cysts are slowly growing.
     - ❖ **In case of soft tissue cysts:**
       - ✓ Painless and slowly growing.
       - ✓ Soft, fluctuant, and dome shape.
       - ✓ It may be doughy in consistency if it contains keratin.
       - ✓ Its color may be normal (e.g. retention cyst), bluish (e.g. extravasated mucocele, ranula, and eruption cyst), or white (e.g. gingival cyst of the newborn due to the presence of keratin).
       - ✓ The cysts may be covered with normal skin (e.g. lymphoepithelial and thyroglossal tract cysts).
3) **Histopathological:**
   - ❖ **Epithelial lining:**
     - ✓ In most cases it is stratified squamous epithelium (keratinized or nonkeratinized).
     - ✓ Respiratory epithelium (pseudo-stratified ciliated columnar epithelium).
   - ❖ **Fibrous connective tissue wall**: it consists of fibroblasts and collagen fibers.

❖ **Content:** obtained through aspiration:
- ✓ Water, electrolytes, degenerated epithelial cells.
- ✓ It may contain keratin (semi-fluid)

4) Radiographically:

❖ **In case of intra-osseous cysts:**
- ✓ Well-defined radiolucent area with radio-opaque margin.
- ✓ It may be unilocular or multilocular (honeycomb or soap bubble appearance).
- ✓ It may be associated with an unerupted tooth.

❖ **In case of soft tissue cysts:**
- ✓ Negative in x-ray.

5) Treatment:

❖ **In case of intra bony cysts:**
- ✓ Inoculation.
- ✓ Marsupilization in case of large cysts.
- ✓ Open and close in case of traumatic bone cyst.

❖ **In case of soft tissue cyst:**
- ✓ Excision (surgical removal).

❖ **No treatment: in case of:**
- ✓ Eruption cyst.
- ✓ Gingival cyst of new born.
- ✓ Static bone cyst.

Odontogenic Cysts

Odontogenic ⟹ **Odonto-**Tooth forming apparatus

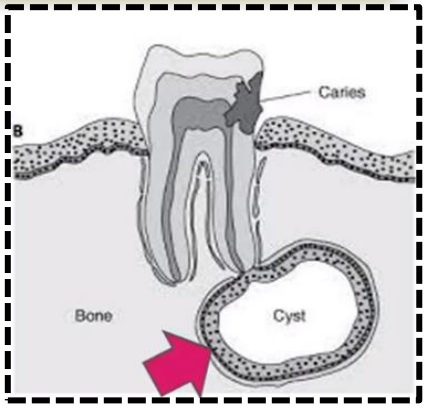
⟹ **Genic**-Origin

❖ Jaw cysts lined by epithelium derived from odontogenic epithelium.

❖ Odontogenic cysts are sub-classified as developmental or inflammatory:
- Inflammatory cysts are ⟹ the result of inflammation.
- Developmental cysts are ⟹ of unknown origin. but they do not appear to be the result of an inflammatory reaction.

Inflammatory Odontogenic Cysts

1-PERIAPICAL CYST (RADICULAR CYST, APICAL PERIODONTAL CYST)



❖ **Definition:**

- It is ***the commonest cystic lesion*** of the jaws and (the most common odontogenic cyst of the oral cavity).
- It is **an inflammatory** odontogenic cyst that occurs at **the periapical area of non-vital tooth.**

❖ **Origin:** Epithelial lining derived from the proliferation of **epithelial rests of Malassez (ERM)** within the periodontal ligament at the apex of the tooth.

❖ **Etiology & Pathogenesis:**

- A periapical cyst develops from ***a preexisting periapical granuloma***, which is chronically inflamed granulation tissue located at ***the apex of a nonvital tooth***.

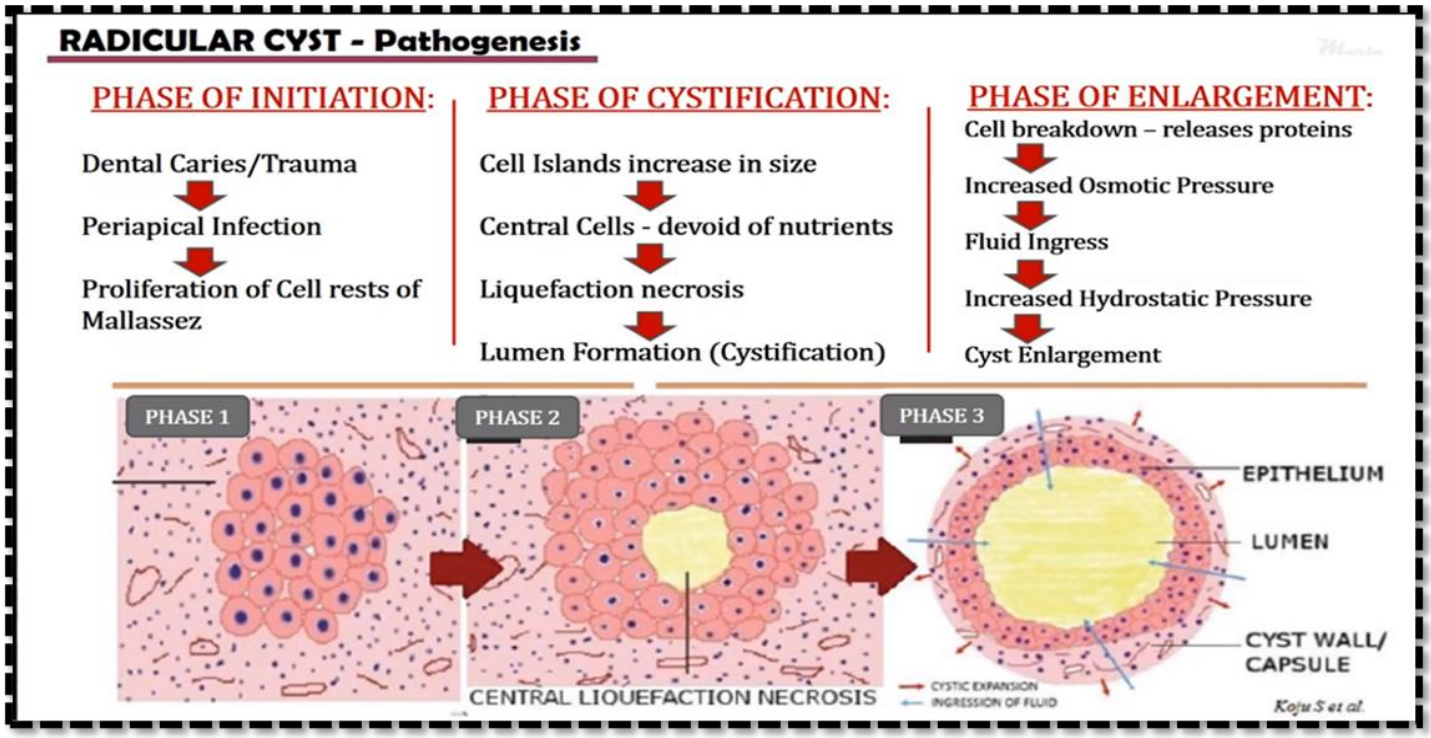
Pathogenesis of Radicular Cysts:

- ✓ Radicular cyst arise from undergoes proliferation of the epithelial rests of malassez in response to the products of inflammation which lead to form of masses.
- ✓ As theses masses enlarge, their central cells become away from the source of nutrition and they degenerated and forming microcyst which enlarge and finely fuse together forming one large macrocyst.

Mechanism of Cyst Enlargement:

- Breakdown of the complex tissue proteins that derived from degenerating epithelial and connective tissue cells into simpler protein resulted in an increase in the osmotic pressure inside the cyst.
- The epithelial lining acts as a semi-permeable membrane through which the fluid is transported from outside to the cystic cavity, which leads to an increase in the osmotic pressure within the cyst.

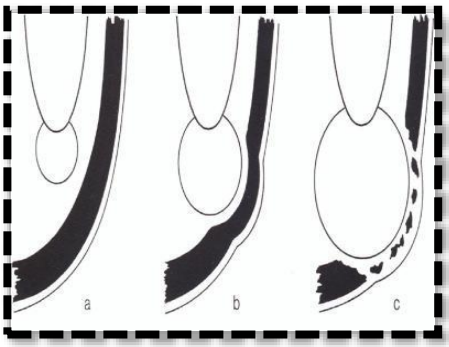
- In addition, cyst lining and fibroblasts in the cyst capsule release bone-resorbing factors such as prostaglandins, especially PGE2 and PGF3 and interleukin-1, which can assist in the process of bone resorption by creating more space for cyst enlargement.



Clinical Features:

- **Incidence & Frequency:** Most common cyst of the jaw, accounting for (60-75%) of all jaw cysts.
- **Age**: Peaks in the 3th and 6th decades. Rarely occurring in the first decade of life.
- **Sex**: Slightly more in **males.**
- **Site predilection**: Most commonly occurs in **maxillary anterior permanent teeth**.
- **Signs & symptoms**:
  - ✓ Painless (asymptomatic) discovered accidentally during routine radiographic examination which is *always associated with the* ***non-vital tooth***.
  - ✓ Slowly enlarging painless swelling on labial or buccal sides.

The swelling usually runs across 3 stages



a. **Initially**, it slowly expands the cortical plate at a rate of about 5mm in diameter per year and produces a bony-hard expansion.

b. **Later**, when the rate of expansion increases, this leads to progressive thinning of the cortical plate to produce a clinical sign of **(egg–shell crackling.**

c. **Finally**, the cyst perforates the cortical plate and presents clinically as a bluish, fluctuant sub-mucosal swelling.

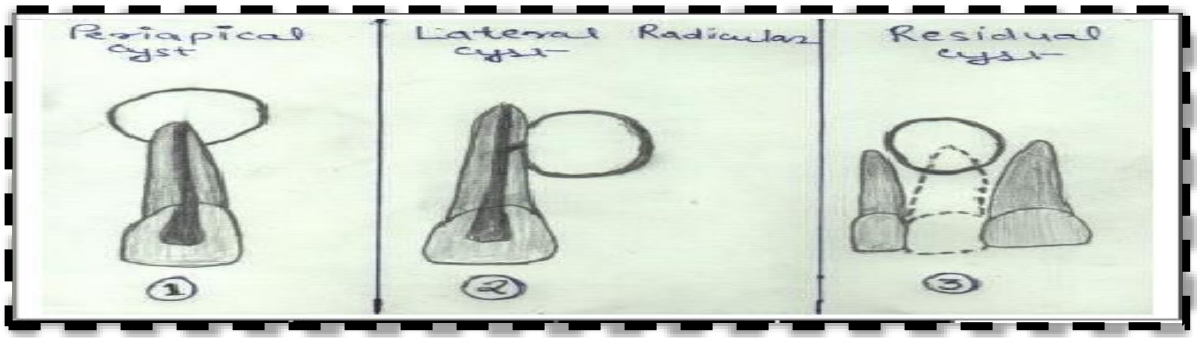
Radiographic Features:

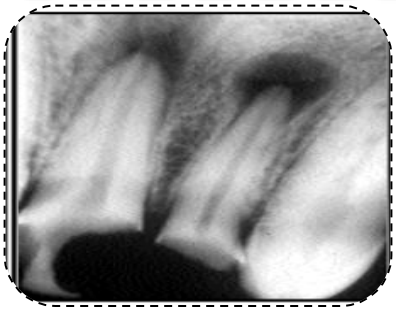
- Round or ovoid well-defined radiolucency area surrounded by a narrow radiopaque margin around the apex of a non-vital tooth.
- The size of the cyst is about 5mm in diameter.
- The opaque margin becomes hazy if the cyst is secondarily infected.
- In long-standing cysts, root resorption of the associated tooth may occur.

Types of radicular cyst radiographically

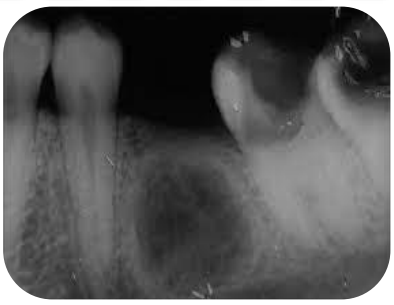
Radicular cysts are subdivided into three types depending on the location of the cyst:

1. **Apical radicular cyst:** most common type of radicular cyst (about 70%). It arises as a result of the extension of inflammation from the pulp to the periapical part of the tooth (around the apex of the root).
2. **Lateral radicular cyst:** appears along the lateral aspect of the root, as the inflammation spreads through a lateral foramen. Radiographically, it appears as discrete radiolucencies along the lateral aspect of the root.
3. **Residual periapical cyst:** If periapical inflammatory tissue is not curetted at the time of tooth removal, it may give rise to an inflammatory cyst called a residual inflammatory cyst. It appears as a round-to-oval radiolucency of variable size within the alveolar ridge at the site of the previous tooth extraction.









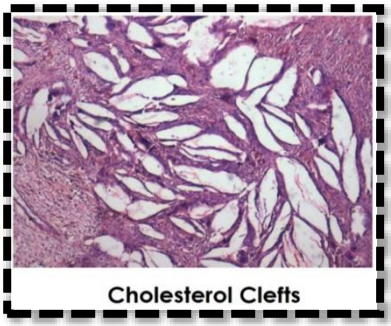
Differential Diagnosis of Periapical Radiolucency:

1. Periapical granuloma.
2. Radicular cyst.
3. Chronic periapical abscess.
4. Periapical fibrous scar.
5. Periapical cemento-osseous dysplasia (early stage)



Histopathological Features:

- Cystic cavity is lined by ***non-keratinized stratified squamous epithelium*** of variable thickness.
- **In newly formed cysts,** the epithelial lining is hyperplastic, exhibiting long anastomosing cords of epithelium forming arcades or rings over vascularized CT and many inflammatory cells.
- **In fully formed cysts (old cysts)** the epithelial lining becomes more regular and flattened, and the CT is more fibrous & less vascular, with fewer inflammatory cells.
- **Connective tissue contains:**
  - ✓ Variable number of inflammatory cells; plasma cells are also present and may be degenerated **(Russell bodies).**
  - ✓ Multinucleated foreign body giant cells MNGC are seen especially in areas of cholesterol clefts.
  - ✓ **Cholesterol clefts** are seen in CT and may erode through the epithelium and protrude into the cyst lumen (mural nodule).
  - ✓ **Foam cells** (lipid-filled macrophages).



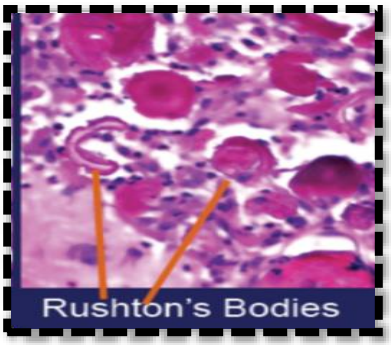
Cholesterol Clefts

- **Cyst contents:**
  - ✓ The cyst contents vary from a watery, straw-colored fluid to a semi-solid brownish paste-like material. The cyst consists of all serum proteins. The soluble protein level is 5-11g/dl.
  - ✓ Water, electrolytes, cholesterol crystals, and breakdown products of degenerating, necrotic epithelial, connective tissue, and inflammatory cells.

Microscopic Variation in Cyst Lining & Wall of Radicular Cyst

1. **Metaplasia** of the epithelial lining to the respiratory epithelium (pseudostratified columnar ciliated cells with goblet cells).
2. **Foci of dystrophic calcification**: Basophilic (haematoxyphilic) calcific deposits may be found in both the epithelium and CT.

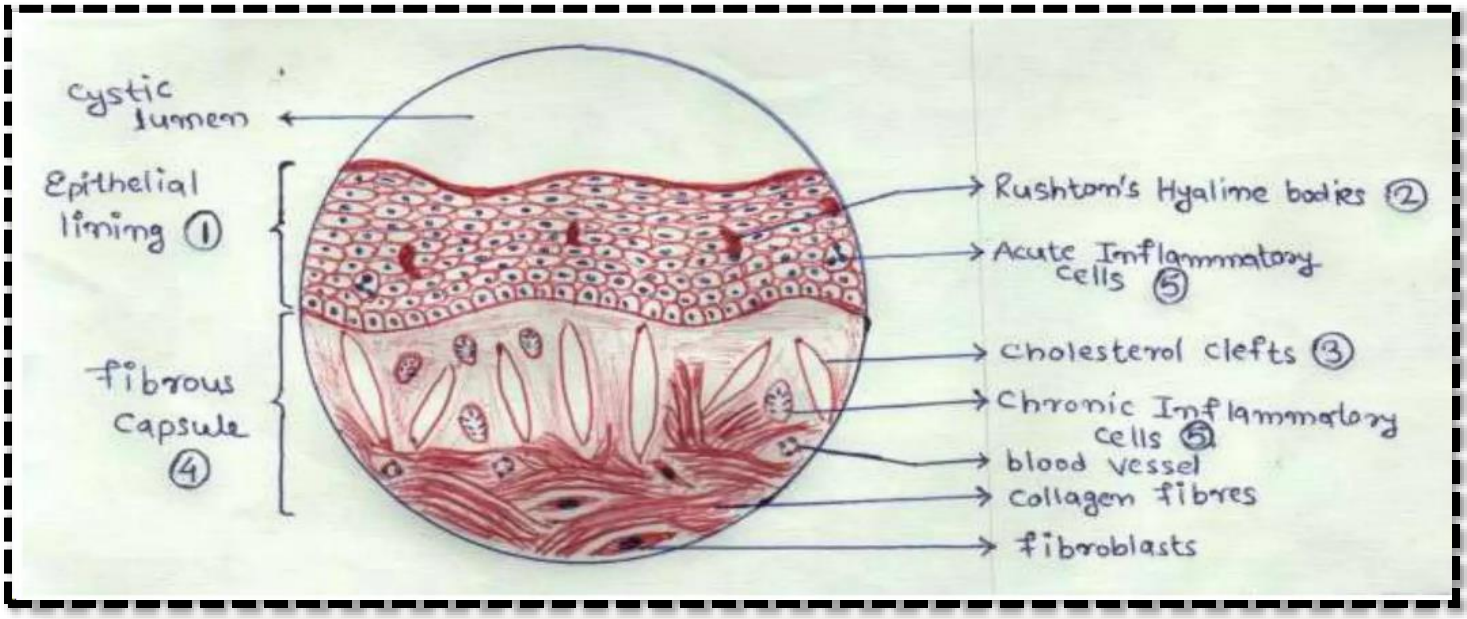
3. **Hyaline bodies or "Rushton bodies":** These are thin, Eosinophilic (acidophilic), linear, hairpin, or slightly curved bodies with concentric lamination. If present, they are only seen in the epithelium.



- **The origin of Hyaline bodies:**
  - Haematogenous origin related to previous hemorrhage.
  - Odontogenic origin.

Treatment and Prognosis

- Root Canal Treatment (RCT) +Apicoectomy +Enucleation.
- Extraction of the involved tooth & curettage of the apical zone.



2-Inflammatory Collateral Cyst

- It's **an uncommon inflammatory odontogenic cyst** that arises on the disto-buccal aspect of the roots of partially or recently erupted teeth.
- They arise as a result of **inflammation in the pericoronal tissues**; it may induce periodontal pocketing, which may lead to hyperplasia and cyst formation in adjacent odontogenic epithelium, which may be either remnants of the reduced enamel epithelium or cell rests of Malassez.
- The cysts were lined with a non-keratinised squamous epithelium with a mixed inflammatory cell infiltrate in the connective tissue

➢ **Two main types of inflammatory collateral cysts are recognized:**

i- Mandibular infected buccal cyst (Buccal bifurcation Cyst)

ii- Paradental cyst

i- Buccal Bifurcation Cyst

Definition:

It is an uncommon inflammatory odontogenic cyst that characteristically develops on the a buccal aspect of **the mandibular first permanent molar**.

Etiology & Pathogenesis:

The pathogenesis of this cyst is *uncertain*, it has been speculated that when the tooth erupts, an inflammatory response may occur in the surrounding follicular tissues that stimulates cyst formation.

Clinical Features:

- It typically occurs in **children from 5 to 11 years of age.**
- The patient has slight to moderate tenderness on the buccal aspect of the mandibular first molar, which may be ***in the process of erupting*.**
- The patient often notes associated clinical swelling and a foul-tasting discharge.
- Periodontal probing usually reveals pocket formation on the buccal aspect of the involved tooth.
-

Radiographic features:

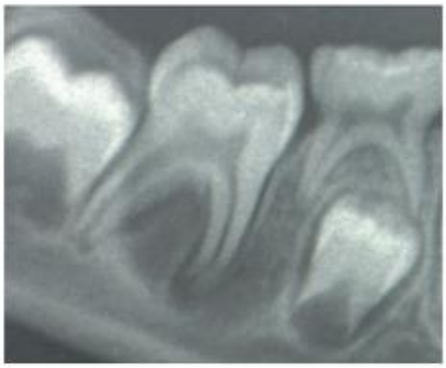
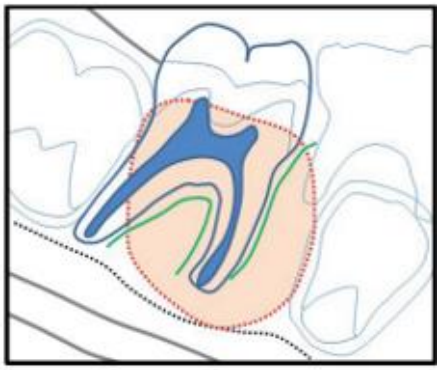
- Typically show a **well-circumscribed unilocular radiolucency involving the buccal furcation and root area of the involved tooth**.
- An *occlusal radiograph* is most helpful in demonstrating the buccal location of the lesion (The root apices of the molar are characteristically tipped toward the lingual mandibular cortex).

Histopathologic Features

- The microscopic features are nonspecific and show a cyst that is lined by nonkeratinizing stratified squamous epithelium with areas of hyperplasia.
- A prominent chronic inflammatory cell infiltrate is present in the surrounding connective tissue wall.

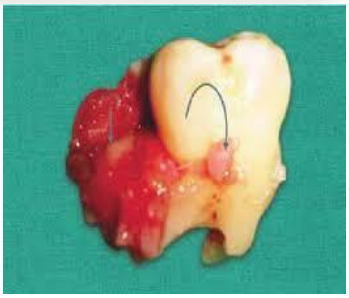
Treatment and Prognosis

The buccal bifurcation cyst is usually treated by enucleation; extraction of the associated tooth is unnecessary.

ii-Paradental Cyst

Definition:

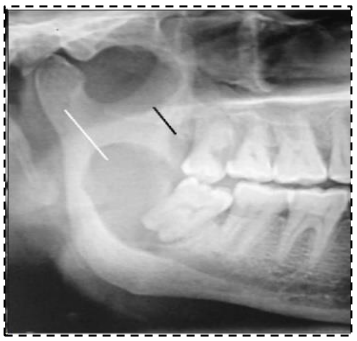


- The term paradental cyst has sometimes been used synonymously for buccal bifurcation cyst.
- Such lesions typically occur distal or buccal to partially erupted mandibular third molars with a history of pericoronitis.

Etiology & Pathogenesis:

- The pathogenesis of the paradental cyst is also uncertain. Some of these lesions have been associated with teeth that demonstrate buccal enamel extensions into the bifurcation area.
- Such extensions may predispose these teeth to buccal pocket formation, which could then enlarge to form a cyst in response to pericoronitis.
-

Clinical features:



Paradental cyst arises alongside a partially erupted third molar involved by pericoronitis

Radiographic features:

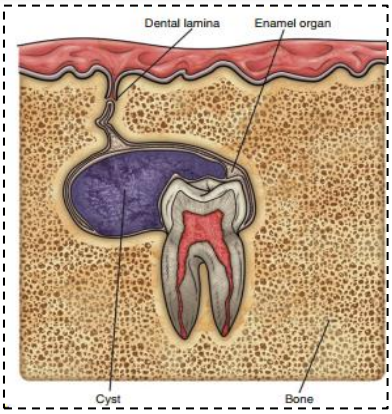
Appears as a well-defined area of radiolucency related to the neck of a partially erupted tooth and the coronal third of the root.

Histopathological features:

A paradental cyst is similar to an inflammatory radicular cyst.

Developmental Odontogenic Cysts

1-DENTIGROUS CYST (Follicular Cyst)



❖ **Definition:**

✓ Is the developmental odontogenic cyst which encloses **a part or entire crown of an unerupted tooth** and is attached to the tooth at ***the cementoenamel junction.***

N.B: Dentigerous cysts, or follicular cysts: the term dentigerous is a Latin word that literally means "tooth-bearing/producing"

❖ **Origin:**

Enamel organ or remnant "Reduced enamel epithelium".

❖ **Etiology & Pathogenesis:**

The pathogenesis of this cyst is **uncertain**, but apparently, it develops by an accumulation of fluid between the *reduced enamel epithelium* and the tooth crown & the separation of the follicle from around the crown of an unerupted tooth.

❖ **Clinical Features**



- **Incidence and Frequency:** Dentigerous cyst is **the second** most common type of odontogenic cyst and **the most common** developmental cyst of the jaws (20% of all epithelium-lined cysts of the jaws).
- **Age:** Most frequently present in patients between **(10 and 30 years)** of age.
- **Sex:** There is a slight **male** predilection.
- **Site:** Dentigerous cysts may occur in association with any unerupted tooth, but most often involve ⟹ **Mandibular third molars.** Other frequent sites ⟹ include maxillary canines, maxillary third molars, and mandibular second premolars.
- **Signs & Symptoms:**

✓ ***Small dentigerous cysts*** are usually asymptomatic and are discovered only on a routine radiographic examination, or when films are taken to determine the reason for the failure of a tooth to erupt.

✓ ***Large cysts*** may be associated with a painless expansion of the bone in the involved area, *extensive* lesions may result in facial asymmetry.

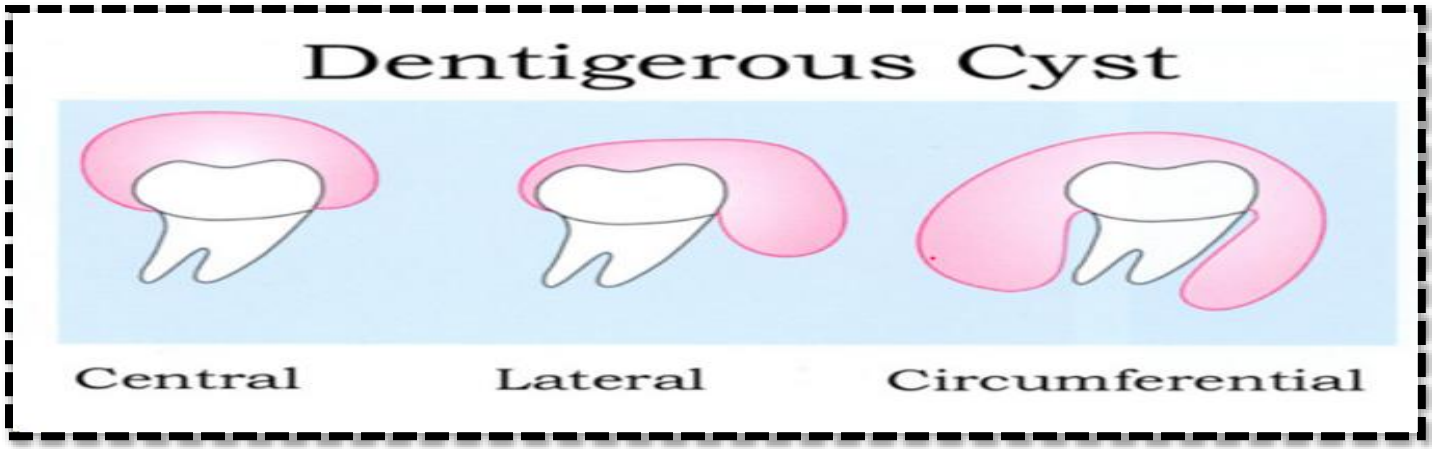
- ✓ Dentigerous cysts may become infected and be associated with pain and swelling (Such infections is associated with a partially erupted tooth or by extension from a periapical or periodontal lesion that affects an adjacent tooth).

❖ **Radiographic Features:**

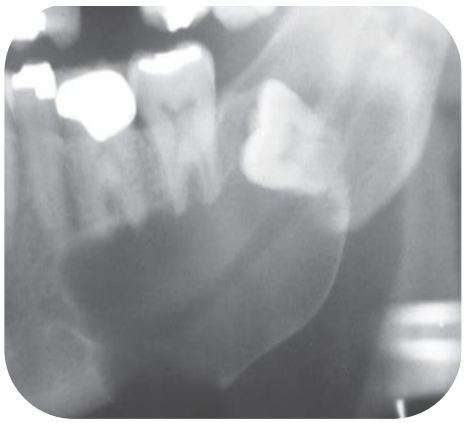
- ✓ Dentigerous cyst typically shows a unilocular radiolucent area that is associated with the crown of an unerupted tooth.
- ✓ The radiolucency usually has a well-defined and often sclerotic border, but an infected cyst may show ill-defined borders.
- ✓ Dentigerous cysts may displace the involved tooth for a considerable distance.
- ✓ Root resorption of adjacent erupted teeth can occur.

The cyst-to-crown relationship shows several radiographic variations:

1. **Central variety:** is the most common, the cyst surrounds the crown of the tooth and the crown projects into the cyst.
2. **Lateral variety:** is usually associated with mesioangular impacted mandibular third molars that are partially erupted. The cyst grows laterally along the root surface and partially surrounds the crown.
3. **Circumferential variant:** the cyst surrounds the crown and extends for some distance along the root so that a significant portion of the root appears to lie within the cyst.



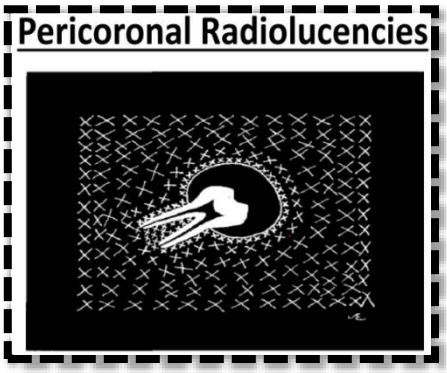






Differential Diagnosis of Peri-Coronal Radiolucency:

1. Dentigerous cyst
2. Unicystic ameloblastoma
3. OKC (in relation to a tooth)
4. Ameloblastic fibroma
5. Adenomatoid odontogenic tumor (AOT)
6. Hyperplastic follicle



❖ **Histopathologic Features:**

- Dentigerous cyst are lined by thin uniform layer (2-10 cell layers) of non-keratinized stratified squamous epithelium without rete ridges resembles the reduced enamel epithelium from which it is derived
- Mucous cells may be present ⟹ due to **metaplasia**.
- The lining is supported by a fibrous connective tissue capsule **free from inflammatory cells** unless gets secondary infected.
- Islands of odontogenic epithelium may be seen.
- **The cyst contains** a protein aqueous, yellowish fluid and cholesterol crystals. The soluble protein is around 5-7g/dl.

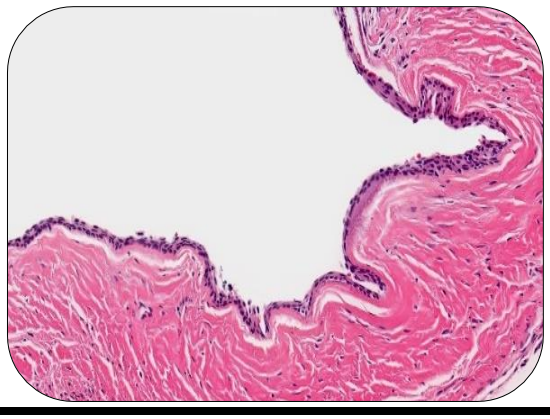
❖ **Treatment and Prognosis:**

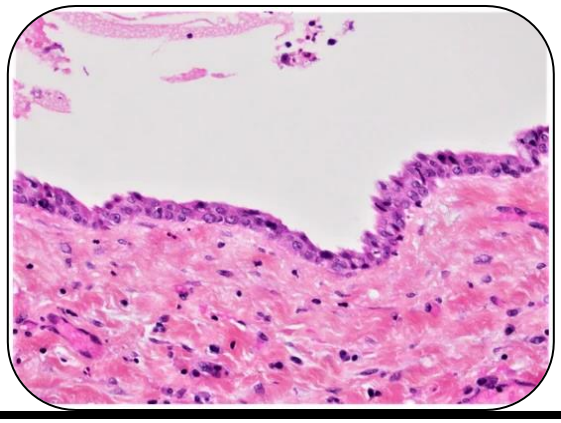
- Enucleation of the cyst together with removal of the unerupted tooth.
- The prognosis is excellent, recurrence seldom is noted after complete cyst removal.

Complications of Dentigerous Cyst

The lining of a dentigerous cyst if not treated may undergo **neoplastic transformation** to:

1. Ameloblastoma.
2. Squamous cell carcinoma.
3. Intraosseous mucoepidermoid carcinomas may develop from mucous cells in the lining of a dentigerous cyst.





2-ERUPTION CYST (Eruption hematoma)



❖ **Definition:**

Eruption cyst is the soft tissue analogue of dentigerous cyst.

❖ **Etiology & Pathogenesis:**

•Accumulation of fluid or hemorrhage between crown of **erupting tooth** and the reduced enamel epithelium of the surrounding dental follicle, possibly as a result of trauma.

•Factors actually impeding eruption through soft tissues are **unknown**, possibly dense fibrous tissue.

❖ **Clinical Features:**

- The eruption cyst appears as a soft, often translucent swelling in the gingival mucosa overlying the crown of an erupting deciduous or permanent tooth.
- Most examples are seen in children younger than age 10.
- The cyst may occur with any erupting tooth but is *most commonly associated with the first permanent molars and the maxillary incisors*.
- Surface trauma may result in a considerable amount of blood in the cystic fluid, which imparts a blue color. Such lesions are referred to as eruption hematomas.

❖ **Histopathological features:**

- The epithelial lining of eruption cyst is similar to that of the dentigerous cyst (non-keratinized stratified squamous epithelium), so the eruption cyst is considered a superficial dentigerous cyst.
- The fibrous capsule shows inflammatory cells possibly as a result of trauma.
- The cystic cavity may contain blood in addition to the yellowish protein fluid as a result of trauma.

❖ **Treatment & prognosis**:

- It may **not be required** because the cyst usually ruptures spontaneously, permitting the tooth to erupt.
- If this does not occur, simple excision of the roof of the cyst generally permits speedy eruption of the tooth.

3-ODONTOGENIC KERATOCYST (OKC)

❖ **Definition:**

It is a developmental odontogenic cyst of with a characteristic lining of parakeratinized stratified squamous epithelium and is a distinctive form of odontogenic cysts due to:

⇒ It may exhibit aggressive clinical behavior
⇒ It has a significant recurrence rate.
⇒ It may an associated with nevoid Basal Cell Carcinoma Syndrome (NBCCS).

❖ **Origin:**

- The cyst arises from **the remnants of the dental lamina**. Since the dental lamina gives rise to tooth primordium therefore this cyst was designated previously as **Primordial Cyst**.
- Origin of this cyst from the extension of basal cells of the overlying oral epithelium has also been suggested.

Growth of Odontogenic Keratocyst



- The odontogenic keratocyst has a destructive pattern of growth, burrowing through cancellous bone in a predominantly **anteroposterior direction**, this distinguishes it from the other odontogenic cysts that tend to expand in ***a unicentric ballooning pattern***.
- The unique pattern of growth of the odontogenic keratocyst, in comparison to the other cysts**, *suggests that different mechanisms of enlargement are involved:***
  - **a.** Active epithelial growth due to a higher rate of mitotic activity.
  - **b.** Cellular activity in the connective tissue capsules due to stimulation of osteoclasts.
  - **c.** Production of bone-resorbing factors including prostaglandins, collagenase, Interlukin1 and Interlukin 6.

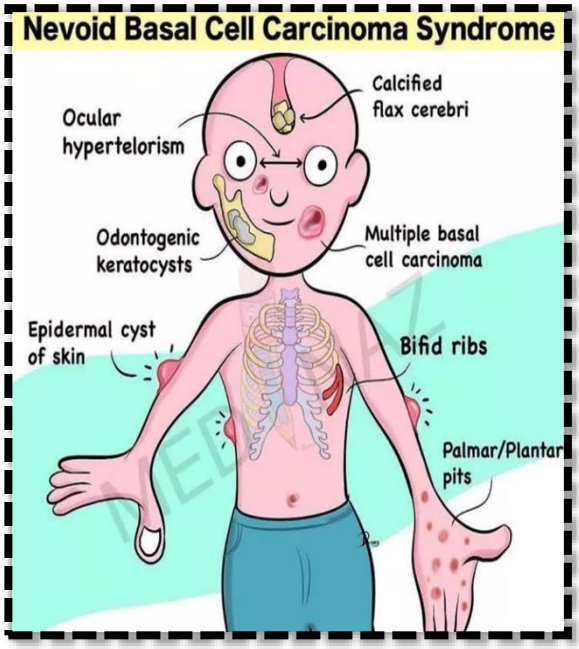
❖ **Clinical features:**

- **Incidence & Frequency:** OKC makes up 3% to 11% of all odontogenic cysts.
- **Age:** about 60% of all cases are diagnosed in people between **10 and 40 years of age**.
- **Sex:** There is a slight **male** predilection.
- **Site:** The **mandible** is involved in 60% to 80% of cases (marked tendency to involve the **posterior body and ascending ramus).**
- **Signs & Symptoms:**

✓ ***Small OKCs*** ⇒ are usually **asymptomatic** and discovered only during a radiographic examination.

- ✓ ***Larger OKCs*** ⟹ may be associated with pain, swelling, or drainage.
- ✓ ***Some extremely large cysts*** ⟹ may cause no symptoms. Why???
  Because (OKCs tend to grow in an *anteroposterior direction* within the medullary cavity of the bone without causing obvious bone expansion. This feature may be useful in differential clinical and radiographic diagnosis because dentigerous and radicular cysts of comparable size are usually associated with bony expansion).
- ➢ It usually arises sporadically and presents as solitary lesions, but **multiple odontogenic keratocysts** may be developed and associated with **Nevoid Basal Cell Carcinoma Syndrome (NBCCS).**

Nevoid Basal Cell Carcinoma Syndrome (NBCCS), (Gorlin-Goltz Syndrome)



- NBCCS is caused by mutation of a tumor suppressor gene in chromosome 9q t which affects signaling that is responsible for the growth of numerous tissues.
- **NBCCS has numerous manifestations including:**

1. Multiple naevoid basal cell carcinoma occurs anywhere in the skin, especially on the face trunk, and neck it is seen in about 90% of cases.
2. Multiple odontogenic keratocysts seen in about 90% of cases
3. Skeletal abnormalities in ribs (bifid ribs), vertebral deformities (kyphoscoliosis), polydactyly, temporal & frontal posing.
4. Central Nervous System disorders including calcified falx cerebri and brain tumours.
5. Mild ocular hypertelorism.

❖ **Radiographic Features:**

- OKC may is **Unilocular or multilocular radiolucency** which characteristically presents as a well-circumscribed radiolucency with smooth radiopaque margins.
- **Large lesions,** particularly in the posterior body and ascending ramus of the mandible, may appear multilocular.
- **An unerupted tooth** is involved in the lesion in 25% to 40% of cases.
- Resorption of the roots of erupted teeth adjacent to OKCs is less common than that noted with dentigerous and radicular cysts.



❖ Histopathologic Features:

- Odontogenic keratocysts contain thick, grey/white cheesy material consisting of keratin debris with little fluid and a low soluble protein level
- The cyst is lined by a regular layer of **parakeratinized stratified squamous** epithelium of some 5-10 cells in thickness.
- The epithelial lining has a well-defined basal epithelial cell layer is columnar or cuboidal cells with hyperchromatic, **palisaded arrangement** which is an important feature in distinguishing OKC from other cysts .
- The fibrous tissue capsule is usually thin and **free from inflammatory cells** and may contain satellite cysts called **(daughter cysts).**
- The epithelium and connective tissue interface are usually flat**,** with **no rete ridge formation**, so, detachment of portions of the cyst-lining epithelium from the fibrous wall is commonly observed.
- The luminal surface shows flattened parakeratotic epithelial cells, which exhibit a wavy or **corrugated appearance.**

  .
- **Small satellite cysts or daughter cysts**, cords, or islands of the odontogenic epithelium may be seen within the fibrous wall, these structures have been present in 7% to 26% of cases in various reported series.

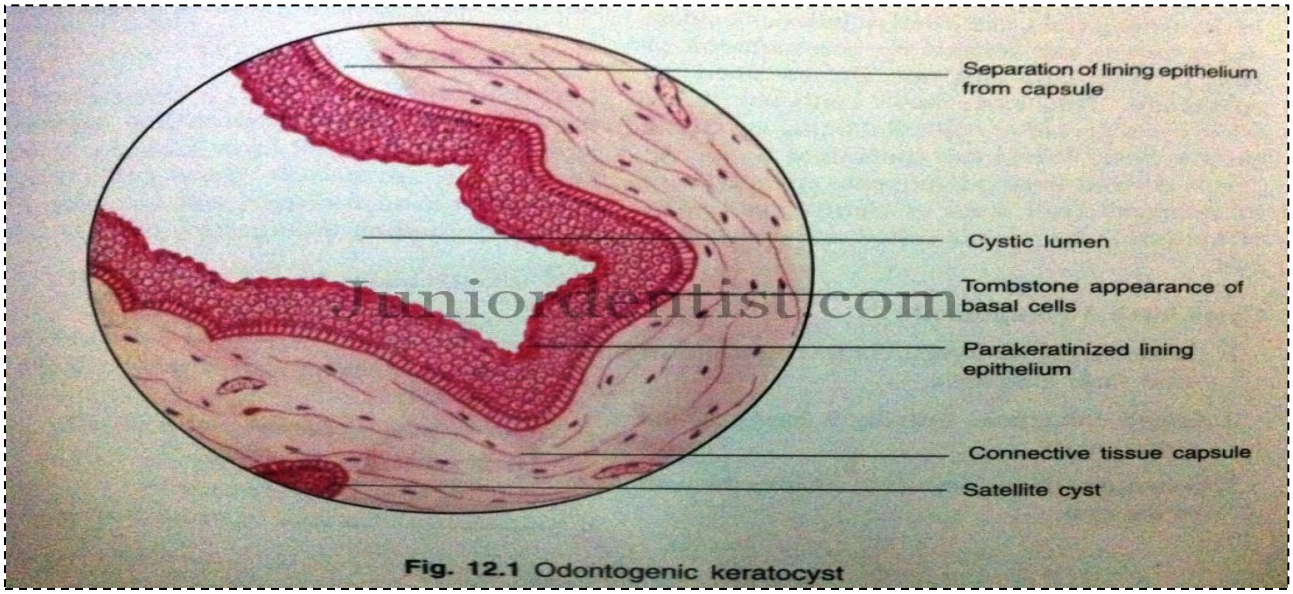


Fig. 12.1 Odontogenic keratocyst

❖ Treatment and Prognosis:

- Surgical excision with peripheral osseous curettage or ostectomy is the preferred method of management.
- It has a high recurrence rate after being removed.
- Follow-up examinations are important for patients with this lesion. Patients should be evaluated for completeness of excision, new keratocysts, and NBCCS.

Recurrence of Odontogenic Keratocyst

- In contrast to other odontogenic cysts, OKCs often tend to recur after treatment 30% or higher recurrence rate associated with OKCs due to:

1) Thin epithelial lining is difficult to enucleate intact.
2) Weak epithelial -connective tissue attachment, this causes the detachment of the epithelium and therefore more difficult to removal during surgery.
3) A"new" cyst that has developed from **satellite cyst or daughter cyst** may be left behind after surgery
4) Scalloped margins in multilocular variety.
5) Extensive lesion involving coronoid/ascending ramus (difficult access).

❖ **Differential Diagnosis:**

- **In case of unilocular radiolucency-** Dentigerous cyst, COC, AOT, Unicystic ameloblastoma.
- **In case of multilocular radiolucency**- Conventional ameloblastoma, Central giant cell tumor, Aneurysmal bone cyst.

4-ORTHOKERATINIZED ODONTOGENIC CYST (OOC)

❖ **Definition:**

- It is a developmental odontogenic cyst that is predominantly lined by Orthokeratinized stratified squamous epithelium.
- It was originally referred to as a type of OKC but both clinically and histopathological differs from parakeratinized odontogenic cyst so they should be placed into a separated category
- Orthokeratinized odontogenic cyst are **not associated** with with NBCCS., **do not have** high recurrence rates and **do not show** aggressive clinical behavior.
- Prominent palisaded basal layer of epithelial lining **is not** present in orthokeratinized odontogenic cyst which is diagnostic features in parakeratinized OKC.

- **Etiology & Pathogenesis:**
- The exact etiology and pathogenesis of OOC are not yet fully understood.
- It is believed to arise from the remnants of the dental lamina or the cell rests of Serres, which are remnants of the enamel organ.

❖ **Clinical Features:**

- **Age:** OOC most commonly affects individuals in the second to fifth decades of life.
- **Sex:** There is a slight predilection for males.
- **Site:** OOC often manifests in the posterior mandible, particularly in the molar and premolar regions. However, it can also occur in the maxilla and other regions of the jaws.

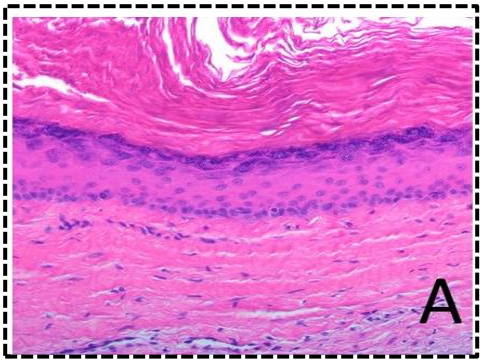
- **Symptoms:** OOC is typically asymptomatic and is often discovered incidentally during routine radiographic examinations. In some cases, patients may present with swelling, pain, or paresthesia if the cyst reaches a large size or impinges on adjacent structures.

❖ **Radiographic Features:**

- OCC: typically, present as well-defined unilocular radiolucent lesions. The borders of the cysts are usually corticated, indicating a slow and non-aggressive growth pattern.
- Most of OOC about 2/3 of cases are **associated with unerupted teeth.**
- Root Resorption: In some cases, OOCs may cause root resorption of adjacent teeth

❖ **Histopathological Features:**

- The hallmark histopathological feature of OOC is the presence of ***orthokeratinized stratified squamous epithelium,*** which is characterized by the absence of a granular layer.
- The granular cell layer is well defined and prominent which is absence in parakeratinized odontogenic cyst.
- The basal cells are cuboidal or flattened and absence palisaded basal cell layer



❖ **Treatment & Prognosis:**

- The treatment of OOC typically involves enucleation or marsupialization, depending on the size and location of the lesion.
- It has a lower recurrence rate than OKC.

The differences between OOC & OKC

Characteristic	OOC	OKC
Epithelial lining	Orthokeratinized stratified squamous epithelium	Parakeratinized stratified squamous epithelium
Granular layer	Absent	Present
Clinical presentation	Asymptomatic	Asymptomatic or associated with symptoms
Site of involvement	It can occur in any region of the jaws	Predilection for posterior mandible
Growth pattern	Slow and non-aggressive	Aggressive and infiltrative
Recurrence rate	Lower recurrence rate	Higher recurrence rate
Association with Gorlin syndrome	Not commonly associated	Frequently associated
Radiographic appearance	Well-defined unilocular radiolucency	Variable, can be unilocular or multilocular
Root resorption	Less commonly associated	More commonly associated
Treatment approach	Enucleation or marsupialization	Enucleation with adjunctive treatments
Prognosis	Generally favorable	Potentially more aggressive

5- LATERAL PERIODONTAL CYST (BOTRYOID ODONTOGENIC)

❖ **Definition:**
- It is an uncommon type of developmental odontogenic cyst (accounts for less than 2% of all epithelium-lined jaw cysts).
- It typically occurs along the lateral root surface of **a vital tooth.**

❖ **Origin:**
- It arises from the rests of the dental lamina
- It represents the intrabony counterpart of the gingival cyst of the adult.

❖ **Clinical Features:**
- **Age:** It most frequently occurs in patients in the 5th- 7th decades of life
- **Site:** 75% - 80% of cases occur in **the mandibular premolar–canine-lateral incisor area**.
- **Signs & Symptoms:** It is most often an asymptomatic lesion that is detected during a radiographic examination.

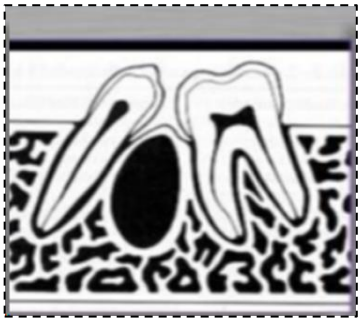
❖ **Radiographic Features:**
- The cyst appears as a well-circumscribed radiolucent area located laterally to the root of *the vital tooth;* most cysts are less than 1.0 cm in diameter.

A variant of the lateral periodontal cyst (Botryoid Odontogenic Cyst)



✓ It has a polycystic appearance (botryoid odontogenic cysts).
✓ They grossly and microscopically show a grape-like cluster of small Individual cysts.
✓ It is possibly the result of cystic degeneration and subsequent fusion of adjacent foci of dental lamina rests.
✓ The botryoid variant often shows a ***multilocular radiographic appearance.***

Differential Diagnosis of lateral Radiolucency:
1. Lateral Periodontal Cyst.
2. OKC that develops between the roots of adjacent teeth.
3. lateral radicular cyst.



❖ **Histopathologic Features:**

- The lateral periodontal cyst has a thin, generally non-inflamed, fibrous wall with an epithelial lining that is only 1 to 3 cells thick in most areas.
- This epithelium usually consists of flattened squamous cells, but sometimes the cells are cuboidal in shape.
- Some cysts show focal nodular thickenings of the lining epithelium, which is composed of glycogen-rich clear cells.
- Clear cell epithelial rests (represents rests of the dental lamina) are sometimes seen within the fibrous wall.

❖ **Treatment and Prognosis:**

- Conservative enucleation of the lateral periodontal cyst is the treatment of choice. Usually, this can be accomplished without damage to the adjacent teeth.
- Recurrence is unusual.

6-GINGIVAL CYST OF THE ADULT

❖ **Definition:**

The gingival cyst of the adult is an uncommon lesion; it is considered to represent *the soft tissue counterpart of the lateral periodontal cyst*, derived from rests of the dental lamina.

- It occurs in the free gingiva, attached gingiva, or sometimes in the interdental papilla

❖ **Clinical Features:**

- **Age:** Most commonly found in patients in the 5th and 6th decades of life.
- **Site:**
  - ✓ Like the lateral periodontal cyst, it shows a striking predilection to occur in ***the mandibular canine and premolar area*** (60% - 75% of cases).
  - ✓ They are almost invariably located on ***the facial gingiva or alveolar mucosa***.
  - ✓ Maxillary gingival cysts are usually found in the incisor, canine, and premolar areas.



- **Signs & Symptoms:**
  - ✓ The cysts appear as painless domelike swellings, usually less than 0.5cm in diameter, often bluish or bluish-gray.
  - ✓ In some instances, the cyst may cause a superficial "cupping out" of the alveolar bone. This is usually not detected on a radiograph but is apparent when the cyst is excised.

❖ **Histopathologic Features:**

Similar to those of the lateral periodontal cyst.

❖ **Treatment and Prognosis:**

Responds well to simple surgical excision; prognosis is excellent.

7- GINGIVAL (ALVEOLAR) CYST OF THE NEWBORN

❖ **Definition:**

- Gingival cysts of the newborn are small superficial keratin-filled cysts that are found on the alveolar mucosa of infants.

❖ **Origin:** These cysts arise from remnants of the dental lamina.

❖ **Appearance:**

- They are common lesions; however, because they disappear spontaneously by rupture into the oral cavity, the lesions are seldom sampled for biopsy.

N.B :{ similar inclusion cysts (e.g. Epstein's pearls and Bohn's nodules) are also found in the midline of the palate or laterally on the hard and soft palate}.

❖ **Clinical Features:**

- Gingival cysts of the newborn appear as small, multiple whitish papules on the mucosa overlying the alveolar processes of neonates (individual cysts no more than 2- 3 mm in diameter).
- The maxillary alveolus is more commonly involved than the mandibular.

❖ **Histopathological Features:**

- These are true cysts lined by stratified squamous epithelium and show a lumen filled with desquamated keratin, occasionally containing inflammatory cells

❖ **Treatment and Prognosis:**

No treatment is indicated because the lesions spontaneously involute after rupture; they are rarely seen after 3 months of age.

8- CALCIFYING ODONTOGENIC CYST (COC) OR GROLIN CYST

❖ **Definition:**

- ✓ It is a rare developmental odontogenic lesion characterized histologically by ghost cells.
- ✓ COC exists either as a cystic or a solid variant; the solid variant is known as Dentinogenic ghost cell tumor.
- ✓ The calcifying odontogenic cyst may **be associated with other recognized odontogenic tumors**, most commonly odontomas. However, adenomatoid odontogenic tumors and ameloblastoma also have been associated with calcifying odontogenic cysts.

❖ **Origin:**

Can arise from the dental lamina or remnant.

❖ **Clinical Features:**

- **Age:** 20-30 years.

- **Sex:** No sex predilection.
- **Site:** Most commonly in the maxilla.

Signs &Symptom

- Asymptomatic, slowly growing, and painless.



❖ **Radiographic Features:**

o Well-defined unilocular or multilocular radiolucent area contains calcified masses or radiopaque masses.

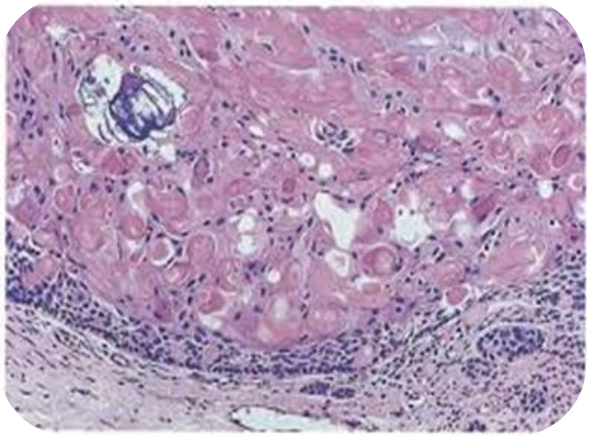
❖ **Histopathology features:**

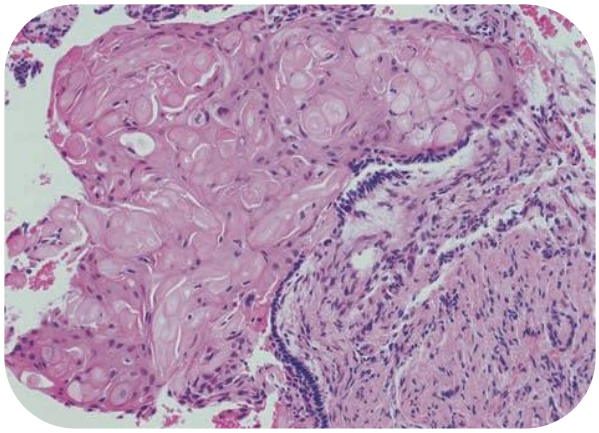
The lesion is characterized by cystic spaces lined by epithelium, which shows:

- Basal layer of columnar cells (Ameloblast-like cells with reversed polarity) and overlying layers of loosely arranged cells (stellate reticulum-like cells).
- The most prominent and **unique microscopic feature** is the presence of **ghost cells.**
- **Ghost cells** are pale eosinophilic swollen cells that have lost their nucleus and nuclear membrane, and these cells undergo calcification.
- **Calcification** is seen as dentinoid, i.e., (dentine like irregular eosinophilic masses) or dystrophic calcification.
- The nature of the ghost cell is controversial. Some believe that this change represents coagulative necrosis or accumulation of enamel protein; others contend it is a form of normal or aberrant keratinization of odontogenic epithelium.
- All these structures can invade the cyst lumen and connective tissues, which elicits foreign body giant cells in C.T.

❖ **Differential Diagnosis:**

1- Odontogenic keratocyst.
2- Ameloblastoma.
3- Adenomatoid odontogenic tumor and ameloblastic fibro-odontoma





9- GLANDULAR ODONTOGENIC CYST (SIALO-ODONTOGENIC CYST)

❖ **Definition:**

- A glandular odontogenic cyst is a rare and **recently recognized** type of developmental odontogenic cyst that can show *locally aggressive recurrence potential*.
- It has some histologic features that suggest a mucus-producing salivary gland tumor **(low-grade mucoepidermoid carcinoma).**

❖ **Origin:**

Although it is accepted as being of odontogenic origin, it also shows glandular or salivary features that presumably are an indication of the pluri-potentiality of odontogenic epithelium.

❖ **Clinical Features:**



- **Age:** The glandular odontogenic cyst occurs most commonly in middle-aged adults with a mean age of 49 years; rarely does it occur **before** the age of 20.
- **Site:** Nearly 85% of cases occur in the mandible. The cyst has a strong predilection for ***the anterior region of the jaws***, and many mandibular lesions will cross the midline.

Signs & Symptoms:

- The size of the cyst can vary from small lesions less than 1cm in diameter to large destructive lesions that may involve most of the jaw.
- Small cysts may be asymptomatic; however, large cysts often produce clinical expansion, which sometimes can be **associated with pain or paresthesia.**

❖ **Radiographic Features:**

- Unilocular or, more commonly *a multilocular radiolucency*. The margins of the radiolucency are usually well-defined with a sclerotic rim.
- More aggressive lesions have shown an ill-defined peripheral border.

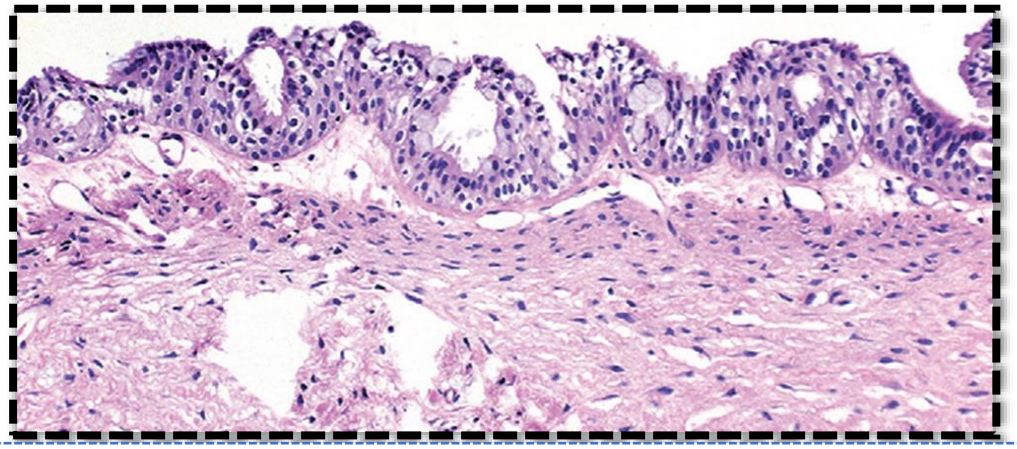
❖ **Histopathologic Features:**

- The cyst is lined by squamous epithelium of varying thickness.
- The interface between the epithelium and the connective tissue wall is generally flat.
- The superficial epithelial cells that line the cyst cavity tend to be cuboidal to columnar and have an irregular and sometimes papillary surface. Occasionally cilia may be noted.
- **Pools of mucinous material** are often present within the epithelium. Cuboidal cells usually line these pools.
- **Glandular duct-like spaces** within the epithelial lining are another characteristic finding
- Mucous cells may or may not be present within the epithelium.

- The fibrous wall is usually devoid of any inflammatory cell infiltrate.

❖ **Treatment and Prognosis:**

- Most cases of glandular odontogenic cyst have been treated by enucleation or curettage.
- The cyst shows a tendency for recurrence (30% or more of all cases).



Glandular odontogenic cyst lined by epithelium showing ductlike features and mucous cells.

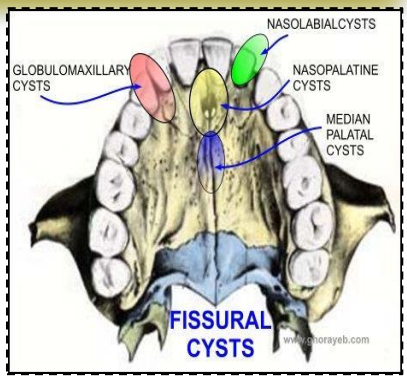
Non-Odontogenic Cysts

➢ **Non-odontogenic cysts** :

Cysts that arise from epithelial inclusion or entrapment in the lines of fusion of the developing facial processes during embryogenesis.

1-NASOPALATINE DUCT (INCISIVE CANAL) CYST

❖ **Definition:**



- The Nasopalatine Duct Cyst (also known as Incisive Canal Cyst) is **the most common** non-odontogenic developmental cyst that arises from remnants of the nasopalatine duct.

❖ **Etiology & Pathogenesis:**

- It arises from epithelial remnants of the nasopalatine duct (which connects the oral and nasal cavities in the embryo) so it is classified as a "fissural cyst".
- It may arise at any point along the nasopalatine canal; most cases arise in the lower part of the canal, but some may arise within ***palatal soft tissues*** at the point of opening of the canal, where the lesions are called ⟹ ***Cysts of the palatine papilla or Cyst of incisive papilla.***

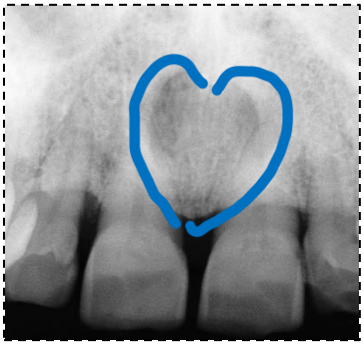
- The stimulus for cyst formation is unknown but:
  - ***The bacterial infection and/or trauma*** is thought to have a role.
  - The mucous glands within the lining may cause cyst formation as a result of mucin secretion.

❖ **Clinical Features:**

- **Frequency & Incidence:** commonest non-odontogenic developmental cyst (occurring in about 1%of the population).
- **Age:** 5th & 6th decades.
- **Sex:** males more than females.
- **Site:** Found in the midline of the anterior maxilla, just behind the central incisors. It may be seen in the soft tissue of the incisive papilla or as a cyst of papilla palatine
- **Signs & Symptoms:**
  - ✓ Asymptomatic (unless infected) & discovered on routine x-ray examination.
  - ✓ It may present as a slowly growing swelling in the anterior region of the midline of the palate and may discharge in the mouth through a sinus giving (a salty taste) and **pain (pressure on nerves).**

❖ **Radiographic Features:**

- A well-defined round, ovoid, or **heart-shaped radiolucency** with a sclerotic rim.
- Usually symmetrical about the midline between the roots of the central incisors, but some are displaced to one side.
- The classic heart shape as a result of *superimposition of the anterior nasal spine*.
- Its radiographic diameter can range from (6mm) to (6 cm).



Differential Diagnosis Radiographically:

1. Normal incisive fossa (not greater than 6 mm wide).
2. Radicular cyst of incisors (non-vital teeth).

❖ **Histological Features:**

- **The epithelial lining:**

The cysts may be lined by a variety of different types of epitheliums:

a. Stratified squamous epithelium, **b.** pseudostratified ciliated columnar (respiratory) epithelium often containing mucous cells, **c.** cuboidal epithelium, may be seen alone or in an combination ⟹ *This is according to relation to the nasal or the oral cavities.*

- **Connective tissue wall:**
  - ✓ Prominent neurovascular bundles from the terminal branches of the long sphenopalatine nerve and vessels helpful diagnostic aid.
  - ✓ Mucous glands & chronic inflammatory cell infiltration.

❖ **Treatment and Prognosis:**
Treated by surgical enucleation, Recurrence is rare.

2-NASOLIBAL CYST

Nasolabial Cyst

❖ **Definition:**

- It is a rare lesion which arises in the soft tissue of the upper lip lateral to midline, just below the ala of the nose.
- It is traditionally regarded as a jaw cyst, although strictly speaking it should be ***classified*** ⟹ ***as a soft tissue cyst***.

❖ **Etiology & Pathogenesis:**

Its pathogenesis is uncertain, although there are two major theories:

- **One theory** ⟹ considers the nasolabial cyst to be a "fissural cyst" arising from epithelial remnants entrapped along the line of fusion of maxillary, medial nasal, and lateral nasal processes.
- **A second theory** ⟹ suggests developing from remnants of the lower part of the embryonic nasolacrimal duct.

❖ **Clinical Features:**



- **Frequency:** rare in occurrence.
- **Age:** most commonly seen in adults (peak prevalence in 4th -5th decades of life).
- **Sex:** a significant predilection for women (F: M ratio = 3: 1).
- **Signs & Symptoms:** slowly growing swelling & occasionally, pain and difficulty in nasal breathing.
  - o Extra orally ⟹ obliterating the nasolabial fold & distorting the nostril.
  - o Intra orally ⟹ bulge in the labial sulcus.
  - o Fluctuant lesion & may arise bilaterally.

❖ **Radiographic Features:**

- Difficult to interpret on radiograph; ***negative on X-ray***.
- May be seen as localized increased lucency of alveolar process above apices of incisors.

❖ **Histological Features:**

- The cysts are usually lined by pseudostratified columnar epithelium but stratified squamous epithelium, mucous cells, and ciliated cells may also be present.
- Supported by CT capsule.
-

❖ **Treatment & Prognosis**: complete surgical excision of cyst.

3-POST SURGICAL CILITED CYST

Nasolabial Cyst

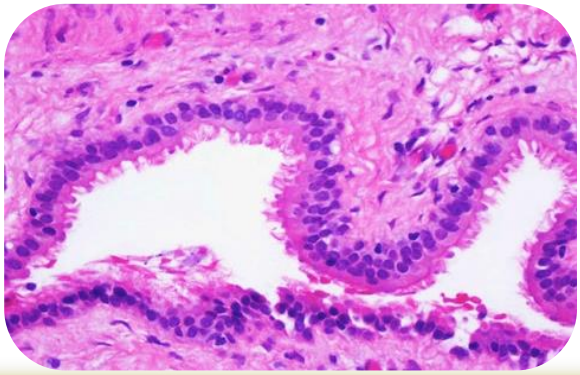
❖ **Definition:**

✓ It is also known as a postoperative maxillary cyst—a rare, non-odontogenic cyst lined by respiratory epithelium, with a fluid-filled cavity that can develop years or even decades after a surgical procedure or trauma involving the upper jaw or sinuses

❖ **Clinical Features:**

- Localized swelling in the cheek or jaw area, **mild pain**, or a feeling of pressure.
- Asymptomatic: Many cases cause no pain at all and are only discovered by accident on routine dental X-rays or CT scans.
- It occurs most commonly in the posterior maxilla but is very rare in the mandible.

❖ **Histopathologically:** the cyst is lined by pseudostratified ciliated columnar and mucous cells.





4-MEDIAN PALATAL CYSTS

Nasolabial Cyst

❖ **Definition:**

- It is very rare and is thought to originate from epithelial remnants at the time of fusion of the palatal shelves of the maxilla.

❖ **Clinical Features:**

- Firm or fluctuant swelling in the midline of the hard palate posterior to the incisive canal.
- Asymptomatic; some complain of pain or expansion.

❖ **Radiographic Features:**

- Well-defined radiolucency is demonstrated in the midline of the hard palate, surrounded by a radiopaque margin and divergence of central incisors

❖ **Histopathological Features:**

 The histological features are similar to those of a nasopalatine duct cyst, and some published cases may represent displaced nasopalatine duct cysts.

❖ **Treatment:** Surgical removal

Non-epithelialized primary bone cysts

1-SIMPLE BONE CYST
(TRAUMATIC BONE CYST, HEMORRHAGIC BONE CYST, SOLITARY BONE CYST, IDIOPATHIC BONE CAVITY)

❖ **Definition:**

It is a benign, empty, or little fluid-containing cavity within bone that lacks an epithelial lining.

❖ **Etiology & Pathogenesis:** *The cause and pathogenesis are uncertain & controversial; several theories have been proposed:*

1. **The trauma-hemorrhage theory** (designated traumatic bone cyst), trauma to the bone that is insufficient to cause a fracture results in an intraosseous hematoma. If the hematoma does not undergo organization and repair, it may liquefy, resulting in a **cystic defect (an empty cavity)**.
2. **Other theories** include inadequate venous drainage, local disturbance in bone growth, ischemic marrow necrosis, and localized alteration in bone metabolism resulting in **osteolysis**.

❖ **Clinical Features:**



- **Age:** children & adolescents with a peak incidence in 2nd decade.
- **Sex:** about 60% of cases occur in males.
- **Site:**
  - Most commonly involve long bones, but occasionally seen in the jaws.
  - In the jaws are restricted to **the mandible premolar & molar areas**.
- ***Signs & Symptoms:***
  - It is asymptomatic & usually found on routine radiographs examination.
  - Only 25% of cases may show expansion of bone.

❖ **Radiographic Features:**



- The lesion presents as a radiolucency of variable size (1 to 10 cm in diameter) & irregular outline.
- Interdental scalloping around & between the roots of standing teeth is seen.
- The margins of the lesion are usually well-defined.
- The involved teeth are vital and do not show root resorption.

❖ **Histological Features:**

- **Surgical exploration:** ***to confirm diagnosis*** reveals a rough bony-walled cavity with no soft tissue lining. It is empty or, in some cases, shows little clear or blood-stained fluid.
- **Histopathologic examination of curettage shows:**
  - ✓ Delicate layer of loose, vascular CT containing extravasated RBCs & deposits of hemosiderin pigment.
  - ✓ No epithelial lining.

❖ **Treatment & Prognosis:**

Rapid healing follows surgical exploration, but even without surgical intervention the cyst will resolve s*pontaneously* with time. Prognosis is excellent.

2-ANEURYSMAL BONE CYST (ABC)

~~Nasolabial Cyst~~

❖ **Definition:**

It is an intra-osseous accumulation of variable-sized blood-filled spaces surrounded by cellular fibrous CT that often is admixed with trabeculae of reactive woven bone.

❖ **Etiology & Pathogenesis:**

- The cause & pathogenesis of the aneurysmal bone cyst are poorly understood.
- Several investigators have proposed that aneurysmal bone cyst arises from a traumatic event, vascular malformation, or neoplasm that disrupts the normal osseous haemodynamics and leads to an enlarging haemorrhagic extravasation.
- It occurs either as **a primary or as a secondary lesion** in a pre-existing intra-bony lesion) as ⟹ (fibro-osseous lesions & central giant cell granuloma).

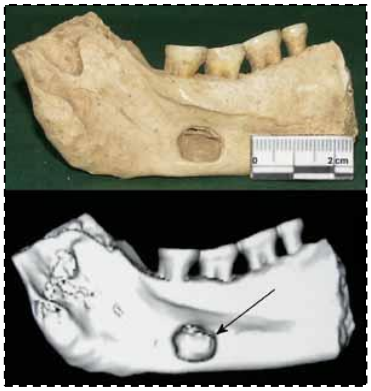
❖ **Clinical Features:**

- **Age:** (most cases arise in children & young adults with a mean age of 20 years).
- **Sex:** No significant sex predilection is noted.
- **Site:**
  - It is located most commonly in the shaft of a long bone or in the vertebral
  - Gnathic aneurysmal bone cysts are uncommon (only 2% reported from the jaws).
  - In the jaws, mostly affects mandible posterior part of the body or the angle.
- **Signs & Symptoms:**

✓ The most common clinical manifestation is a swelling that has usually **developed rapidly**; **Pain** is often reported.

✓ The lesion is known to cause facial deformity, migration of teeth, malocclusion, cortical thinning, perforation, and pathologic fracture.

✓ Maxillary ABCs may cause nasal obstruction, bleeding, and proptosis.



✓ **At the time of surgery,** when the periosteum and bony shell are removed, dark venous blood frequently wells up and venous-like bleeding may be encountered. The appearance at surgery has been likened to that of **a "blood-soaked sponge."**

❖ **Radiographic Features:**

- Usually shows a multilocular, soap bubble appearance.
- May also show a unilocular appearance.
- Displacement of teeth and root resorption.
- Cortical expansion, thinning and perforation.
- **"Ballooning" or "blow out"** expansion of bone (the cortex literally expands like a blown balloon).

❖ **Histopathologic Features:**

- Blood-filled spaces of varying size, surrounded by cellular fibroblastic tissue containing **multinucleated giant cells**, and trabeculae of osteoid and woven bone.
- The blood-filled spaces are not lined by endothelium.

❖ **Treatment and Prognosis:**

Curettage or enucleation, sometimes supplemented with cryosurgery.

Differential Diagnosis of Aneurysmal bone cyst:

- Radiographically should differentiated from ⇒ Multilocular Radiolucent Lesions.
- Microscopically should differentiated from ⇒ Central Giant Cells Lesions.

3-STAFNE, S BONE CYST (LINGUAL MANDIBULAR SALIVARY GLAND DEPRESSION; LATENT BONE CYST; STATIC BONE CYST)

Nasolabial Cyst

❖ **Definition:**

This condition represents a focal concavity of the cortical bone on the lingual surface of the mandible, containing ectopic normal salivary gland tissue.

❖ **Clinical Features:**

- A striking male predilection is observed (80% to 90% of all cases seen in men).
- It is symptomless & found by chance in routine radiograph examinations.
- Stafne defects typically **remain stable in size**; hence the name static bone cyst.

❖ **Radiographic Features:**

- It appears as a round or oval, well-demarcated radiolucency between the premolar region & angle of the mandible.
- It is usually located beneath the inferior dental canal and is bilateral as a saucer-shaped depression on the lingual aspect of the mandible.

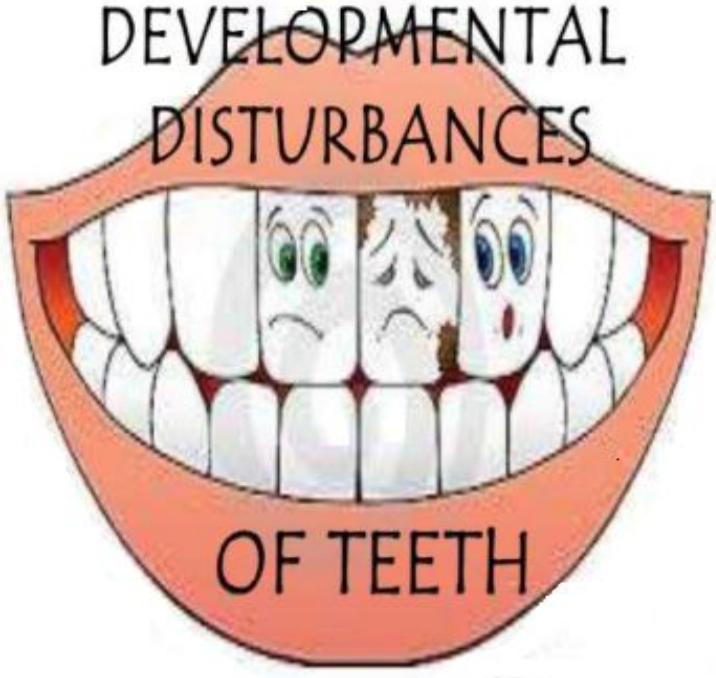
N.B.: **The diagnosis** can usually be made on a clinical basis by the typical radiographic location and lack of symptoms; clinical diagnosis can be confirmed by CT scans, MRI, or sialography.

❖ **Histopathologic Features:**

- Because of **the typical radiographic appearance,** biopsy is **usually not necessary.**
- If biopsy is performed, normal submandibular gland tissue is usually seen.

❖ **Treatment and Prognosis:**

No treatment is necessary for patients with Stafne defects of the posterior mandible, and the prognosis is excellent.



Developmental Disturbances of Teeth

University of Tripoli
Dental Faculty
ORAL PATHOLOGY
3rd year dental students
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(2026)

Developmental Disturbances of Teeth

I. Disturbances in number of teeth (defective initiation):

- *Anodontia.*
- *Hyperdontia.*

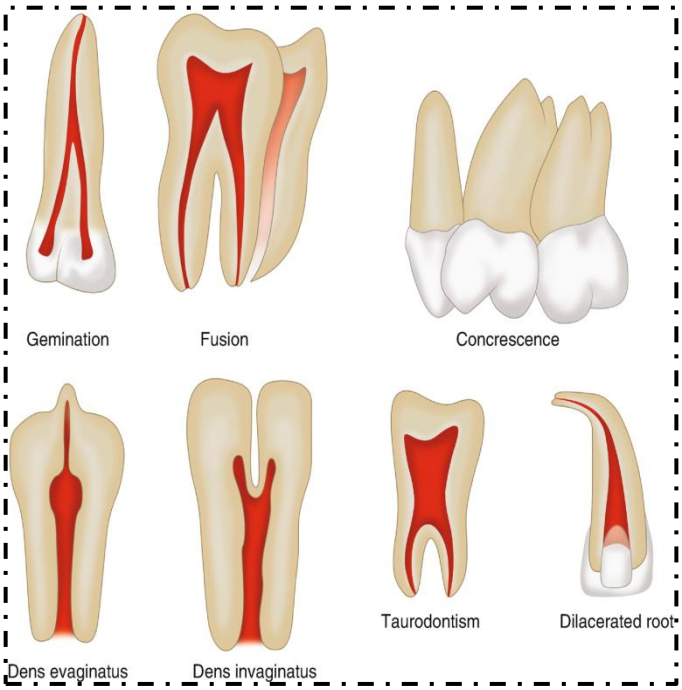
II. Disturbances in size & shape of teeth (defective in morpho & histo-differentiation):

1. Disturbances in size of teeth:

- *Macrodontia.*
- *Microdontia.*

2. Disturbances in shape of teeth:

- *Germination.*
- *Fusion.*
- *Concrescence.*
- *Talon cusp.*
- *Dilaceration.*
- *Supernumerary root.*
- *Enamel pearl.*
- *Dens evaginatus.*
- *Dens invaginatus*
- *Taurodontism.*
- *Congenital syphilis.*



III. Disturbances in structure of teeth (defective apposition & calcification):

- Disturbances in structure of enamel:
  - *Hereditary defects (e.g amelogenesis imperfecta).*
  - *Acquired or environmental defects.*
- Disturbances in structure of dentine:
  - *Dentinogenesis imperfecta.*
  - *Dentine dysplasia.*
- Disturbances in structure of enamel & dentine:
  - *Regional odontodysplasia.*
- Disturbances in structure of cementum:
  - *Hypocementosis.*
  - *Hypercementosis.*

I. Disturbances in number of teeth:

- Anodontia ⟶ decrease in number of teeth.
- Hyperdontia ⟶ increase in number of teeth.

Classification of Anodontia:

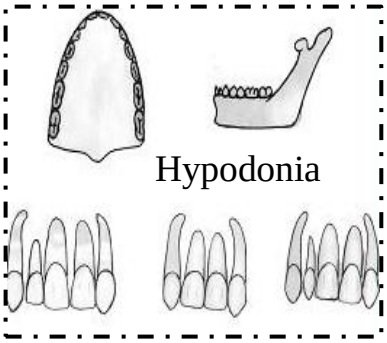
1. **Complete anodontia** ⟶ its congenital absence of all teeth.
2. **Partial anodontia**:
   a) *Hypodontia* ⟶ *its lack of development of one or more teeth*
   b) *Oligodontia* ⟶ *its lack of development of six or more* teeth.
3. **False anodontia** ⟶ occurs as a result of tooth loss (extraction or trauma).
4. **Pseudo-anodontia** ⟶ the clinically absent teeth are developed but fail to erupt (impacted or embedded teeth).



Complete Anodontia:

- ✓ Total absence of all teeth is extremely rare.
- ✓ It often associated with a syndrome known as Hereditary Hypohidrotic Ectodermal Dysplasia.

Partial Anodontia:



- ✓ Relatively common condition in permanent dentition. The most frequently missing teeth are 3rd molar, maxillary lateral incisor, 2nd premolar.
- ✓ Congenitally missing primary teeth is uncommon, but when occurring, usually involve the maxillary lateral incisor.
- ✓ Etiology-unknown, hereditary (familial) & racial tendency ⟶ absence of mandibular permanent central incisors are more common in Swedish & Japanese population.
- ✓ It may be associated with other defects and syndromes such as:
  a) Cleft lip and palate.
  b) Crouzon syndrome.
  c) Down's syndrome.
  d) Hereditary Hypohidrotic Ectodermal Dysplasia.



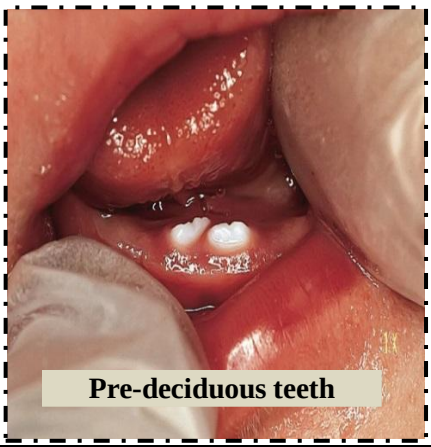
Hereditary Hypohidrotic Ectodermal Dysplasia (Streeter's syndrome):

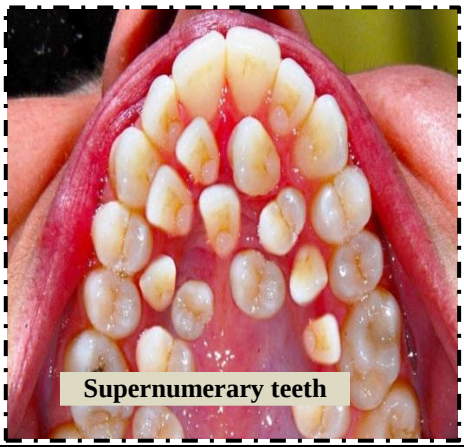
- ✓ It is a hereditary defect characterized by congenital absence of ectodermal structures.
- ✓ **Etiology:** inherited as X-linked recessive trait (affect mainly males & rare in females).
- ✓ **Clinical features**:
  - ➢ Affected patients have thin, smooth dry skin lack of sebaceous glands.
  - ➢ Partial or total absence of sweat glands (hypohidrotic or anhidrosis) leads to hyperthermia and the patient cannot tolerate hot weather.
  - ➢ The hair follicles may be defective (hypotrichosis) or absent, the patient is completely bald or possess fine scanty hair.
  - ➢ Finger nails are usually defective.
  - ➢ Depressed bridge of the nose saddle nose.
- ✓ **Oral manifestation:**
  - ➢ Total or partial anodontia of both dentitions.
  - ➢ The few teeth that are present are often retarded in eruption with conical crown.
  - ➢ The palate is frequently high shaped & cleft palate may be present.
  - ➢ Reduction of dimension of alveolar process.
  - ➢ Salivary gland hypoplasia with xerostomia may be present.
  - ➢ Lips are protuberant.

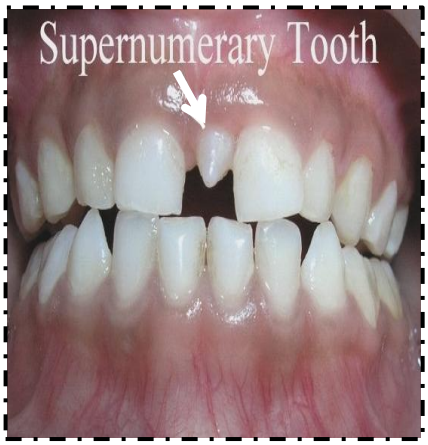
Classification of Hyperodontia:

1) **Pre-deciduous dentition (congenital teeth)** ⟶ natal & neonatal teeth.
2) **Post-permanent dentition:**

   Eruption of several teeth after extraction of the permanent teeth particularly after insertion of denture.
3) **Supernumerary teeth.**


Pre-deciduous teeth


Supernumerary teeth



Supernumerary Teeth:

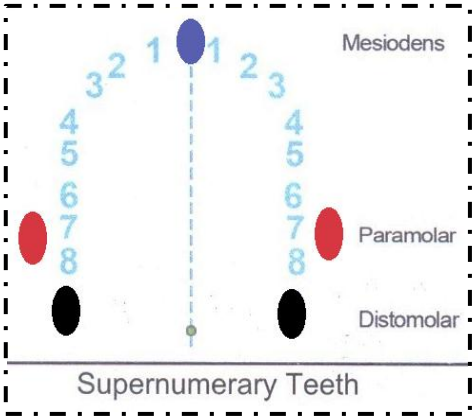
- ✓ These are teeth additional to those of the normal series.
- ✓ **Sex**: F > M. Permanent teeth > deciduous teeth.
- ✓ **Site**: maxilla (anterior & molar region) > mandible (premolar region).

Etiology:

- ✓Some supernumerary teeth may have hereditary tendency.
- ✓They may be associated with ***cleft lip & cleft palate, cleidocranial dysplasia and Gardner's syndrome.***

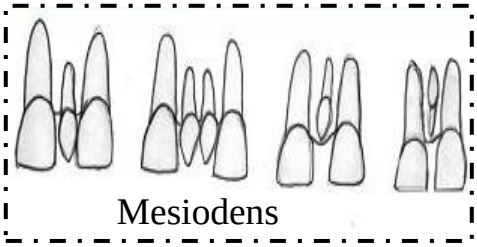
Pathogenesis:

- ✓ It may arise from additional tooth germ.
- ✓ Division of the tooth germ of the normal series.



Types of Supernumerary teeth:

1) ***Rudimentary Supernumerary teeth:*** *are extra teeth that morphologically differ from normal teeth and may be small and conical or peg shaped. Several terms have been used to describe supernumerary teeth depending on their location:*
   a) ***Mesiodens***: a supernumerary tooth in the midline between maxillary central incisors.
   It is the most common type of supernumerary teeth and usually has conical crown and short roots.

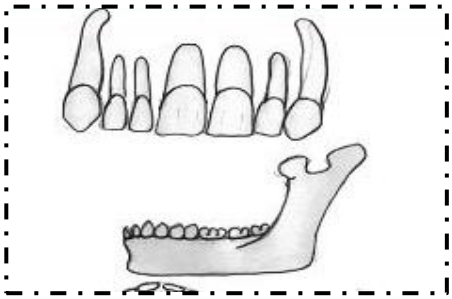

Mesiodens

   b) ***Paramolar****:* it located buccally or lingually to a 2nd or 3rd molars. It may be separate or fused to the adjacent molar, appearing as an extra cusp.
   c) ***Distomolar or distodent:*** a fourth molar develops distal to the 3rd molar.

2) ***Supplemental Supernumerary teeth:*** *are extra teeth that morphologically resemble a tooth of normal series, e.g. upper lateral incisor & lower premolar region.*





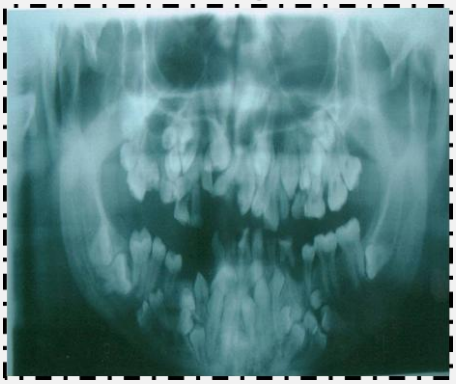


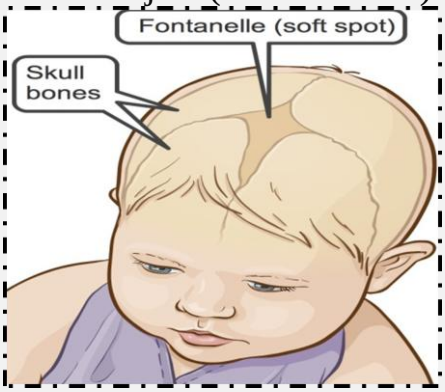
Clinical consideration (complications):

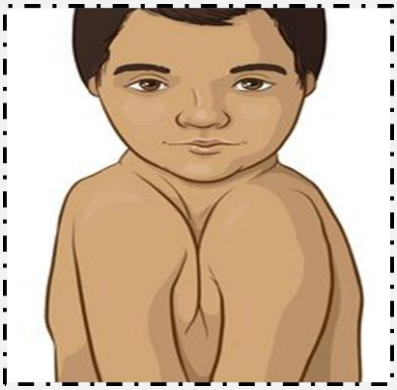
1) Displacement of adjacent normal teeth → crowding or formation of diastema.
2) Creation of stagnation areas (food debris tends to lodge) → caries.
3) Delay or prevent eruption of normal teeth.
4) Impacted supernumerary teeth can develop dentigerious cyst around the crown.

Cleidocranial Dysplasia:

- ✓ It is a condition that primarily affects the development of the bones and teeth.
- ✓ **Etiology:** inherited as an autosomal dominant.
- ✓ **Signs and symptoms**:
  - Underdeveloped or absent collar bones (clavicles). As a result, their shoulders are narrow and sloping, can be brought unusually close together in front of the body.
  - Delayed closing of the spaces between the bones of the skull (fontanels).
  - ***Dental abnormalities include***:
    - Delayed shedding of the primary teeth.
    - Delayed eruption of the permanent teeth.
    - Many unerupted supernumerary teeth and multiple dentigerious cyst.
    - The maxilla is narrow with high-arched palate → open bite.
    - Crown → conical or peg-like shape with enamel hypoplasia.
    - Root → short, hooked & curved with absence of cellular cementum.
    - Misalignment of the teeth and jaw (malocclusion).



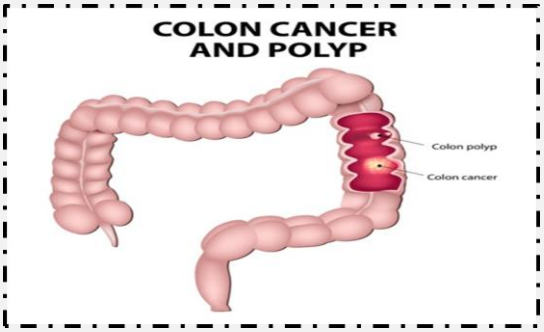




Gardner's syndrome:

It is a rare familial disorder transmitted as an autosomal dominant trait. The Gardner's syndrome is characterized by:

1. Multiple osteomas of the skull and long bones.
2. Multiple fibrous tumours.
3. Multiple epidermal/sebaceous cysts of the skin.
4. Multiple impacted supernumerary & permanent teeth.
5. Multiple intestinal polypoid lesions specific at large intestine → they have a highly malignant potential.





II. Disturbances in size & shape of teeth:

1) Disturbances in size of teeth:

A. **Macrodontia:** is used to describe teeth larger than normal.

Classification:

a) *True generalized macrodontia* ⟶ all teeth that are larger than normal. It occurs in pituitary gigantism.

b) *Relative generalized macrodontia* ⟶ result from disproportion between size of the tooth and size of the jaw (**i.e. normal size teeth in small jaw**).

c) *Localized or Focal macrodontia* ⟶ affecting single or group of large teeth in relation to an otherwise dentition. The most involved maxillary central incisors. It occurs in facial hemihypertrophy.





B. **Microdontia:** is used for teeth smaller than normal.

Classification:

a) *True generalized microdontia* ⟶ all teeth that are smaller than normal. It occurs in pituitary dwarfism and Down's syndrome.

b) *Relative generalized microdontia* ⟶ result from disproportion between size of the tooth and size of the jaw (**i.e. normal size teeth in large jaw**).

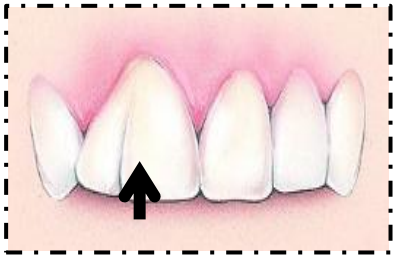
c) *Localized or Focal microdontia* ⟶ affecting single or group of small teeth in relation to an otherwise dentition. The most commonly affected teeth are permanent maxillary lateral incisors (Peg lateral) & 3rd molar.





2) Disturbances in shape of teeth:

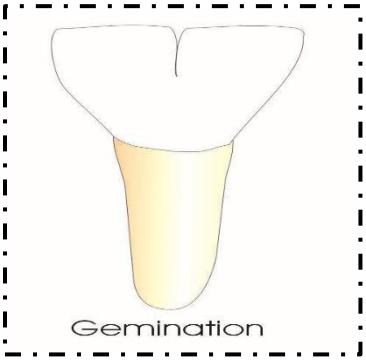
A. Gemination:



It is the term applied when **one tooth** appears as two teeth joined together due to **incomplete division** of single tooth germ.

Clinical features:

- Incomplete formation of 2 teeth resulting in a single large tooth structure with two completely or incompletely separated crowns that have single root and root canal.
- Tooth count is normal.
- Occur both in primary & permanent dentition, the maxillary incisors are the most teeth frequently affected.
- •Exhibit a hereditary tendency.



B. Fusion:

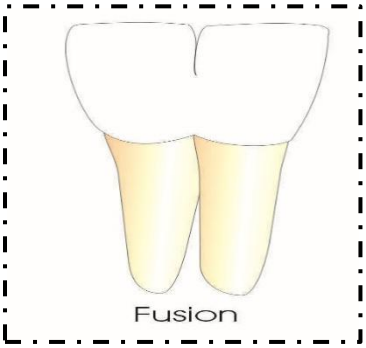
It is the term applied when **two teeth** joined together which results from the union of two adjacent teeth germs.

Types:

- ✓ ***Complete fusion*** ⟶ union of the crown (dentine and/or enamel) & the roots
- ✓ ***Incomplete fusion*** ⟶ union of the roots only.

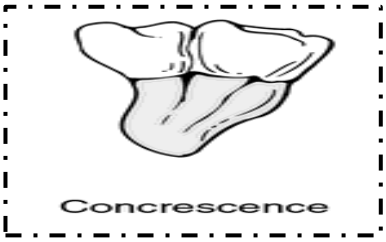
Clinical features:

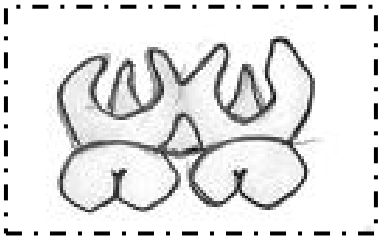
- Tooth may have separated or fused roots canals.
- Tooth count reveals a missing tooth, unless that one of the fused teeth was a supernumerary.
- Seen in deciduous as well as permanent dentition.
- Higher frequency in anterior and maxillary region.



C. Concrescence:

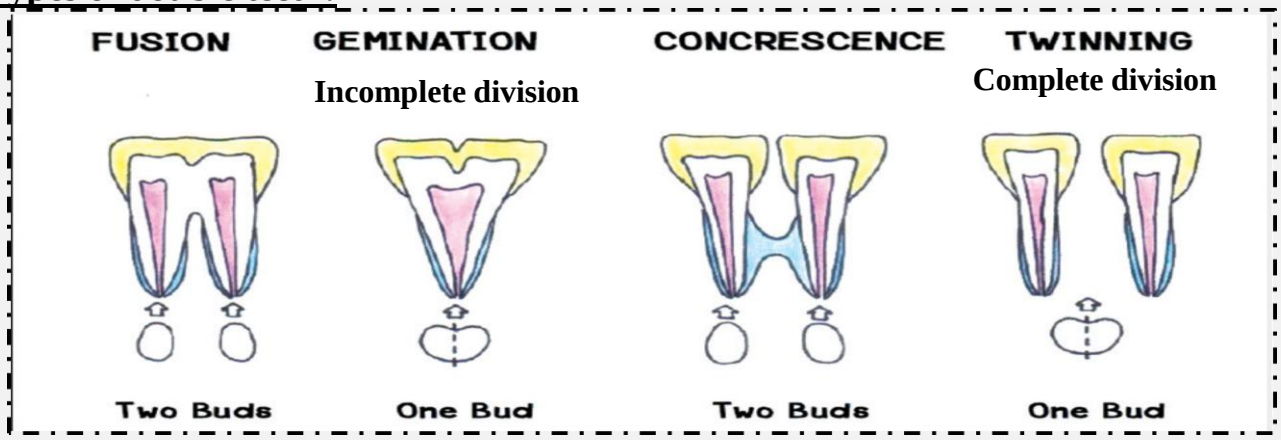
- It is the term applied when two teeth are joined together along their roots by **cementum,** due to trauma injury or crowding leading to loss of interdental bone.
- Fusion which occurs after root formation has been completed.
- It may occur before/after teeth have erupted.
- Concrescence is more common in permanent than primary dentitions, particularly the roots of maxillary 2nd & 3rd molars.



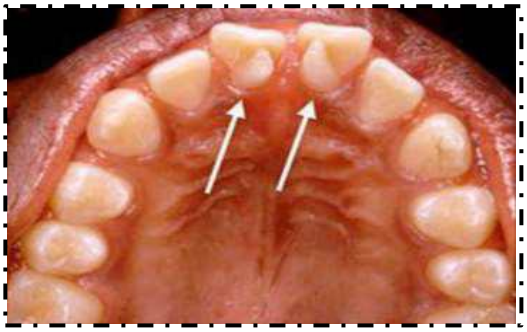


Double teeth (connated teeth) are terms used to describe two teeth joined together.

Types of double teeth:



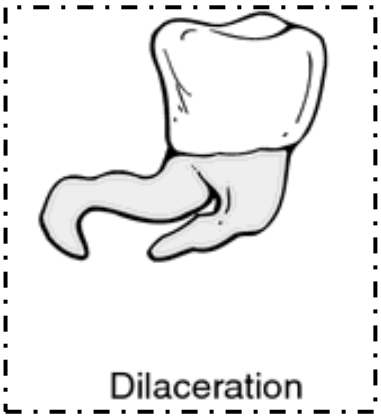
D. Talon's Cusp:

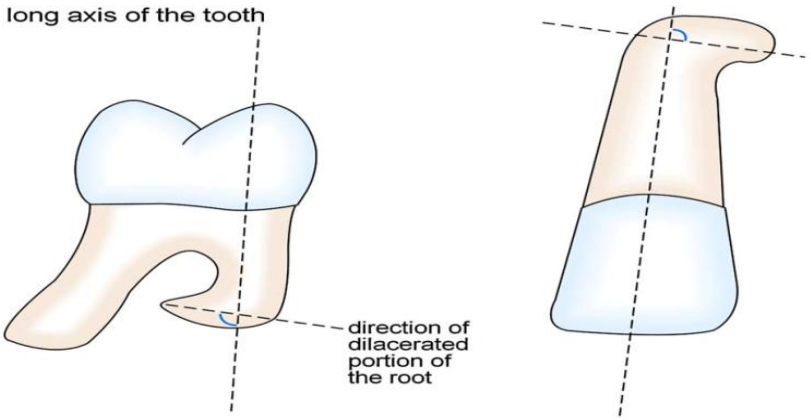


- It is an extra cusp resembling an eagle's talon.
- It projects lingually from the cingulum area of maxillary and mandibular permanent incisor.
- It is composed of normal enamel and dentin & contains a horn of pulp tissue.

E. Dilaceration:

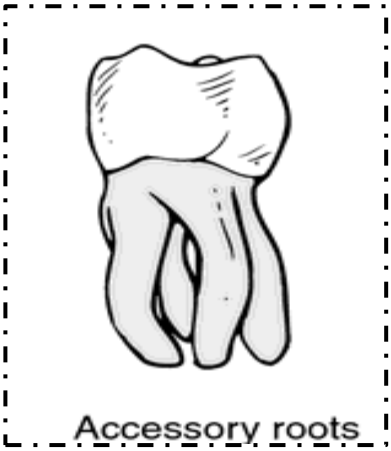
- Refer to an angulation or a sharp bend or curve in root or crown of a formed tooth.
- Due to trauma in the period in which the tooth is forming, the position of calcified portion of tooth is changed & remain of tooth is formed at an angle to normal axis of tooth.
- Most common in maxillary incisors.



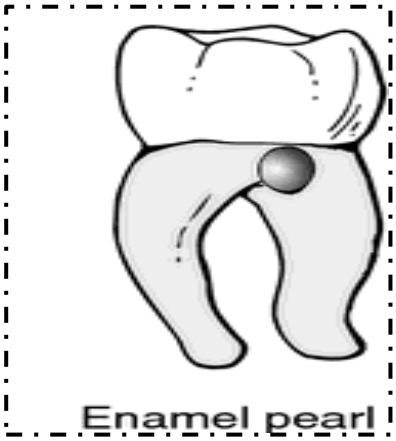


F. Supernumerary Roots:

- Extra roots are most commonly seen in mandibular premolar, canine & maxillary and mandibular 3rd molars.



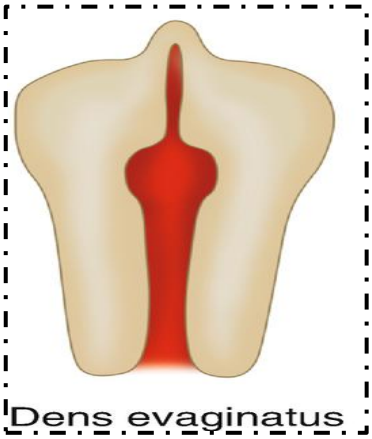
G. Enamel Pearl or Droplet(Enameloma):



- It is a small, spherical enamel projection located on the root surface near bifurcation or trifurcation area. Usually found on maxillary molars.
- **Histologically:** ⟶ it may consist of enamel only or enamel, dentine & pulp.
- **Etiology:**

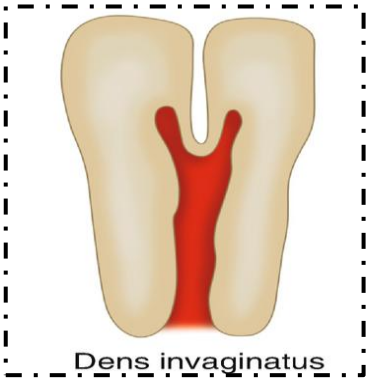
  Due to proliferation of group of *displaced ameloblasts* or localized *activity of odontoblastic layer* located between Herwig's root sheath and developing dentine which induce enamel formation.

H.Dens Evaginatus (Occlusal Enamel Pearl/ Leong's Premolar):



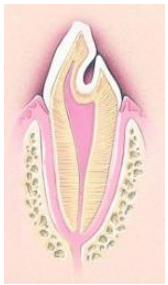
- Accessory cusp or globule of enamel on occlusal surface between buccal and lingual cusps of premolars.
- Unilateral or bilateral.
- A pulp horn may extend into this extra cusp.
- **Etiology:** due to proliferation and evagination of enamel epithelium and subsequent proliferation odontogenic mesenchyme during tooth development.
- **Clinical significance:** the extra cusp may contribute to displacement of the teeth and/or pulp exposure with subsequent infection following occlusal wear or fracture.

I. Dens Invaginatus (Dens in Dente):

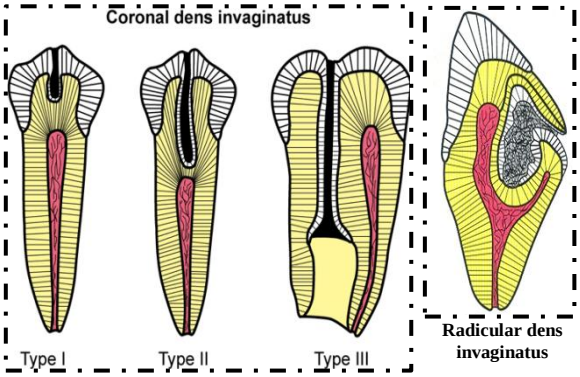


- It is a deep surface invagination of the crown / root that is lined by enamel.
- **Etiology:** it is caused by invagination of all layer of enamel organ into dental papilla before calcification has occurred.

Forms:

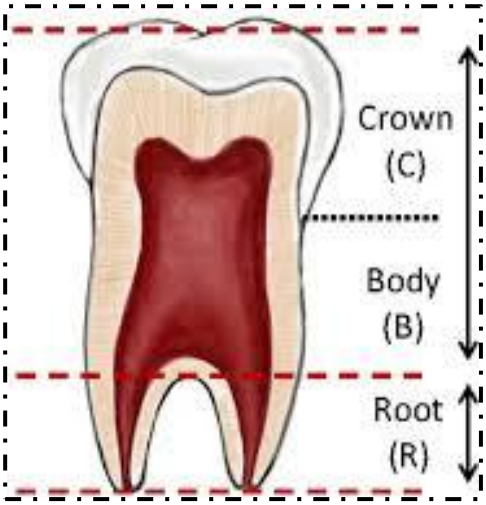


- **Mild form** ⟶ it appears as deep accentuation of the lingual pit of permanent maxillary incisors (usually upper lateral incisors bilaterally).
- **Sever form** ⟶ the invagination extends to the apex of the root causing expansion of affected root "dilated odontome".



- **Clinical significance:**

  Food debris tends to lodge in the pit causing caries and infection of pulp and periapical disease.



J. Taurodontism (bull-like tooth):

Definition:

It is peculiar dental anomaly in which the body of tooth is enlarged at expense of roots.

Etiology:

It is hereditary condition that believed to occur due to failure of Hertwig's root sheath to invaginate at proper horizontal level to form the furcation area.

Clinical features:

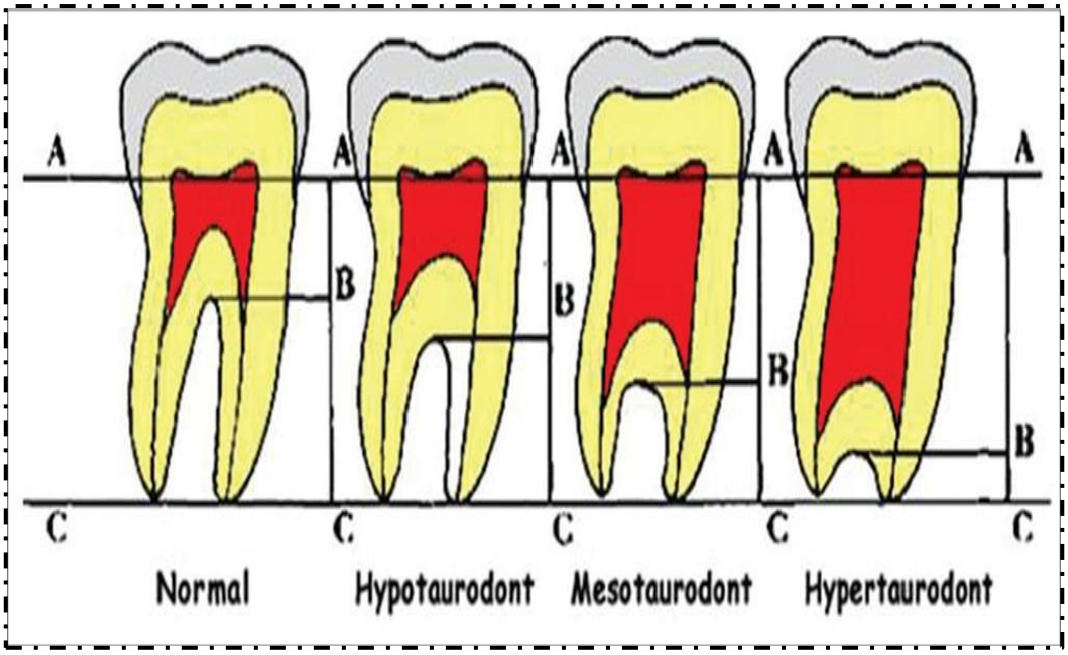
- ✓ Both permanent & primary teeth may be affected.
- ✓ It may be unilateral or bilateral affected multi-rooted teeth (molars).
- ✓ It may be seen in association Down's syndrome, Klinfelter's syndrome & amelogenesis imperfecta.

Radiographic features

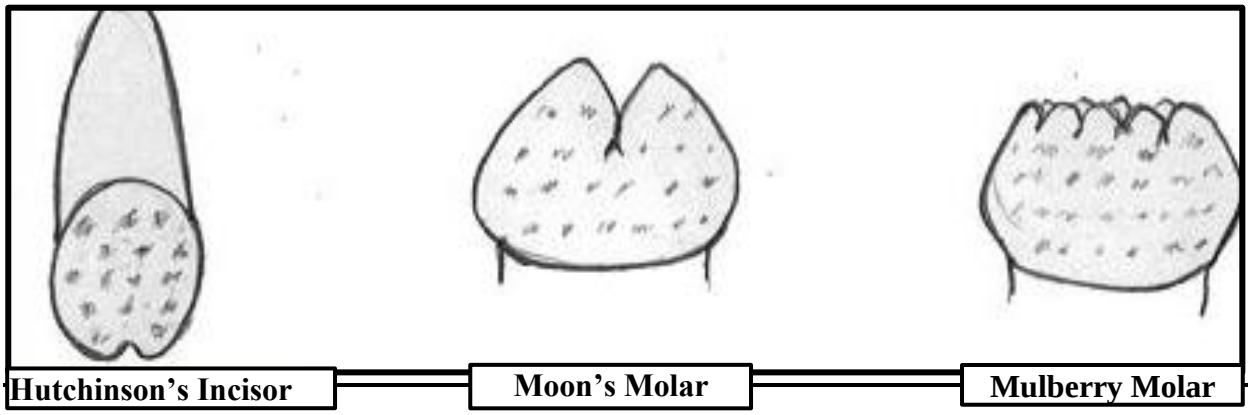
- ✓ Involved tooth tend to be rectangular in shape.
- ✓ The pulp chamber is extremely large.
- ✓ The pulp lacks the usual cervical constriction.
- ✓ Bifurcation or trifurcation are apically displaced & roots are exceedingly short.

Classified into:

1. **Hypotaurodont** ⟶ bi or trifurcation occurs near the cervical area of the roots.
2. **Mesotaurodont** ⟶ bi or trifurcation occurs at the middle area of the roots.
3. **Hypertaurodont** ⟶ bi or trifurcation occurs at the apices of the roots.



K.Dental manifestation of Congenital syphilis:



1. Hutchinson's Incisor:

- ✓ The *permanent incisors* are barrel-shaped; their mesial and distal surfaces taper toward the incisal edge rather than cervical margin giving a "screw driver" appearance.
- ✓ The incisal edges usually have a central notch.
- ✓ The mesial and distal incisal edges are rounded off.

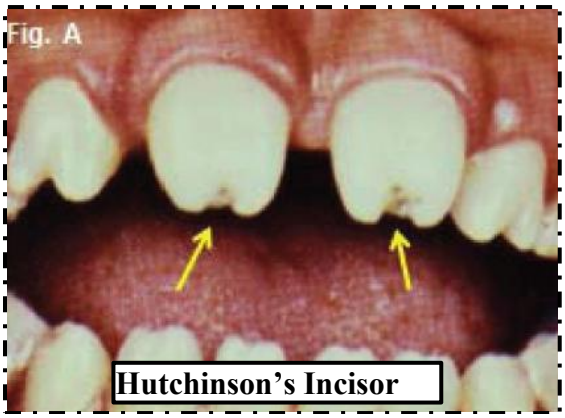
2. Moon's Molar:

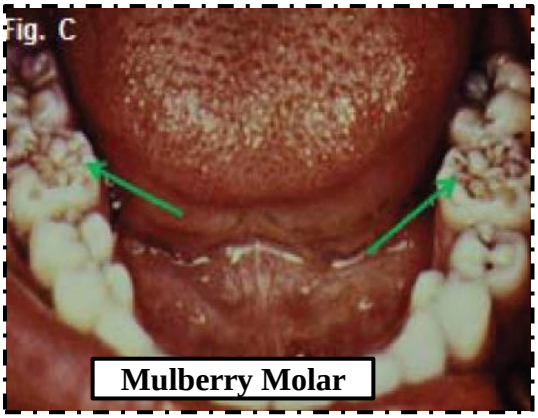
- ✓ The *1st permanent molars* show a constricted occlusal surface and rounded angles.
- ✓ They are dome-shaped.

3. Mulberry Molar:

- ✓ The occlusal surfaces of *1st permanent molars* may be rough, pitted and exhibit multiple irregular tubercles replacing their normal cuspal pattern ⟶ giving the appearance of a mulberry.

4. **Enamel hypoplasia.**





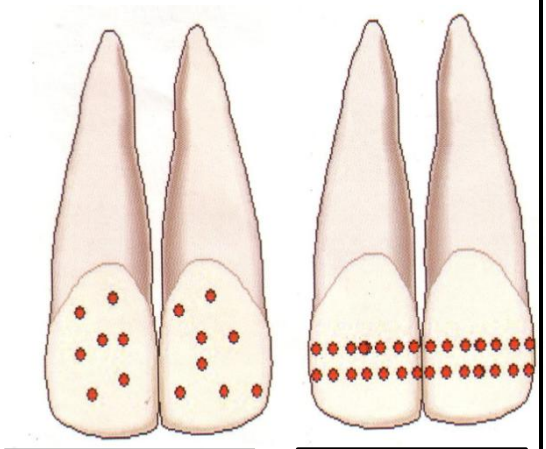
III. Disturbances in structure of teeth:

1) Disturbances in structure of enamel:

Types:

1) **Hereditary defects**.
2) **Acquired or environmental defects**.

Hereditary defects	Acquired defects
Affects both deciduous and permanent dentitions	Affects one dentition Usually permanent dentition
Usually affect all teeth	Usually affect single or group of teeth
Affect enamel or dentine	Affect both enamel and dentine
Produce diffuse or vertical or haphazard pits or defect.	Produce horizontally oriented pits or defect



Hereditary defect **Acquired defect**

a) Amelogenesis imperfecta (Hereditary Enamel Dysplasia):

Definition:

It is a group of inherited disorder of enamel formation affects both deciduous and permanent dentitions with complex inheritance pattern.

Pattern of inheritance; depending on the altered gene (**AMELX, ENAM & MMP20**):

- ✓ *Autosomal Dominant*
- ✓ *Autosomal Recessive*
- ✓ *X-linked recessive*

Enamel is developed in 3 stages:

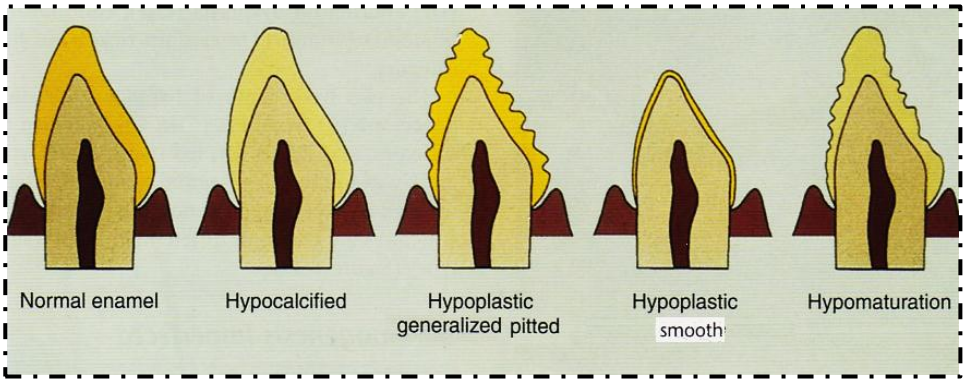
1) **Formative Stage**; it is a stage of organic matrix deposition by ameloblast giving normal thickness of the enamel.
2) **Calcification Stage**; mineralization of organic matrix is done by formation of crystals and removal of the majority of original proteins.

- In the early stages of calcification, the enamel is dull, white and soft (opaque enamel).
- In the late stages of calcification, the enamel becomes hard and translucent.

3) **Maturation Stage**; during this stage, the crystals enlarge and mature. The remnants of original proteins are removed.
   - ✓ If the disturbance occurs in the 1st stage, resulting in deficiency in the normal volume of the matrix. This is called **enamel hypoplasia.**
   - ✓ If the disturbance occurs in the 2nd stage, resulting in deficiency in the normal volume of the minerals. This is called **hypocalcified enamel**.
   - ✓ If the disturbance occurs in the 3rd stage, resulting in enamel crystallites remain immature. This is called **hypomaturation enamel.**

3 Basic Types:

a) Hypoplastic type.
b) Hypocalcified type.
c) Hypomaturation type.



Hypoplastic amelogenesis imperfecta:

- Defective formation of organic matrix, but calcification and maturation are normal ⟶ enamel will be of thickness less than normal, but normal hardness and translucency (*quantitative defect*).
- Teeth tend to be small in size with loss of contact points.
- There are two forms:
    1) Generalized roughness and pitting form produced by areas of hypoplasia distributed over the enamel surface.
    2) Smooth form; whole enamel affected.


Hypoplastic amelogenesis imperfecta

Hypocalcified amelogenesis imperfecta:

- Enamel matrix is formed in a normal quantity, but the formative enamel fails to calcify properly ⟶ enamel will be of normal thickness only on eruption, but it is very soft and easily lost.
- During eruption, enamel appears normal in size and shape, but it soft, weak and white opaque to yellowish-brown colour. Later, the enamel is rapidly lost by abrasion and attrition exposing the dentine.
- Gross attrition is feature of hypocalcified type and the teeth may be worn down to the gum level.
- Associated occasionally with anterior open bite.



Hypocalcified Amelogenesis Imperfecta

Hypomaturation amelogenesis imperfecta:

- Enamel matrix is laid down and begins to calcify. The defect in maturation of enamel crystal structure ⟶ enamel will be soft pigmented and **chips easily**.
- Affected tooth are normal in shape but exhibit a mottled, opaque white brown yellow discoloration.
- Enamel is soft, open bite and attrition occurs but not as severely as in of hypocalcified type.



Hypomature Amelogenesis Imperfecta

b) Acquired or Environmental Enamel Hypoplasia:

1) ***Local (focal) factors:***
    a. Infection.
    b. Trauma.
    c. Irradiation.

2) . Systemic factors:
    a. Nutritional deficiency (Vitamin A, C & D).
    b. Hypocalcemia.
    c. Exanthematous disease (measles, chicken pox).
    d. Congenital syphilis.
    e. Ingestion of chemicals (fluoride, tetracycline & cytotoxic drugs).
    f. Birth injury, premature birth and RH haemolytic disease.
    g. Chronic diseases such as congenital heart disease and endocrine diseases.

Turner's tooth:

Acquired localized or focal enamel hypoplasia involving only one or two teeth is known as Turner's tooth.

Pathogenesis: it results from trauma or infection during tooth formation:

- Traumatic injury or extraction of deciduous tooth causes hypoplasia of permanent tooth.
- Abscess of deciduous tooth causes hypoplasia of permanent tooth.



A single spot lesion on a permanent central incisor, corresponding to an injury of the primary tooth which was its predecessor.

Site: most common site of permanent maxillary incisors or permanent premolars.

Clinically:

- It appears as yellowish or brownish discoloration with pitting or irregularity of the enamel surface.
- Sometimes the tooth appears smaller than normal.

Dental Fluorosis (Mottled Enamel):

Fluoride intake in drinking water *above the normal limit* that recommended by World Health Organization (WHO) of 0.6-0.7 ppm in hot climate countries and 1.2 ppm for cold climate countries, *at the time of teeth formation* may produces mottling of teeth (dental fluorosis).

Pathogenesis: increased fluoride levels interfere with ameloblastic function, which adversely affect both enamel matrix and calcification.

Clinically: mottling ranges from white or chalky patches to brown pitted enamel.

2) Disturbances in structure of dentine:

a) Dentinogenesis Imperfecta:

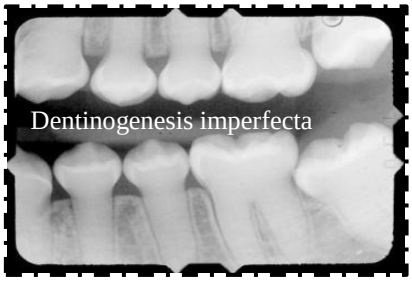
- ✓ It is an inherited disorder of dentine formation, transmitted as autosomal dominant.
- ✓ **Types:**
  1) Type I (associated with Osteogenesis Imperfecta).
  2) Type II (not associated with Osteogenesis Imperfecta).
  3) Type III (Brandywine Type; Shell tooth).
- ✓ **Clinical Features:**



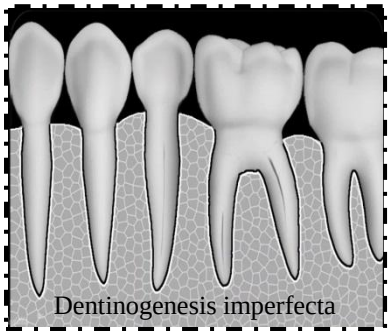
  - Deciduous teeth are more severely affected than permanent teeth in type I.
  - In type II & III both deciduous and permanent dentitions are affected.
  - On eruption the teeth have ***opalescent amber-like*** appearance then become translucent and finally bluish -grey to yellowish-brown colour.
  - The enamel of the affected teeth is normal but it is rapidly lost and the teeth show rapid and sever attrition.

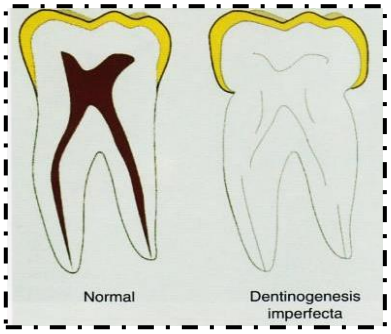






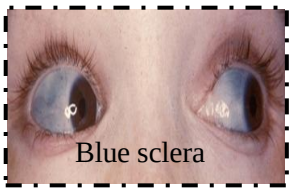
- ✓ **Radiological Features:**
  - Small, bulbous crowns with cervical constriction.
  - Thin, blunt roots.
  - Partial or even complete obliteration of the pulp chamber & root canal.





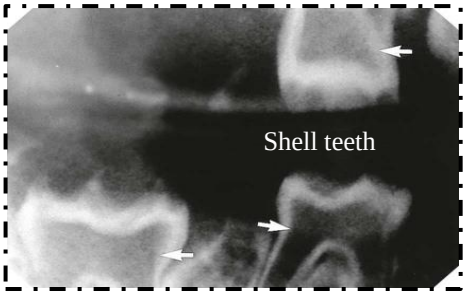
1) Dentinogenesis imperfecta type I (associated with Osteogenesis Imperfecta):

- **Etiology**: mutation in genes that code for type I collagen.
- Osteogenesis imperfecta is characterized by generalized osteoporosis of bone with marked tendency for bones to fracture, hyper-extensible joints, blue sclera and deafness.



2) Dentinogenesis imperfecta type II (not associated with Osteogenesis Imperfecta) (Hereditary Opalescent Dentine):

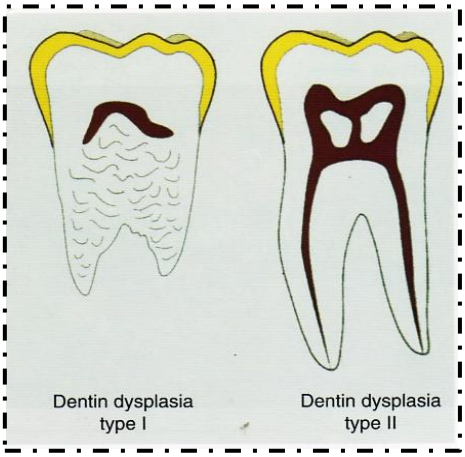
- **Etiology**: mutation in DSPP gene, encoding Dentine Sialoprotein & Dentine Phosphoprotein.
- Clinical and radiological features are similar to type one, except they are some that severe form and is not associated with osteogenesis imperfecta.



3) <u>Dentinogenesis imperfecta type III (Brandywine Type; Shell teeth</u>):

- ✓ Discovered in Brandywine isolate in state of Maryland, USA.
- ✓ This is very rare abnormality, similar to dentinogenesis imperfecta type I & II except that the pulp chamber is not obliterated, the dentine being thin and forming a shell around the wide pulp ⟶ "Shell tooth" appearance on radiographs.
- ✓ The earliest formed dentine layer (pre-dentine) is formed normally but further dentine formation ceases leaving a huge pulp chamber and root canals with multiple pulp exposure.

b) Dentinal dysplasia:



- ✓ Rare hereditary disorder of dentine formation transmitted as autosomal dominant, characterized by normal enamel but atypical dentine formation with abnormal pulpal morphology.

<u>Types:</u>

1) Type I (radicular dentine dysplasia, rootless teeth).
2) Type II (coronal dentine dysplasia).

1) <u>Type I (radicular dentine dysplasia; rootless teeth)</u>:

<u>Clinical Features</u>:



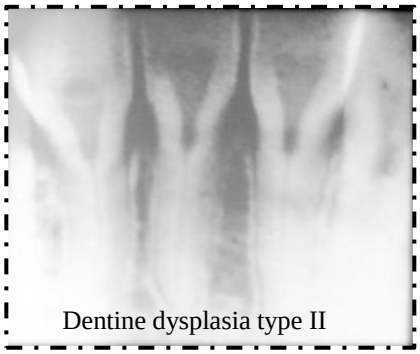
- ✓ Both deciduous and permanent dentitions are affected.
- ✓ The crowns are normal in colour and shape.
- ✓ The roots are very short and conical.
- ✓ Extreme mobility or premature tooth loss may occur because of short roots and periapical inflammatory lesions.

<u>Radiological Features</u>:

- ✓ The affected teeth have short, conical roots with obliteration of pulp chambers and root canals of deciduous and permanent teeth with periapical radiolucencies.

2) <u>Type II (coronal dentine dysplasia)</u>:

Significant difference in colour in both dentitions.



- ✓ Primary teeth have similar colour of dentinogenesis imperfecta type II and obliterated pulp chamber
- ✓ Permanent teeth have normal colour, **thistle-tube or flame-shaped** pulp chamber and multiple pulp stones, but normal root length.

3) Disturbances in structure of enamel & dentine:

- ***Regional Odontodysplasia* (Ghost Teeth):**

It is a rare developmental (non-hereditary) disorder of unknown etiology associated with abnormalities in both enamel, dentine, in which one or several teeth in a localized area are affected and usually unilateral.

Clinical Features:

- ✓ It can affect either deciduous or permanent dentitions (anterior part of the maxilla).
- ✓ Central incisors, lateral incisors and canines are most affected.
- ✓ One or two quadrants of same arch may be affected.
- ✓ The affected teeth are irregular in shape and all elements of tooth are hypoplastic and hypocalcified.
- ✓ Affected teeth exhibit delay or total failure in eruption.
- ✓ Teeth are susceptible to caries and fracture.



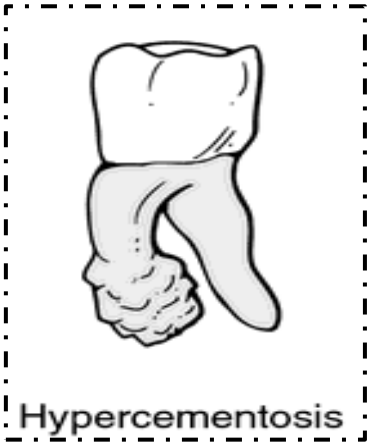
Radiological Features:

- ✓ Show marked reduction in radio-density with loss of distinction between enamel and dentine therefore it described as ghost tooth.
- ✓ The pulp chamber is extremely large.

4) Disturbances in structure of cementum:

a) Hypercementosis: ⟶ it increase in the thickness of cementum.

- ✓ Hypercementosis may affect one or several teeth and may be associated with root ankylosis.
- ✓ Hypercementosis may occur idiopathic or may occur with certain conditions:
  - Periapical inflammation.
  - Mechanical stimulation.
  - Functionless and unerupted teeth.
  - Paget's disease of bone.

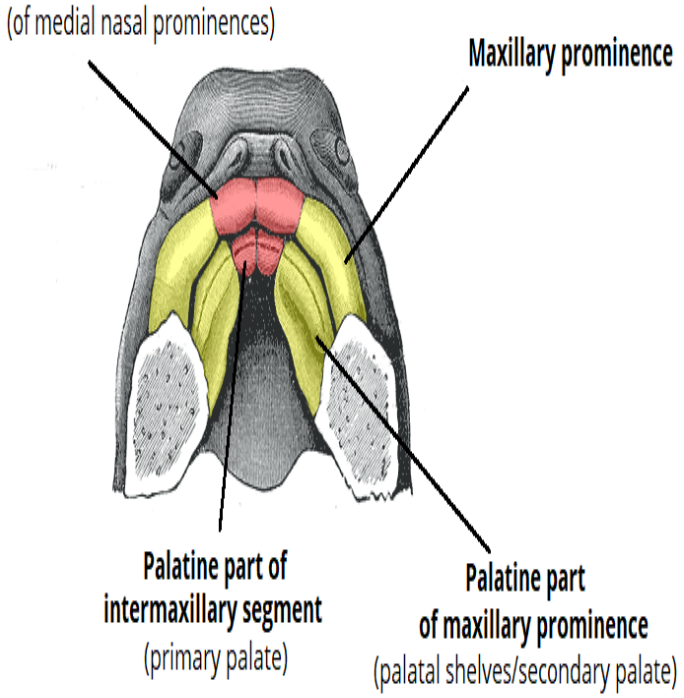


b) Hypocementosis:

Aplasia and hypoplasia of cementum may be seen in association with cleidocranial dysplasia, hypophosphatasia and some skeletal abnormalities.

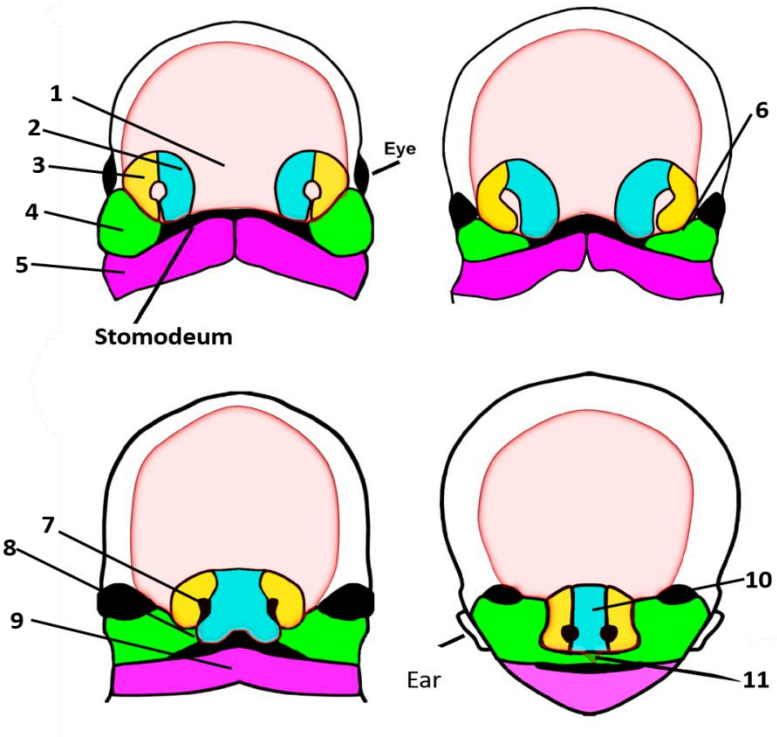
☺☺Thank you☺☺

Developmental Disturbances of Oral & Para-oral Structures



University of Tripoli
Faculty of Dentistry
ORAL PATHOLOGY
3rd year dental students (26-10-2026)

Dr. Asmaa Salem Shanab



Development of Face

1. Frontonasal process
2. Medial nasal process
3. Lateral nasal process
4. Maxillary process
5. Mandibular process
6. Fusion of lateral process with maxillary process
7. Nasal pit
8. Fusion of maxillary process with medial nasal process
9. Fusion of mandibular processes
10. Intermaxillary process (fused medial nasal processes
11. Philtrum of nose

(2026)

Developmental Disturbances of Oral & Para-oral Structures:

I. Developmental Disturbances of the Tongue:

- *Aglossia.*
- *Microglossia.*
- *Macroglossia.*
- *Ankyloglossia.*
- *Cleft (Bifid) tongue.*
- *Fissured tongue.*
- *Median Rhomboid glossitis.*
- *Lingual Varices.*
- *Geographic Tongue.*
- *Hairy Tongue.*
- *Lingual Thyroid nodule.*

II. Developmental Disorders of Lips:

- *Glandular Cheilitis.*
- *Granulomatous Cheilitis.*
- *Congenital Lip Pits & Fistula.*
- *Double Lip.*
- *Cleft lip.*

III. Developmental Disturbances of the Palate:

- *Torus palatinus.*
- *Cleft palate.*

IV. Developmental Disturbances of the Face

- *Oblique facial cleft.*
- *Transverse facial cleft.*
- *Macro & Micro-Stomia.*
- *Facial Hemihypertrophy.*
- *Facial Hemiatrophy.*

V. Developmental Disturbances of Salivary Glands:

- *Aplasia.*
- *Atresia.*
- *Aberrancy.*

VI. Developmental Disorders of the Jaws:

- *Agnathia.*
- *Micrognathia.*
- *Macrognathia.*
- *Exostoses and Tori.*

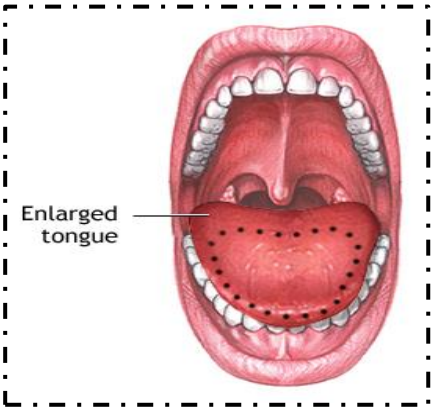
VII. Developmental Disturbances of Oral Mucosa:

- *Fordyce's granules.*
- *Leukoedema.*

.

I. Developmental Disturbances of the Tongue:

1) **Aglossia:** ⟶ it means complete absent of tongue at birth.
2) **Microglossia:** ⟶ it means abnormal small or rudimentary tongue.
   - ⦿ It usually associated with syndromes like *Oromandibular-limb hypogenesis syndrome & Pierre Robin syndrome.*

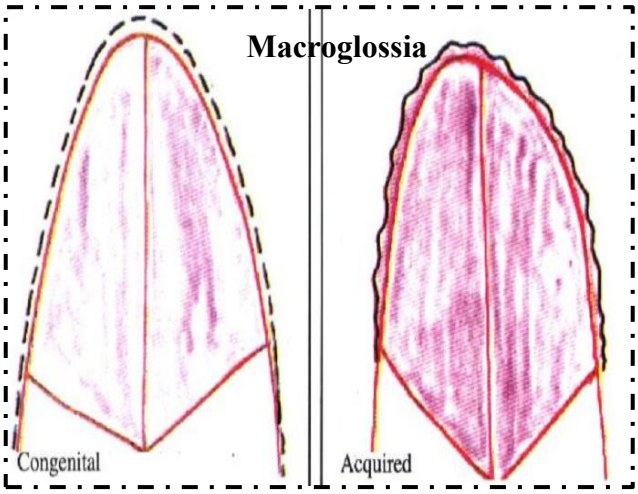


3) **Macroglossia:** ⟶ it means abnormal large tongue.

Etiology:

A. Congenital macroglossia:
- Haemangioma & Lymphangioma.
- Cretinism (congenital hypothyroidism).
- Down's syndrome.
- Neurofibromatosis.
- Hemi-hyperplasia.

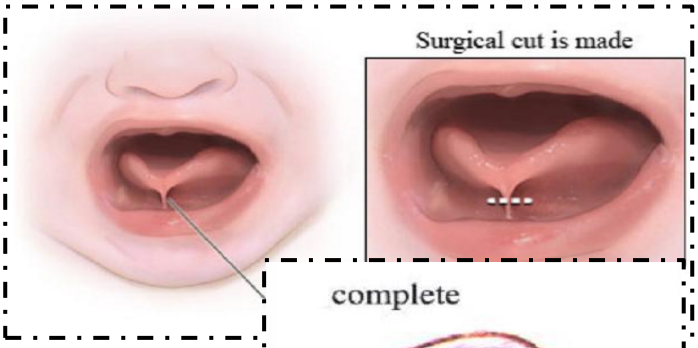
B. Acquired Macroglossia:
- Amyloidosis.
- Myxoedema (adult hypothyroidism).
- Acromegaly (adult hyperpituitarism).
- Tumours as papilloma, fibroma, carcinoma, or sarcoma.
- Chronic granulomas as actinomycosis, syphilis, or T.B.
- Edentulous state.



Clinical significance:
- Protruding & scalloped borders of the tongue, because they are compressed against the teeth & interdental spaces.
- Malocclusion and food impaction leading to caries and periodental diseases.
- Other symptoms include; noisy breathing, drooling of saliva, difficulty in eating, speech and airways problems.

Differentiating Features:
- ⦿ **Cretinism** : ⟶ smooth, diffuse generalized enlargement.
- ⦿ **Lymphangioma:** ⟶ pebbly appearance with multiple vesicle like blebs.
- ⦿ **Amyloidosis & Neurofibromatosis**: ⟶ multinodular appearance.
- ⦿ **Down's Syndrome:** ⟶ papillary fissured surface.
- ⦿ **Hemifacial Hyperplasia & Neurofibromatosis:** ⟶ unilateral enlargement.
- ⦿ **Edentulous:** ⟶ lateral spreading of tongue.

4) Ankyloglossia (Tongue Tie):

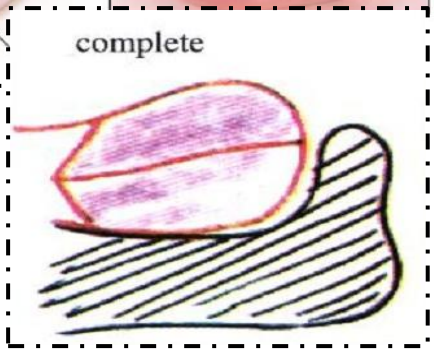


Definition:

It is a developmental malformation, resulting in the tongue movements are restrained.

Types:

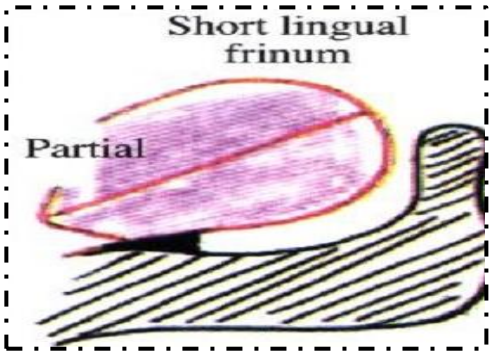
A. Complete ankyloglossia: occurs as a result of fusion between the tongue and the floor of the mouth.

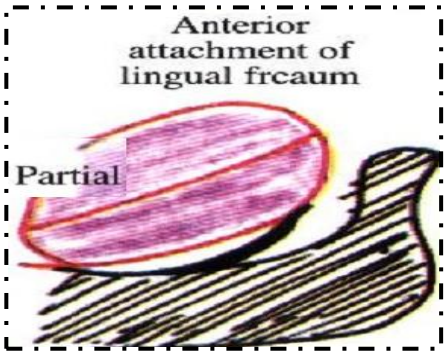


B. Partial ankyloglossia (commonest type):

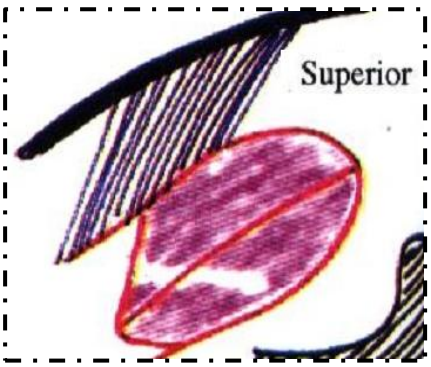
a. Abnormal short lingual frenum , or

b. The lingual frenum attached to the tip of the tongue.





C. Superior ankyloglossia: it is associated with cleft palate. It is due to the dorsal surface of the tongue is connected by a membrane to one side of the palate.



D. Acquired ankyloglossia: it may be due to infection, malignant neoplasm, or irradiation.

Clinical significance:

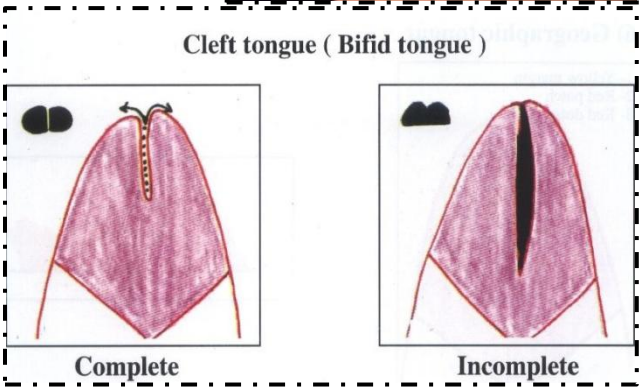
- Males > females.
- Limited mobility of the tongue.
- May cause difficulty in swallowing & speech.
- Recurrent tongue biting, poor sucking and inability to chew some food.
- Notched or "heart-shaped" tongue when it is protruded.
- Surgery (frenectomy) is recommended if there are functional problems.



5) Cleft Tongue (Bifid Tongue):

Definition:

It is a developmental congenital malformation in which a complete or partial cleft or fissure is present on the tongue from the tip extending backward.



Types:

A. Complete cleft tongue:

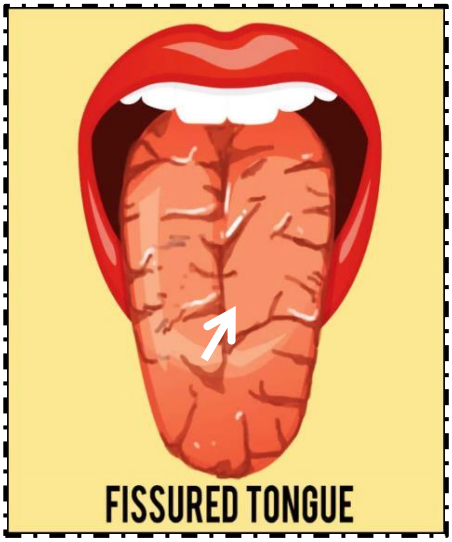
- A rare condition.
- It is due to lack of fusion between the two halves of the tongue.

B. **Partial cleft tongue**:

- Is more common condition.
- It appears as a deep groove in the midline of the dorsal surface of the tongue . It is due to incomplete fusion between the two halves of the tongue.

Clinical significance:

- It is insignificant except that food debris and micro-organism may collect in the base of the cleft causing irritation ⟶ regular cleaning of the tongue should be carried out.
- It may be associated with **Oro-facial-Digital Syndrome.**



6) Fissured Tongue (Scrotal Tongue):

Definition:

It is a malformation manifested clinically by multiple grooves and furrows on the dorsal surface of the tongue.

Etiology:

A. **Congenital:** hereditary plays an important role.

B. **Acquired:** due to chronic trauma, vitamin deficiency, tobacco smoking, psychological stress & infection as syphilis.

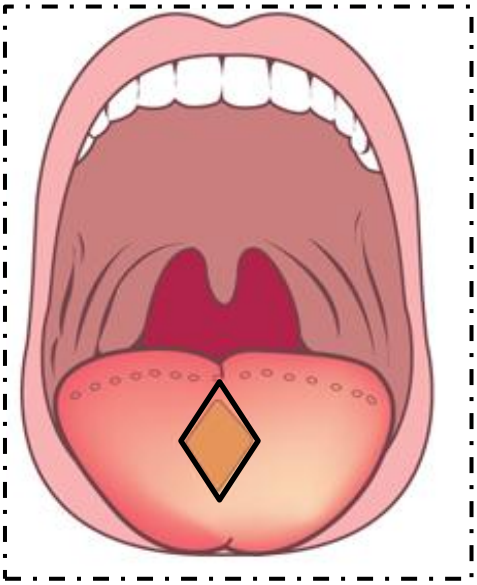
Clinical significance:

- The incidence, number, width and depth of fissures increase with age.
- It is usually painless except when food debris tends to collect in the groove and produce irritation ⟶ patients may complain of mild burning soreness.
- Strong association noted between geographic tongue (same gene).
- May be asssociated with syndrome such as **Melkersson Rosenthal syndrome**.

Treatment & Prognosis:

- A benign condition, no specific treatment is required.
- Patient should be encouraged to brush his tongue as food entrapped in the deep fissures of tongue may act as a source of irritation.

7) Median Rhomboid Glossitis:



Median rhomboid glositis

Definition:

It is benign lesion of the tongue, characterized by rhomboid or oval in shaped changes occur in the tongue mucosa in the midline, just anterior to the foramen caecum.

Etiology:

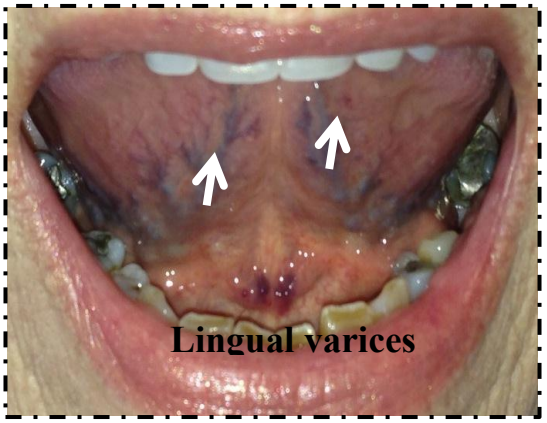
- ✓ Classically was thought to be a congenital abnormality due to failure of the tuberculum impar to retract or withdraw before fusion of the lateral halves of the tongue.
- ✓ Recently, it was suggested that it is due to a localized chronic fungal infection specifically Candida albicans.

Clinical Features:

- **Age:** common in adult.
- **Sex:** males > females.
- **Shape:**
  - ✓ It appears as an ovoid ,diamond, rhomboid shaped reddish patch or plaque on the dorsal surface of the tongue, immediately anterior to circumvallate papillae.
  - ✓ The surface is red and completely devoid of filiform papillae and usually smooth.
  - ✓ It could be flat or slightly raised on the dorsal surface of the tongue.
- **Symptoms:** it is generally asymptomatic, but pain and ulceration are reported.

8) Lingual Varices:

- ✓ Abnormally dilated and tortuous veins.
- ✓ Seen in 2/3^{rds} of people older than 60 years of age (indicating age related degeneration).
- ✓ Asymptomatic condition.

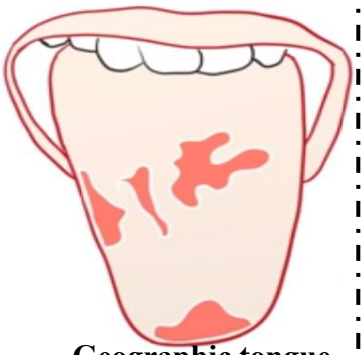


Lingual varices

9) Geographic Tongue (Erythema Migrans, Benign Migratory Glossitis):

Definition:

It is an idiopathic benign lesion characterized by formation of red depapillated areas on tongue in ‘’map” like configuration.



Geographic tongue

Etiology:

- ✓ The exact etiology is unknown.
- ✓ May be related to immunological reaction, allergic, emotional stress, hereditary (familial) factors, infection and nutritional deficiencies.
- ✓ In patient with geographic tongue there is a high frequency of history of asthma & eczema.



Clinical Features:

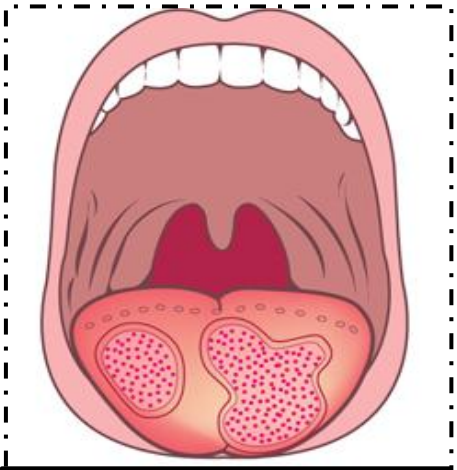
- **Age:** debilitated children and young adult.
- **Sex:** female predilection.
- **Site:**
  - Lesions confine to the dorsal surface & lateral border of the tongue.
  - Rarely occur on the ventral surface of the tongue.
  - May also affect extra-lingul sites (buccal and labial mucosa & soft palate) which called ⟶ **Migratory stomatitis.**
  - Lesions are not constant in size, shape and location.
- **Shape:**

✓ Multiple, irregularly shaped red patches represents:

- Loss filiform papillae ,(fungiform papillae persist as small elevated red dots).
- Loss of keratin.
- Thinning of epithelium.

✓ Surrounded by elevated whitch or yellowish margins represents:

- Regenerating filiform papillae.
- Hyperkeratosis.
- Hyperplasia of epithelium.



Geographic tongue

- **Symptoms**

✓ Asymptomatic but patient may complain of burning sensation that may worsen with hot, spicy or acidic foods.

✓ The depapillated areas remain for short time in one location and then heals and appears in another location ⟶ *term migration*.

✓ The condition may persist for weeks or months then regress spontaneously, only to recur in another location at a later date.

✓ People with geographic tongue often have fissured tongue.

Treatment: no treatment required. Reassure the patient.

10) Hairy Tongue: Lingua Villosa, Black Hairy Tongue:

Definition:

- ✓ An overgrowth of filiform papillae on the dorsum of tongue, giving the tongue a hairy appearance.
- ✓ Represents probably an increase in keratin production (actual hypertrophy of the filiform papillae) with lack of normal keratin desquamation.

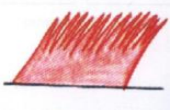


Hairy tongue

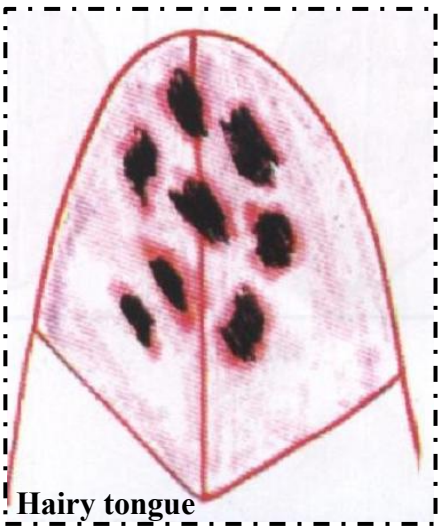
Etiology:

- Unknown.
- Micro-organisms (fungal and bacterial overgrowth).
- Systemic disturbances (anaemia & gastric upset).
- Heavy smoking and extensive x-radiation.
- Use of certain drugs like penicillin and sodium peroxide mouth wash that stimulate growth of filiform papillae.
- Change in the local oral enviroments (whether in saliva or in microbial flora).

Clinical Features:



- The papillae are elongated and have appearance of hair that may cover the whole dorsum of the tongue or only a part of it.
- The elongated papillae acquired discoloration ranging from yellowish, to brown to black. The discoloration is depending on extrinsic factors such as tobacoo, certain food, medicines or chromogenic organism.
- Asymptomatic or may cause tickling of soft palate lead to gagging, or bad taste.



Hairy tongue

Treatment & Prognosis:

- Benign condition with no serious sequelae.
- Elimination of predisposing factors.
- Maintenance of oral hygiene (brushing of tongue)
- Topical application keratolytic agent.

N.B. Because of similarity in names, care must be taken to differentiate this condition from **hairy leukoplakia (HL)**. HL occurs on *lateral borders of tongue*, is caused by *Epstein-Barr virus* and is usually associated with **HIV** infection or other immunosuppressive conditions.

11) Lingual Thyroid Nodule:

Definition:

- ✓ It is developmental anomaly in which follicles of thyroid tissue are found in the substance of the tongue anterior to foramen caecum.
- ✓ It is considered as an ectopic condition.

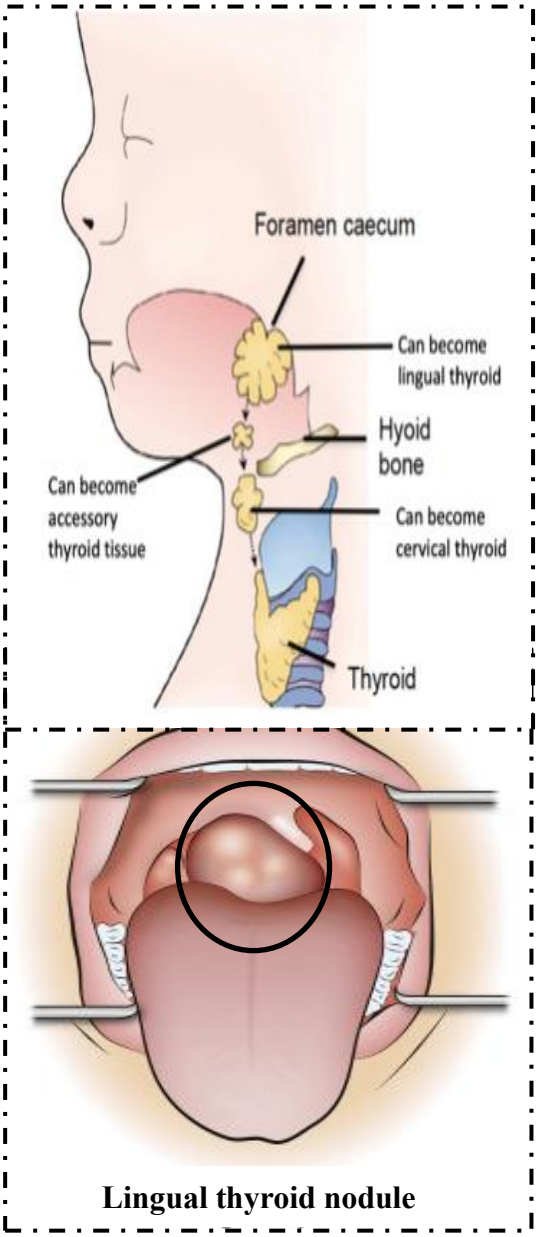
Pathogenesis:

- ✓ Failure of thyroid analoge to migrate to its predestined position.
- ✓ Differentiation of the epithelial tract into thyroid follicle takes place while it still lies in the substance of the tongue.

Clinical Features:

- ➢ **Age:** from birth to 60 years old.
- ➢ **Sex:** female > male.
- ➢ **Sign and symptom**:

- It presents as a nodular mass in or near the base of the tongue, usually in midline with smooth surface of vascular or normal colour.
- It may range from small, asymptomatic lesions to large masses that block airway. Common symptoms are dyspnea, dysphagia, dysphonia, heamorrhage with pain or feeling of fullness in the throat.



Lingual thyroid nodule

Diagnosis:

- ▪ Biopsy is often avoided as the mass may be highly vascular and may represent the patient's only functioning thyroid tissue.
- ▪ Diagnosis is established by thyroid scan using iodine isotopes or technetium 99m.
- ▪ CT & MRI: size and extent of lesion.

Complications:

- ✓ Enlargement in cases of thyroid insufficiency (goiter).
- ✓ Adenoma or adenocarcinoma of thyroid type may develop in it.

Treatment:

- ✓ **Asymptomatic patients:** ⟶ no treatment, periodic follow up.
- ✓ **Symptomatic patients**:⟶ suppressive therapy with thyroid supplements.
- ✓ **In case of airway obstruction**; surgery with auto-transplantation to another body site.
- ✓ In males **over 30 years**; prophylactic surgical excision as there is a risk of malignancy.

II. Developmental Disturbances of the Lips:

1) Cheilitis Glandularis (Glandular Cheilitis):

Definition: it is uncommon condition in which the lower lip is enlarged.

Etiology: unknown but may be due to:

- ✓ Chronic exposure to sun, wind and dust.
- ✓ Use of tobacco.
- ✓ Hereditary factor.
- ✓ Emotional disturbance or allergic reaction.


Cheilitis glandularis

Clinical features:

Age: young adult.
Sex: male predilection.
Site: lower lip.

Appearance:

- Lower lip ⟶ enlarged (painless), firm and everted exposing the openings of the labial salivary glands.
- The labial mucous glands ⟶ enlarged and nodular.
- The orifices of salivary ducts ⟶ inflamed and dilated giving the labial mucosa a red macular appearance.
- Viscid mucous secretion.
- *Malignant transformation* has been reported.

2) Cheilitis Granulomatosa (Granulomatous Cheilitis; Miescher syndrome):

Definition: it is a chronic swelling of the lips due to granulomatous inflammation.

Etiology: unknown.

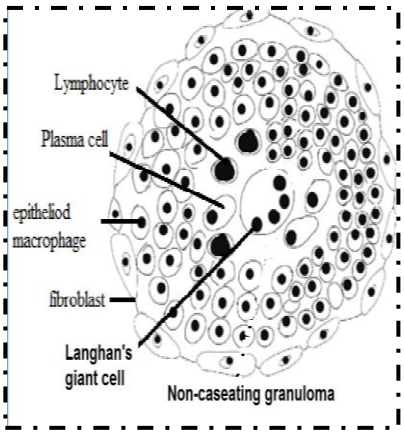

Cheilitis granulomatosa

Clinical features:

Age: children & adult.
Sex: female > male.
Site: affecting usually one lip but sometimes both.
Usually the upper lip is affected.

Appearance:

- Diffuse, painless swelling one or more lips.
- The swelling is usually soft & no pitting on pressure.
- Surrounding skin normal in colour or erythematous.
- May be associated with **Melkersson Rosenthal syndrome.**

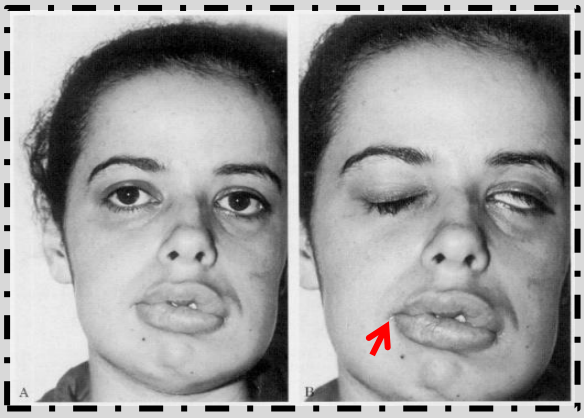


Histopathological features:

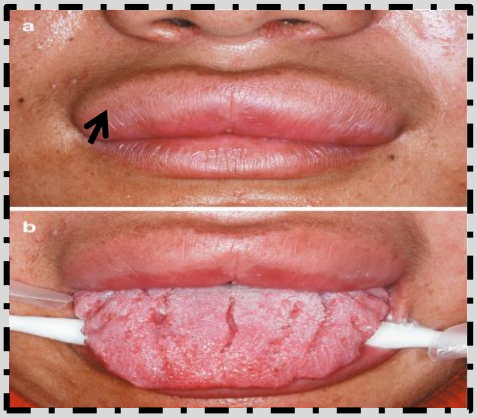
- It consists of chronic inflammatory cell infiltration particularly peri and paravasular aggregation of lymphocytes, plasma and histocyte.
- There is focal non-caseating granuloma formation with epithelioid cells and langhan's giant cells.

Melkersson Rosenthal syndrome: it is a triad of:

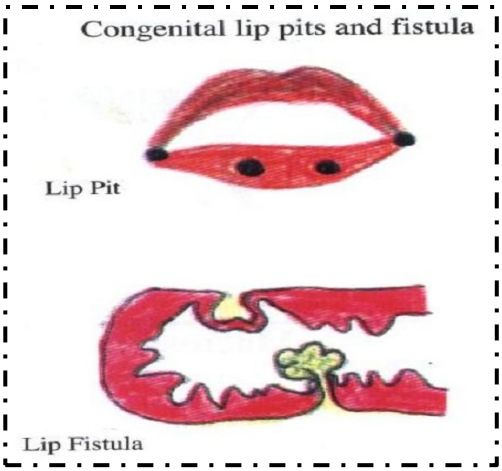
1. *Cheilitis granulomatosa.*
2. *Recurrent facial paralysis.*
3. *Fissured tongue.*







3) Congenital Lip Pits & Fistula:



Hereditary Malformations of the lips, may occur alone or in association with oral clefts.

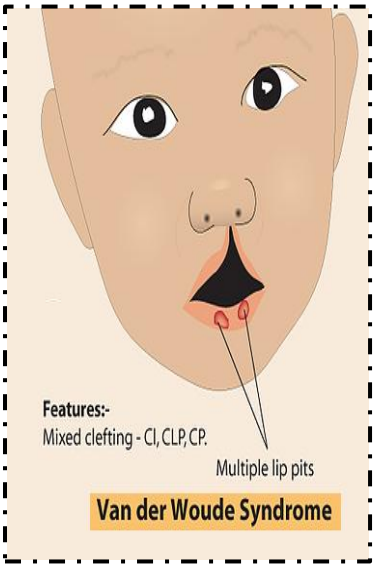
- **Lip pits** are congenital blind, epithelial-lined depression.
- **Lip fistulas** are congenital epithelial-lined depressions, connected with minor salivary glands & shows secretions.

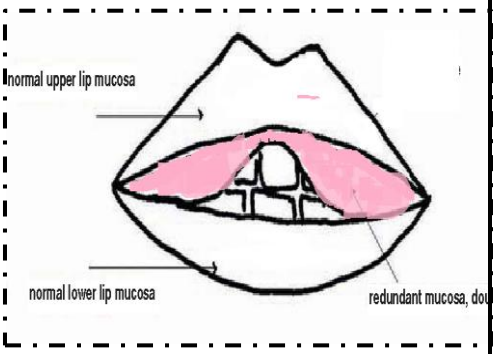
Etiology: (often follow a hereditary pattern).

- Notching of the lip at early stage of it is development, with fixation of the tissues at the base of the notch.
- Or from failure of complete union of the embryonic lateral sulci of the lip.

Clinical features:

- Common sites : lower lip> upper lip.
- ***Unilateral or bilateral*** depression on vermillion border of lip.
  - Near the mid-line ⟶ **paramedian lip pits.**
  - At the corner of the mouth⟶**commissural lip pit.**
  - If express fluid ⟶ **lip fistula**.
- Lip may appear swollen.
- It may be associated with **Van der Woude syndrome** (*paramedian lip pits with cleft lip ± cleft palate*)





4) Double Lip

- Rare, redundant fold of tissue on the mucosal side of the lip.
- May be congenital or acquired (from trauma or oral habits such as lip sucking)
- More common in upper lip, sometimes both lips affected. Seen on smiling.
- It may be associated with **Ascher's syndrome.**

Ascher's Syndrome: it is a triad of:

1. *Double lip.*
2. *Blepharochalasis (edema of upper eyelids).*
3. *Non-toxic enlargement of thyroid gland.*



III. Developmental Disturbances of the Palate:

1) Torus Palatinus:

Definition: it is a developmental frequently hereditary malformation of the hard palate.

Clinical Features: it appears as:

- Single or multiple bony swelling about the midline of the hard palate.
- Flat, fusiform , lobular , or nodular in shape.
- Covered by normal stretched (thin) mucosa.


Torus Palatinus

Histological features :

It consists of

- Compact bone only ,or
- Core of cancellous bone covered by a layer of compact bone.

2) Cleft palate:

Cleft Lip and Palate:

- Most common orofacial clefts.
- Congenital developmental cleft in which there is a division or separation of parts of the lip or palate that is formed during the early months of development of the fetus.
- 3-8% of clefts are associated with syndromes.

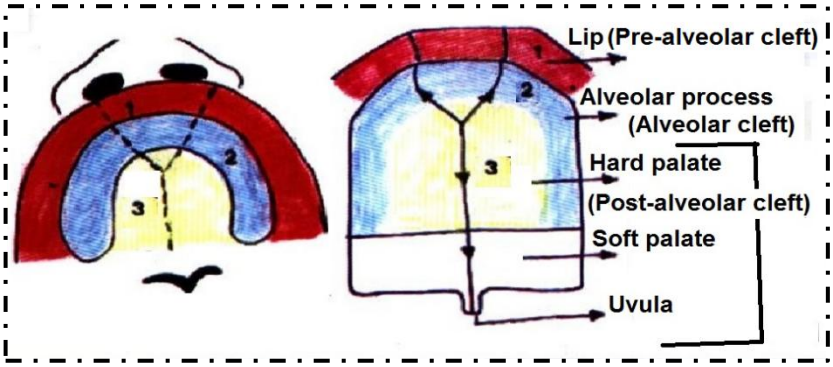


Etiology: *multifactorial, combination of*:

- **Genetic abnormalities** (inherited / spontaneous mutation).
- **Environmental factors** (nutritional disturbance, infection, drugs, radiation, emotional or traumatic stress).

Classification:

1) Pre-alveolar cleft (cleft lip).
2) Alveolar cleft.
3) Post-alveolar cleft (cleft palate).



Cleft Lip / CL:

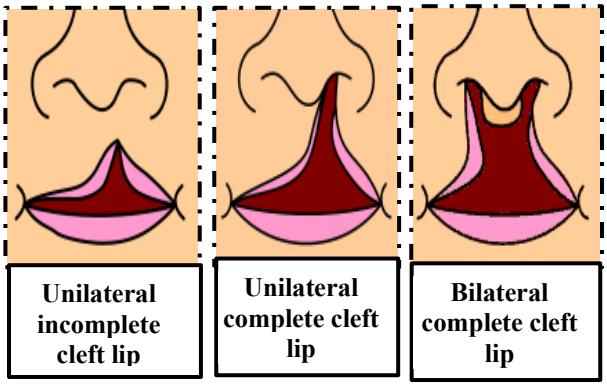
Types:

A. Maxillary:

a) Unilateral (complete / incomplete).

b) Bilateral (complete / incomplete / mixed).

- It occurs due to failure fusion of maxillary processes with medial nasal processes on one side or both sides.



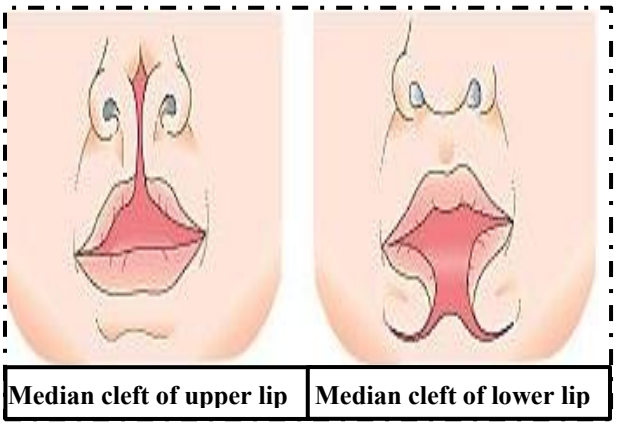
- Incomplete CL ⟶ when it does not extend up to the nostril.
- Complete CL ⟶ when the cleft extends to the floor of the nose.
- 80% of CL are unilateral & commonly on left side (70%).

c) Median cleft of upper lip:

- It is rare defect, result from incomplete fusion of two medial nasal process in the midline.

B. Mandibular:

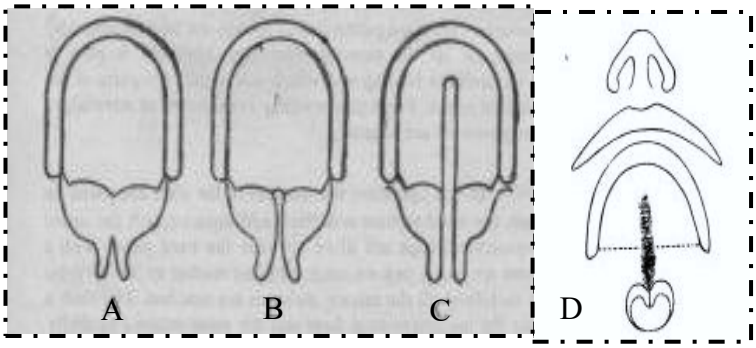
- ***Median cleft of lower lip***:
  - It is very rare defect due to failure of fusion of two mandibular process at the midline.



Cleft Palate / CP:

Definition:

- It is a congenital anomaly in which there is a fissure at roof of mouth which occurs due to failure fusion of lateral palatine processes, primary palate and the nasal septum.
- Cleft palate may be associated with syndromes such as **Down's syndromes & Pierre Robin syndrome**.



Types:

A. Cleft of the uvula.
B. Partial (soft palate only).
C. Complete (soft and hard palate).
D. Submucosal cleft palate: surface mucosa is intact but underlying musculature of soft palate is defective.

Cleft of the Lip, Alveolus & Palate / CLP:

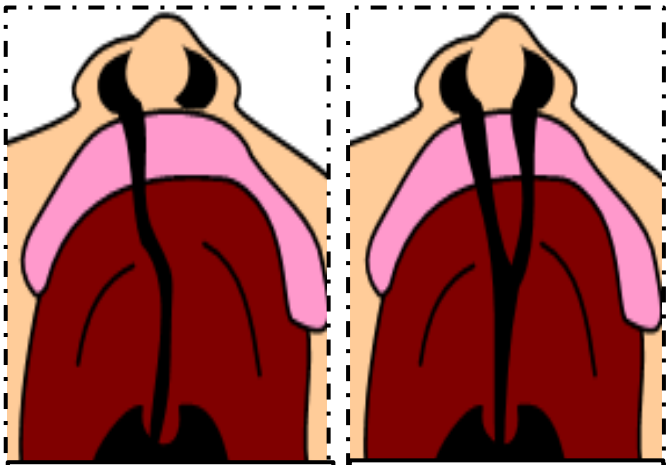
Types:

A. Unilateral Complete CLP:

Cleft extending from the uvula to the incisive foramen in the midline then deviating to one side and meeting the avleolar process in area of the lateral incisor and then into the lip.

B. Bilateral Complete CLP:

Cleft extending from the uvula to the incisive foramen in the midline then deviating forwards bilaterally to the avleolus and lip, the premaxilla remains suspended from primitive nasal septum.

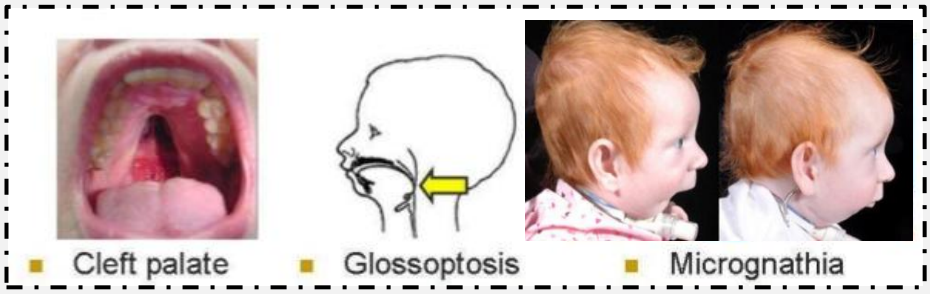


Unilateral complete cleft lip & palate | **Bilateral complete cleft lip & palate**

Pierre Robin Syndrome:

It is a triad of:

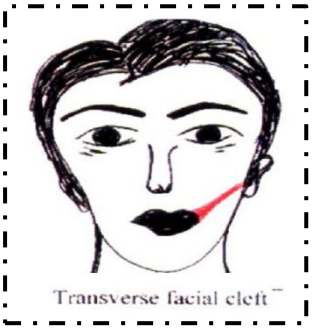
1) Mandibular micrognathia.
2) Glossoptosis.
3) Cleft Palate.



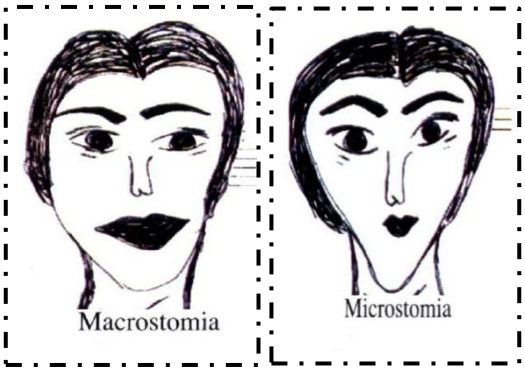
IV. Developmental Disturbances of the Face:



1) **Oblique facial cleft:**
   - A congenital developmental cleft.
   - It runs from inner canthus of the eyes to the ala of the nose or upper lip , along the path of the naso-lacrimal duct; due to failure of fusion between maxillary & lateral nasal processes.



2) **Transverse facial cleft:**
   - A congenital development cleft.
   - It runs from the angle of mouth to the ear.
   - It is due failure of fusion between the maxillary & mandibular processes.



3) **Macro & Micro-Stomia:**
   - **Macro-stomia** : means abnormal large mouth, & is due to premature arrest of fusion between the maxillary & mandibular processes.
   - **Micro-stomia** : means abnormal small mouth, & is due to excessive union between the maxillary & mandibular processes.

4) **Facial Hemihypertrophy:** → it is a congenital malformation in which one half of the face and jaws is abnormally larger than the other.

5) **Facial Hemiatrophy:** → it is progressive shrinkage and degeneration of all tissues below the skin on only one side of the face e.g. **Romberg syndrome**.

V. Developmental Disturbances of the Salivary Glands:

1) **Aplasia:** → it is a congenital absence of one or more of the major salivary glands. It results in xerostomia.
2) **Atresia:** → it is congenital absence or occlusion of one or more of the ducts of the major salivary glands. It may result in xerostomia & formation of retention cyst.
3) **Aberrancy:** → it is a congenital malformation, in which normal secreting salivary glands tissue develops in abnormal site (ectopia) . It leads to formation of retention cyst & neoplasm like central mucoepidermoid carcinoma.

VI. Developmental Disturbances of the Jaws:

1) Agnathia: ⟶ congenital absence of jaw , classified into :

A. **Complete agnathia**: absence of the maxilla or the mandible.

B. **Partial agnathia**: absence of a part of one jaw.

2) Micrognathia: ⟶ abnormal small jaw, classified into:

A. **Relative micrognathia:** the retrusion of the jaw is due to abnormal (post-normal) relationship of the jaw to the base of the skull.

B. **True micrognathia:** the jaw is smaller than normal & it is divided into*:*

a) Congenital true micrognathia:usually hereditary.

b) Acquired true micrognathia: it is due to environmental factors as infection or trauma or irradiation, which disturb the growth centers of the jaw.

3) Macrognathia: ⟶it means abnormal large jaw. It is classified into:

A. **Relative macrognathia**: the protrusion of the jaw is due to abnormal (antero – normal) relationship of the jaw to the base of the skull , & normal of the jaw but small teeth.

B. **True macrognathia**: jaw is larger than normal & it is divided into:

a) Congenital true macrognathia: it is due to hereditary factors.

b) Acquired true macrognathia: it is due to bone diseases as Paget's disease of bone (enlarged maxilla), acromegaly (enlarged mandible).

4) Exostoses: ⟶ it is term used clinically to describe a variety of bony outgrowths (non-neoplastic lesion).

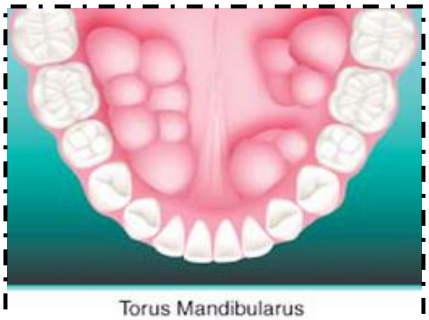
❖ **A torus is an exostosis which occurs at a characteristic site:**

⦿ The midline of the palate (**Torus Palatinus**).

⦿ The lingual surface of the mandible, usually in the premolar region above the mylohyoid line (**Torus Mandibularis**). Mandibular tori are frequently bilateral.



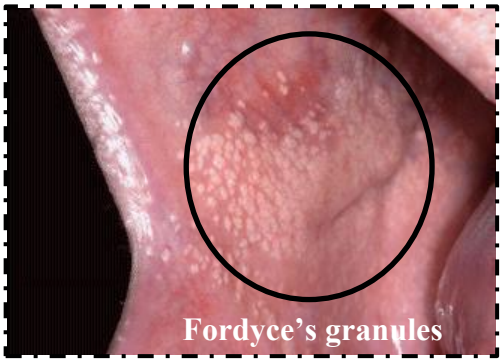
Torus Palatinus



Torus Mandibularus

VII. Developmental Disturbances of Oral Mucosa:

1) Fordyce's Granules (Spots):



Developmental anomaly characterized by ectopic collections of sebaceous glands in the oral mucosa.

Pathogenesis:

- This can be due to inclusion of ectoderm elements in the oral cavity having some of the potentialities of skin during embryonic life.

Clinical features

- It appears as small yellow spots , single or multiple separated or forming large plaque bilaterally in symmetrical pattern on the buccal mucosa opposite to molars , on the lips or on the retromolar area.



Histological features:

- These ectopic sebaceous glands are identical to those seen in the skin but are not associated with hair follicles.
- The glands open directly on to surface or by short keratinized duct.

2) Leukoedema:

It is a common oral mucosal condition that can be regarded as a **variation of normal** rather than a disease.

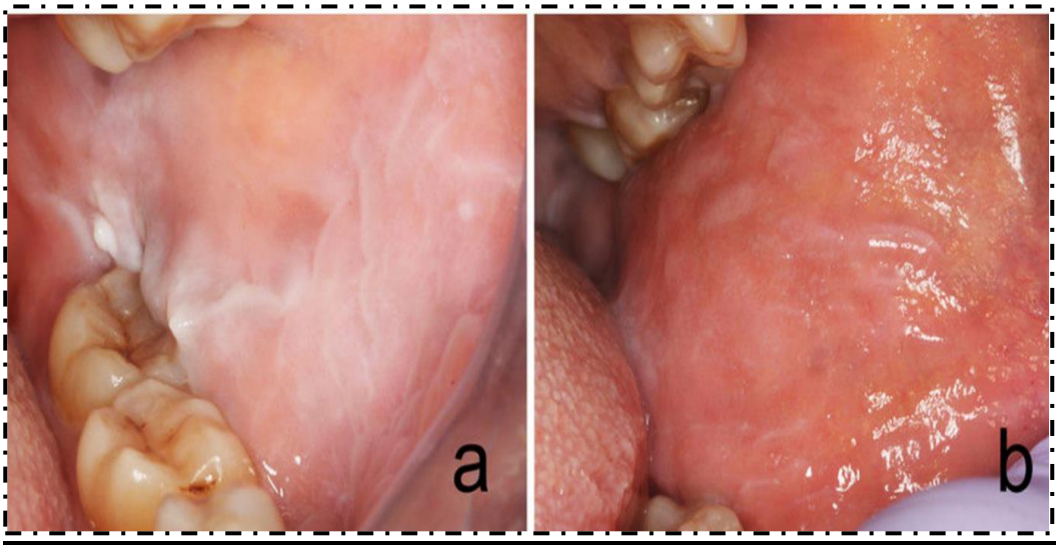
Etiology and Pathogenesis:

- The cause of leukoedema ➔ **unknown**.
- Factors such as genetic factors, smoking, chewing tobacco, alcohol ingestion, and bacterial infection have been implicated in the pathogenesis of leukoderma.

Clinical Features:

- **Age:** 2nd & 3rd decades of life.
- **Sex:** Black males (due to the presence of background pigmentation that makes oedematous changes noticeable).
- **Site:** Bilateral, **symmetrically** distributed lesions on the buccal and labial mucosa are the most common sites.
- **Appearance:**
  - **In the early stage**: ➔ appears as asymptomatic, diffuse, translucent, grey-white, milky, opalescent lesions of the mucosa with a slightly folded appearance.
  - In **the exaggerated (late) stage**: ➔ appears as a whitish cast with surface textural changes, including wrinkling or corrugation may be seen.

- The lesion tends to to **disappear on stretching** ➔ it is not **hyperkeratinzation** (↑↑keratin production) **but** rather it represents **intracellular oedema** of the prickle cell layer (accumulation of fluid within epithelial cells).



Leukoedema

Histopathological Features:



- The epithelium is thickened by **acanthosis** with broad rete ridges.
- **Marked intracellular edema** of spinous cells, resulting in extremely large and pale vacuolated cells (the cytoplasm appears lost and nuclei appear absent or pyknotic).

Treatment& Prognosis:

Treatment is not necessary because **asymptomatic** so considered a normal variation **(developmental condition)** and benign no malignant potential exists.

Differential Diagnosis:

White sponge nevus, hereditary benign intraepithelial dyskeratosis, chronic cheek biting, and lichen planus.

☺☺*Thank you*☺☺

Oral epithelial tumors

DR. Haithem Elghariani

University of Tripoli ⁎ Faculty of Dentistry ⁎ Department of oral surger,pathology and medicine 2026.

Benign and Malignant Epithelial Non-Odontogenic Tumors

The cells of origin of benign and malignant tumours in oral epithelium:-

- ✓ The oral mucous membrane is divided into the stratified a squamous epithelium and the underluing connective tissue layer.
- ✓ The epithelium in turn is formed into ridges that protrude toward the lamina propria ,these ridges called epithelial ridges.

The oral mucosa is classified into:-

1.Keratinized or masticatory mucosa. Such as gingival, hard palate and dorsum of the tongue
2. Non-keratinized or lining mucosa. Such as lip, cheek, ventral surface of tongue, alveolar mucosa, floor of the mouth and soft palate.

- ✓ The stratified squamous epithelium contains three types of cells. These cells are important in origin of benign and malignant tumours.

1.Squamous cells (prickle cells)
2. Basal cells
3. Melanocytes

Terms used for histopathological change in the stratified squamous epithelium:-

1.Keratosis:-

Is used for a keratinized epithelium that is not normally keratinized.

2. Hyperkeratosis:-

Means increase in the thickness of keratin layer.

3. Orthokeratosis:-

The layers of the keratin (squames) are anucleated (absence of nucleus), flattened

and have homogenous acidophilic cytoplasm.

4. Parakeratosis:- The layers of the keratin (squames) are nucleated, flattened and have eosinophilic cytoplasm.

5. Basal cell hyperplasia:-
Increase in the number of basal cells.

6. Acanthosis:- It is a hyperplasia of the prickle cell layer causing increase in the epithelium thickness and producing broadening and lengthening of rate ridges.

7. Epithelial atrophy:-
Decrease in the size (thinning) of the epithelium that usually associated with loss of rate ridges.

8. Cellular atypia:-
Is a group of premalignant or malignant cells.

A. *Benign Epithelial Non-Odontogenic Tumors*

Benign Epithelial Lesions Associated With Human Papillomavirus:-

1) **Squamous Papilloma.**
2) **Verruca Vulgaris (Common Wart).**
3) **Verruciform Xanthoma.**
4) **Keratoacanthoma** (“Self-Healing "Carcinoma; Pseudo-Carcinoma).

B. Malignant non-odontogenic Epith-tumors.

1) Squamous cell carcinoma Scc and its variants:

a) Verrucous carcinoma.
b) Spindle cell carcinoma.
c) Adenosquamous carcinoma.
d) Nasopharyngeal carcinoma.
e) Basaloid squamous carcinoma.

2) Basal cell carcinoma (Rodent ulcer).

3) Melanoma.

A.Benign Non-Odontogenic Epith-Tumors

1) Squamous Papilloma:-

The squamous papilloma is a benign, HPV-induced proliferation of stratified squamous epithelium, resulting in a papillary or verruciform mass.
HPV types 6 and 11 are identified most commonly, with an extremely low virulence and infectivity rate.

Clinical Features:-

» Soft, painless, usually pedunculated, exophytic nodule with numerous fingerlike surface projections a "cauliflower" or wart like appearance.
» Sex:- No significant gender predilection or slight male predominance.
» Age:- any age, but 30 to 50 years old.
» Sites: palate, tongue, and lips, but any oral surface may be affected.
» Size:- solitary and enlarges rapidly to a maximum size of about 0.5 cm - 3.0 cm in greatest diameter.

Histopathologic Features:-

» The papilloma is characterized by a proliferation of keratinized

stratified squamous epithelium arranged in fingerlike projections with fibrovascular connective tissue cores.

» The connective tissue may show inflammation.
» Some papillomas demonstrate basilar hyperplasia and mitotic activity, which can be mistaken for mild epithelial dysplasia.
» Koilocytes (virus altered) epithelial cells with crenated, pyknotic [small and dark] nuclei surrounded by clear halos) sometimes high in the spinous cell layer.

Treatment and Prognosis:-

» Conservative surgical excision, including the base of the lesion, is adequate treatment and recurrence is unlikely.

2) Verruca Vulgaris (Common Wart):-

» Verruca vulgaris is a focal, benign, HPV-induced hyperplasia of stratified squamous epithelium.
» HPV 2 is present most often, although other HPV types may be found as well.

» It is contagious and can spread to other parts of skin or mucosa by autoinoculation.
» It infrequently develops on oral mucosa but is
» extremely common on the skin.

Clinical Features:-

» appears as a painless papule or nodule with papillary projections. It may be pedunculated or sessile.
» Cutaneous lesions may be pink, yellow, or white; oral lesions are almost always white.
» Age:- In children.
» Site:- skin of the hands is the most commonly involved site.
» Oral mucosal lesions (vermilion border, labial mucosa, or anterior tongue).
»
» Size:- enlarges rapidly (usually <5 mm).
» Multiple or clustered lesions are common.

Histopathologic Features:-

» Verruca vulgaris is characterized by a proliferation of hyperkeratotic stratified squamous epithelium arranged in fingerlike, pointed projections with connective tissue cores.
» Chronic inflammatory cells often infiltrate the supporting connective tissue.
» Elongated rete ridges tend to converge toward the center of the lesion, producing a "cupping" effect.
» A prominent granular cell layer.
» Abundant koilocytes often are seen in the superficial spinous layer .

Treatment and Prognosis:-

Two-thirds will disappear spontaneously within(2 years), especially in children.

Treated by chemical drugs, or by Surgical excision.

3) Verruciform Xanthoma:-

It is a hyperplastic condition of the epithelium, with a characteristic subepithelial accumulation of lipid-laden histiocytes. The cause is unknown.

Clinical Features:-

- » Appears as a well-demarcated, soft, painless, sessile, slightly elevated mass with a white color and a papillary or roughened (verruciform) surface.
- » Typically is seen in whites.
- » Age:- 40 to 70 years of age.
- » Sex:- Slight male predilection.
- » Site:- occur on the gingiva and alveolar mucosa.
- » Size:- smaller than 2 cm; no oral lesion larger than 4 cm has been reported

Histopathologic Features:-

- » Verruciform xanthoma demonstrates papillary, acanthotic surface epithelium covered by a thickened layer of parakeratin.
- » Clefts or crypts between the epithelial projections are filled with parakeratin, and rete ridges are elongated to a uniform depth.
- » The most important diagnostic feature is the accumulation of numerous large macrophages with foamy cytoplasm, which typically are confined to the connective tissue papillae.
- » These foam cells, also known as xanthoma cells,

4) Keratoacanthoma:-

("Self-Healing "Carcinoma; Pseudo-Carcinoma; Keratocarcinoma)

- » Self-limiting, epithelial proliferation.
- » Strong clinical and histopathologic similarity to well differentiated squamous cell carcinoma.
- » Intraoral lesions have been reported, but they are rare.

Clinical Features:-

- » Asymptomatic appears as a firm, well-demarcated, sessile, dome-shaped nodule with a central plug of keratin.
- » Age:- occurs before 45 years of age.
- » Sex:- male.
- » Site:- Almost 95% sun-exposed skin, and 8% of all cases involve the outer edge of the vermilion border of the lips

The evolution of keratoacanthoma can be divided into three phases :

a) growth phase: rapid enlargement is typical, with the lesion usually attaining a diameter of 1 to 2 cm within 6 weeks.
b) stationary phase: lesion stabilized, which usually is of similar duration as the growth phase.
c) involution phase: most lesions regress spontaneously within 6 to 12 months of onset, frequently leaving behind a depressed scar.

Histopathologic Features:-

- » Dyskeratosis (abnormal keratin production) typically is seen in the form of deeply located individually keratinizing cells and keratin pearls similar to those found in well-differentiated squamous cell carcinoma.
- » Late-stage lesions show considerably more keratinization in the deeper aspects of the tumor than do early lesions.
- » Chronic inflammatory cell response with migration of eosinophils or neutrophils into the epithelium with resultant microabscess formation

Treatment and Prognosis:-

- » Surgical excision without waiting for spontaneous regression is prefered.
- » Early treatment may prevent significant scarring. Approximately 4% to 8% of lesions recur after excision.

B. *Malignant non-odontogenic epithelial tumors*

Cancer:- is the end product of an unregulated proliferation of cells resulting from the accumulation of sequential genetic alterations (mutations) in a precursor cell.

Etiology of Oral Cancer:-

1. Tobacco Smoking:- Tobacco smoke contains more than 70 carcinogens including nitrosamines, arsenic, benzo[a]pyrene, and benzene.
2. Smokeless Tobacco:- 50% of all oral cancers in smokeless tobacco users occur at the site where the tobacco is habitually placed.
3. Betel Quid (Paan):- it is a combination of natural substances, chewed for their psychostimulating effects.
4. Alcohol:- The risk generally appears to be dose dependent and time-dependent.
5. Occupational Exposures and Environmental Pollutants:- Some studies have reported an increased oral cancer risk for workers in the wood products industry chronically exposed to certain chemicals, such as phenoxyacetic acids.
6. Radiation:- the x-irradiation decreases immune reactivity and produces chromosomal abnormalities.

7. Vitamin / Mineral Deficiencies and Dietary Factors:- Iron deficiency may cause impaired cell-mediated immunity and Vitamin-A deficiency produces excessive keratinization of the skin and mucous membranes.
8. Bacteria:- The potential for microflora of the oral cavity to contribute to carcinogenesis. Studies suggest that oral bacteria may interact with tobacco and alcohol.
9. Candida:- Hyperplastic candidiasis frequently is cited as an oral precancerous condition. Because this lesion appears as a white plaque that cannot be rubbed off, it also has been called *candidal leukoplakia.*
10. Oncogenic Viruses:- (tumor producing) viruses may play a major role in a wide variety of cancers. Viral integration into the host's genetic material may result in abnormal cell growth and proliferation.
11. Immunosuppression:- Persons with HIV infection and those who are undergoing immunosuppressive therapy for malignancy or organ transplantation are at increased risk for oral squamous cell carcinoma and other head and neck malignancies.

Epithelial dysplasia:-

It is not a clinical term but a histopathological term used to distinguish between the normal and abnormal epithelium that may has a premalignant or malignant transformation.

The features of epithelial dysplasia:-

1.Nuclear and cellular pleomorphism.

2. Alteration in nuclear/cytoplasmic ratio.

3. Prominent nucleoli.

4. Nculear hyperchromatism.

5. Abnormal mitosis.

6. Abnormal keratinization. (Individual cell keratinization within the epithelium).

7. Basal cell hyperplasia.

8. Disturbed polarity of basal cell layer. The basal cells have no longer show regular polarity.

9. Disturbed maturation of group of cells resulting in the formation of intra-epithelial keratin pearls.

10. Drop-shape rate ridges.

11. Irregular epithelial stratification.

Classification of epithelial dysplasia:-

- Not all these features are necessary seen in one case.
- Therefore, the epithelial dysplasia may be classified into three sub-types depending on the degree and the site of dysplasia.

1.Mild epithelial dysplasia:-

- This term is used to describe a minor degree of epithelial dysplastic changes involving the lower one/third of the epithelium.
- That usually associated with some inflammatory conditions.

2. Moderate epithelial dysplasia

- This term is used to describe an epithelial dysplastic changes involving the lower two/third of the epithelium.
- That usually indicating a premalignant lesion.

3. Sever epithelial dysplasia (Carcinoma in-situ):-

- This term is used to describe an epithelial dysplasia involving the whole or almost the whole thickness of the epithelium (top-to-bottom dysplastic changes), carcinoma-in- situ or intra-epithelial carcinoma.
- But the basement membrane is still intact and there is no invasion of the lamina propria.

- Carcinoma in-situ in the oral mucosa usually present clinically as leukoplakia, erythroplakia, candidal leukoplakia, or oral submucous fibrosis.
- In some patients it may progress to invasive carcinoma while in others it may remain static for a long period or it may regress with time.

B. Malignant non-odontogenic Epith-tumors:-

1) **Squamous cell carcinoma Scc and its variants:**

- Verrucous carcinoma.
- Spindle cell carcinoma.
- Adenosquamous carcinoma.
- Nasopharyngeal carcinoma.

» Basaloid squamous carcinoma.

2) Basal cell carcinoma (Rodent ulcer).

3) Melanoma.

a) Squamous cell carcinoma:-

» Is A malignant neoplasm of stratified squamous epithelium that is capable of locally destructive growth and distant metastasis.
» Beginning as an epithelial dysplasia and progressing until the dysplastic epithelial cells reach the basement membrane and invade into the connective tissue.
» Accounts for about 3% of all cancers in the U.S.A Sixth most common cancer in males & the 12th in females.
» More than 90% of all oral malignancies are squamous cell carcinoma.
» The risk of intraoral carcinoma increases with increasing age.
» The overall male to female ratio is 3:1.

Clinical features:-

It has many clinical presentations:

A. Exophytic mass.
B. Endophytic (Ulcerating).
C. Leukoplakic (white patch).
D. Erythroplakic (red patch).

✓ The leukoplakic and erythroplakic forms are mostly early cases that have not yet produced a mass or ulceration.

✓ An exophytic lesion typically has a surface that is irregular, fungating, papillary, or verruciform.
» Its color may vary from normal to white or red, depending on the amount of keratin and vascularity.
» The surface is often ulcerated, and the tumor feels hard (indurated) on palpation.

✓ The endophytic growth pattern has a central, depressed, irregularly shaped ulcer with a surrounding “rolled” border of pink, red, or white mucosa.
» The rolled border results from invasion of the tumor downward and laterally under adjacent epithelium.

» Perineural invasion may cause paresthesia.
» Destruction of underlying bone, when present, may be painful or completely painless; it appears on radiographs as a “moth-eaten” radiolucency with ill-defined or ragged margins (an appearance similar to osteomyelitis).

» Scc 90% of cases are located on the lower lip.
» Tongue (posterior lateral and ventral surfaces), floor of the mouth, soft palate, gingiva, buccal mucosa, and hard palate.
» the floor of the mouth lesions are the most likely to arise from a pre-existing leukoplakia or erythroplakia.

Histopathologic Features:-

» It is arises from dysplastic epithelium, with invasive islands and cords of malignant epithelial cells.

❖ *Depending on the degree of differentiation and amount of keratin the lesion is considered :-*

1) Well differentiated:- the malignant epithelium is obviously squamous in type and consists of masses of prickle cells with limiting layer of basal cells around the periphery and keratin pearls formation among the masses of infiltrating cells. Each keratin pearl consists of central area of keratin surrounded by prickle cells. , i.e. a tumor that closely resembles its tissue of origin

2) Moderate differentiated:- the keratin pearls are little and the abnormal mitosis present more than in well differentiated tumours.

3) Poorly differentiatedor anaplastic carcinoma:- enlarges rapidly and metastasizes early New small blood vessels (angiogenesis) and chronic inflammatory cellular infiltrate of variable density are present in the connective tissue.

Histopathologic grading:-

❖ It depends on the degree to which these tumors resemble their parent tissue, and production of their normal product (keratin).

Grade I: where islands of malignant epithelial cells show much, keratin pearls or individual cell keratinization, i.e. low grade, or well differentiated carcinoma.
Grade III and IV: where much cellular and nuclear pleomorphism and little orthokeratin production, i.e. high grade or poorly differentiated, or anaplastic carcinoma.

Grade II: A tumor with a microscopic appearance somewhere between the previous two extremes is called moderately differentiated carcinoma.

Clinical grading or staging of oral cancer:-

- ❖ is the process of determining the extent to which a cancer has developed by spreading.

TNM Clinical Staging System of Oral Cancer

Table1 TNM staging system for oral squamous cell carcinoma

T- Tumor

T_1- Tumor less than 2cm in diameter

T_2- Tumor 2-4 cm in diameter

T_3- Tumor greater than 4 cm in diameter

T_4- Tumor invade adjacent structure

N- Node

N_0- No palpable nodes

N_1- Ipsilateral palpable nodes

N_2- Contralateral or bilateral nodes

N_3- Fixed palpable nodes

M- Metastasis

M_0- No distant metastasis

M_1- Clinical or radiographic evidence of metastasis

Stage Grouping

Stage 0	**Tis**	**N_0**	**M_0**
Stage I	**T_1**	**N_0**	**M_0**
Stage II	**T_2**	**N_0**	**M_0**
Stage III	**T_1**	**N_1**	**M_0**
	T_2	**N_1**	**M_0**
	T_3	**N_0,N_1**	**M_0**
Stage IV	**T_4**	**N_0. N_1**	**M_0**
	Any T	**N_2,N_3**	**M_0**
	Any T	**Any N**	**M_1**

1. Squamous cell carcinoma and its variants:

» Verrucous carcinoma
» Spindle cell carcinoma
» Adenosquamous carcinoma
» Nasopharyngeal carcinoma
» Basaloid squamous carcinoma.

a) Verrucous Carcinoma *(Snuff Dipper's cancer)*

A diffuse, largely exophytic, superficial spreading, highly keratinized, warty form of well differentiated squamous cell carcinoma that is unlikely to metastasize.

Clinical Features:-

» The tumor usually occurs in males and tends to affect individuals over 60 years of age.
» Most intraoral cases involve the gingiva, alveolar mucosa, and buccal mucosa; however, the hard palate and floor of the mouth can also be involved.
» Most lesions arise in people who chronically use chewing tobacco or snuff, where the tobacco is placed habitually
» It grows slowly, appears as a diffuse, well demarcated, painless, thick plaque with papillary or verruciform surface projections.

Histopathologic Features:-

- » It has a misleading benign microscopic appearance.
- » The lesion has a wide elongated rete ridges, with abundant keratin (parakeratin production).
- » The surface has papillary or verruciform projections.
- » The parakeratin fills the numerous spaces between the surface projections.
- » The tumor cells show normal maturation pattern with no significant degree of cellular atypia.
- » Intense chronic inflammatory cells in the connective tissue.

b)Spindle Cell Carcinoma

- » It is an uncommon variant of squamous cell carcinoma.
- » The epithelial cells lose their cohesive character and polygonal shape and resemble malignant fibroblasts (spindle cells).
- » These lesions can be mistaken for fibrosarcoma.

Clinical Features:-

- » It occurs primarily in males and most often affects the lower lip and tongue. Occasionally, the alveolar mucosa or gingiva is involved.
- » This subtype of squamous cell carcinoma is often less aggressive than other forms of poorly differentiated carcinoma.
- » It appears as a pedunculated, polypoid mass, but occasionally it may appear as a sessile, nodular or fungating mass.
- » The surface often is ulcerated. Pain and paresthesia are prominent features.
- » The tumor grows rapidly, tends to metastasize early, and typically is diagnosed at a late stage.

Histopathologic Features:-

- » The malignant cells streaming from the tips of tapered and elongated epithelial rete pegs adjacent to the ulcerated zone.
- » In some lesions, in addition to the spindle cell component, areas of recognizable squamous cell carcinoma with foci of keratin formation are sometimes observed.
- » An inflammatory cell infiltrate consisting of lymphocytes and neutrophils or eosinophils is often present.
- » Immunohistochemical stain reveals cytokeratin and may or may not vimentin positive reaction, in contrast with mesenchymal tumor cells which are positive for vimentin not cytokeratin.

c) Adenosquamous carcinoma

» Rare, aggressive carcinoma of the mucosa consisting of a mixture of malignant squamous and glandular cells.

Clinical Features:-

» is a rare, aggressive carcinoma that occurs within the oral and nasal cavities and in the larynx.
» Within the oral cavity, it occurs primarily on the floor of the mouth and the hard palate.

Histopathologic Features:-

» It is appears to be simultaneously derived from the surface mucosa and the adjacent minor salivary gland ducts.
» The histologic appearance resembles a high-grade mucoepidermoid carcinoma.

d) Nasopharyngeal Carcinoma

» An aggressive form of squamous cell carcinoma located in the nasopharynx and having varying degrees of differentiation.
» The lesions in this site are usually less differentiated than those found in the oral cavity.
» The categories proposed of nasopharyngeal carcinoma were (1) keratinizing carcinoma, (2) Nonkeratinizing carcinoma, and (3) undifferentiated carcinoma.

Clinical Features:-

» It accounts for 0.25% of all cancers and occurs in all age groups but more commonly in males by a 3:1 ratio.
» Nasopharyngeal carcinoma is usually found during an investigation of an asymptomatic lateral neck mass, because the lymph nodes of the lateral neck are the usual sites of initial metastases.
» Other presentations are nasal obstruction, unilateral otitis media, epistaxis, and otalgia.

Histopathologic Features:-

» Show either
» Keratinizing squamous cell carcinoma Or non-keratinizing carcinoma Or undifferentiated (anaplastic) carcinoma.
» Keratinizing nasopharyngeal carcinomas exhibit all the features of the usual well-differentiated or moderately differentiated squamous cell carcinoma found elsewhere in the oropharynx.
» Undifferentiated carcinoma is the most common form. It differs from other carcinomas by having neoplastic cells that are often difficult to recognize as being of epithelial origin.
» They are also distinguished by the presence of large aggregates of nonneoplastic lymphocytes surrounding the epithelial component.
» Because of the intermingling of the two types of tissues, this form of the nasopharyngeal carcinoma was previously designated as lymphoepithelioma.

e) Basaloid Squamous Cell Carcinoma

» Rare, aggressive form of poorly differentiated squamous cell carcinoma consisting of medullary patterns of cells with central areas of necrosis.

Clinical Features:-

» It is an aggressive malignancy that occurs primarily at the base of the tongue, larynx, piriform sinus, and tonsil.
» Males are affected far more often than females; the gender ratio is approximately 7:1.
» Most of the reported cases have been in habitual smokers, many of whom also consumed significant amounts of alcohol.
» The average age of the patients affected by this tumor is approximately 60 years.

Histopathologic Features:-

» basaloid squamous carcinoma has two microscopic components.
» The first is a superficial, well differentiated or moderately differentiated squamous cell carcinoma.
» The second, deeper component is an invasive basaloid epithelium arranged in islands, cords, and gland like lobules.

b) Basal Cell Carcinoma *(Rodent Ulcer)*

- » A locally invasive, slowly spreading epithelial malignancy.
- » Arises from the basal cell layer of the skin and its appendages, It does not arise on mucous membranes; however, it can involve mucous membranes by directly spreading from adjacent skin.
- » 80% of cases occur in the skin of the head& neck region, due to chronic exposure to UV radiation.
- » Multiple naevoid basal cell carcinomas are characteristic feature of naevoid basal cell carcinoma syndrome.

Clinical Features:-

- » It starts as a slightly elevated painless papule that slowly enlarges and eventually develops a central depression with an umblicated appearance,
- » Crusted ulcer with an elevated smooth rolled border where history of intermittent bleeding followed by healing.
- » If untreated, the tumor enlarges and invades adjacent tissues and structures by direct extension (Rodent ulcer) but rarely metastasizes.

Histopathologic Features:-

- » The basal cell carcinoma is characterized by a proliferation of Uniform, dark staining basaloid cells with oval nuclei and relatively little cytoplasm.
- » Arranged in well demarcated islands and strands appear to arise from the basal layer (i.e. budding of the periphery of these islands), invade into the CT.
- » Palisading of the peripheral cells with a clear zone between the epithelial islands and the connective tissue.
- » Some lesions show areas of keratin production.

Nevoid Lesion (Nevus)

- » Nevus is a general term that refers to any congenital lesion of various cell types as congenital vascular lesions.

Clinical Features:-

- » Nevi usually appear shortly after birth or childhood, at any site, but
- » intraoral lesions are rare.

» Occurs on the hard palate, less on the buccal or gingival mucosa,
» alveolar ridge and the vermilion border.
» They appear as elevated papules, may be non-pigmented.

Histopathologic Features:-

» Several subtypes are seen, depending on the location of the cells:

a- Junctional nevus:

» Cells at the junction of epithelium and connective .Rare in the oral cavity.
» They have malignant potentiality (specially the skin lesions).

b- Intradermal or intramucosal:

» Cells in the dermis or submucosa.
» It is the most common in the oral cavity.

c- Compound nevus:

» - Are seen in combination of zones, rare orally.

d- Blue nevus:

» Are spindle-shaped.
» Located deep in the connective tissue.

c)Melanoma
(Malignant Melanoma)

» Malignant melanoma may be predisposed by:-
» Excessive exposure to ultra-violet light.
» Congenital or acquired melanocytic naevi and other genetic and racial factors.

Clinical Features:-

» Is rare and slightly more common in males than females.
» More than 70% of cases affect the posterior maxillary alveolar ridge and hard palate.
» Present as a nodular or papillary lesion of dark-brown or bluish-black in colour.
» Spread rapidly into the surrounding tissue results in extensive bone destruction, loosening of teeth, lymphatic and blood spread to distant organs.
» In most patients, malignant melanoma has a very poor prognosis.

✓ Four types or melanoma:-

a) Superficial Spreading Melanoma
b) Nodular Melanoma.
c) Lentigo Maligna Melanoma.
d) Acral Lentiginous Melanoma

Histopathologic Features:-

» The tumour usually consists of typical spindle-shaped melanocytes with cytoplasmic vaculation and cytological criteria of malignancy.